MAT 2039 · EXAM GUIDE

# Differential Equations, calmly

Every method from weeks 1–6 explained step by step, with the 2023 midterm fully solved, a cheat sheet and practice tests.

Covers Weeks 1–6 and the 2023–2024 midterm · Practise online: brtzl.github.io/study-shelf

# What's inside


**1 Integration toolkit** 5

Every method in this course ends with "now integrate both sides". This short refresher collects exactly the derivatives, integrals and tricks the lecturer's solutions use, so the calculus never stops you in the middle of a DE.

**2 Differential equations and their solutions** 11

After this chapter you can look at any equation and say at once: ODE or PDE, which order, linear or non-linear. You will also know exactly what a solution is, and the difference between general, particular and singular solutions.

**3 Initial value problems and the existence–uniqueness theorem** 18

Here you learn to pick the one solution that passes through a given point, and to use Theorem 1 to say whether such a solution exists and is unique, without solving anything.

**4 Separable equations** 22

The first real solving method: move all the y's to one side and all the x's to the other, then integrate. You will also learn the step most students forget, which is checking for lost (singular) solutions.

**5 Exact equations** 28

Some equations are secretly the total differential of a function F(x,y). Spot them with one test, rebuild F in two ways, and the answer is simply F(x,y) = C.

**6 Integrating factors** 33

When an equation fails the exactness test, you can often repair it by multiplying by a well-chosen $\mu(x)$ or $\mu(y)$. This chapter shows how to decide which one, how to compute it, and how to finish. It is exactly midterm Question 2.

**7 First-order linear equations** 39

The most reliable recipe of the course: put the equation in standard form, multiply by $\mu = e^{\int p}$, and the left side becomes one derivative. You will also see when to swap the roles of x and y.

**8 Bernoulli equations** 44

A Bernoulli equation is a linear equation spoiled by one power of y. After this chapter you can spot it, kill the power with u = y^(1−n), solve the linear equation, and remember the solution y = 0 that the substitution hides.

**9 Homogeneous equations (y = ux)** 50

Some equations only care about the ratio y/x. After this chapter you can test for that, substitute y = ux to get a separable equation, and collect every solution — including the ones lost when you divide.

**10 Shifting and clever substitutions** 56

Equations with linear expressions like (x − 2y + 1) are almost homogeneous — a shift of the origin (or the substitution z = ax + by) fixes them. After this chapter you can do that, and you can use a hint like u = sin y to reduce an equation to a linear one.

**11 Riccati and Clairaut equations** 61

Two named equations with their own tricks. Riccati: if someone hands you one particular solution, a single substitution gives all the others. Clairaut: y = xp + f(p) is solved by straight lines y = cx + f(c), plus one special curve, the singular solution.

12 **Orthogonal trajectories** 66
Given a family of curves, find the curves that cross every one of them at a right angle. It is a three-step recipe: get the family's slope, flip it to −1/slope, solve the new (usually separable) equation.

13 **Applications: cooling, falling bodies, decay and mixtures** 71
Turn a word problem into a first-order differential equation, solve it with methods you already know, and use the numbers in the story to pin down the constants. Question 1 of the midterm is always one of these.

14 **Higher-order linear equations and reduction of order** 82
Learn the language of n-th order linear equations (operator, IVP, linear independence, Wronskian, fundamental set, general solution) and the reduction-of-order trick that turns one known solution into a second one — that trick is midterm Question 5.

15 **Which method? Recognising the equation** 93
The exam never tells you which method to use. This chapter is a checklist you can run in your head in under a minute for any first-order equation.

16 **The 2023 midterm, fully solved** 97
The real MAT 2039 midterm of 17.11.2023: five questions, 20 points each. For each one you get how to recognise it, a clean solution that follows the lecturer's handwritten steps (corrected where the handwriting has a mistake), and the traps that cost points.

17 **Cheat sheet** 103
Two dense pages to print and review before the exam

18 **Self-test** 105
30 multiple-choice questions and one full mock midterm

19 **Answers** 111
With a short explanation for every question

# How to use this guide

Each chapter follows the slides in order: the idea in plain words, the rule, the lecturer's examples worked step by step, then the traps.

| Colored box | What it tells you |
|---|---|
| ! Red | A trap: where students lose points. |
| ★ Yellow | A tip or memory trick. |
| → Teal | A key idea worth pausing on. |
| ✓ Purple | Learn this: very likely on the exam. |
| i Grey | A side note, including places where the slides have a small mistake. |

## The midterm, in one picture

The 2023 midterm had 5 questions × 20 points, and each one tested a different method. The exam never tells you which method to use, so Chapter 15 ("Which method?") is the most important chapter for the exam.

| Question | Type | Chapters |
|---|---|---|
| 1 | Word problem → model → solve | 13 |
| 2 | Not exact → integrating factor | 5–6 |
| 3 | Substitution → Bernoulli / linear | 7–10 |
| 4 | "Find all solutions" (don't lose any) | 4, 8–9 |
| 5 | Reduction of order + general solution | 14 |

## A plan if you have a week

| Day | Read | Then do |
|---|---|---|
| 1 | Chapters 1–4 | Practice: Basics, IVP, Separable. |
| 2 | Chapters 5–7 | Solve the exact / integrating-factor examples again without looking. |
| 3 | Chapters 8–12 | Practice: Bernoulli, Homogeneous, Riccati, Clairaut. |
| 4 | Chapters 13–14 | Practice: Applications and Higher-order. |
| 5 | Chapter 15 + cheat sheet | Mock midterm 1 on the web page, timed. |
| 6 | Chapter 16 | Real 2023 midterm, timed, then compare. |
| 7 | Your mistakes | Mock midterms 2–4. |

→ **The web page: brtzl.github.io/study-shelf**

Every question here plus more, timed mock midterms with step-by-step solutions, and flashcards are at **https://brtzl.github.io/study-shelf/**. It works on a phone and saves your progress.

CHAPTER 1

# Integration toolkit

Every method in this course ends with “now integrate both sides”. This short refresher collects exactly the derivatives, integrals and tricks the lecturer's solutions use, so the calculus never stops you in the middle of a DE.

Collected from the worked solutions in Week+3.pdf and Week+4.pdf

You already know calculus; it is just rusty. You do **not** need everything from Calculus I. The differential-equation methods in this course keep using the same dozen or so facts. Learn these well and the DE part becomes the only thing you have to think about.

→ **How a DE solution really works**

Each method (Bernoulli, homogeneous, Riccati, ...) is a **substitution** that turns a scary equation into one of two easy kinds: **separable** (put all $u$'s on one side, all $x$'s on the other, integrate) or **linear** (multiply by an integrating factor, integrate). So the last step is always an integral.

## 1. Derivatives you must know instantly

| $f(x)$ | $f'(x)$ | $f(x)$ | $f'(x)$ |
|---|---|---|---|
| $x^n$ | $nx^{n-1}$ | $\sin x$ | $\cos x$ |
| $\ln\lvert x\rvert$ | $\dfrac{1}{x}$ | $\cos x$ | $-\sin x$ |
| $e^{ax}$ | $ae^{ax}$ | $\tan x$ | $\sec^2 x = 1+\tan^2 x$ |
| $\dfrac{1}{x} = x^{-1}$ | $-\dfrac{1}{x^2}$ | $\arctan x$ | $\dfrac{1}{1+x^2}$ |

Three rules do the rest:

- **Chain rule** — the heart of every substitution. If $u = y^{-1}$ and $y$ depends on $x$, then $\dfrac{du}{dx} = -y^{-2}\dfrac{dy}{dx}$. Never forget the $\dfrac{dy}{dx}$ at the end.
- **Product rule** — $(\mu u)' = \mu u' + \mu' u$. The integrating-factor trick is this rule read backwards.
- **Implicit differentiation** — $\dfrac{d}{dx}(y^2) = 2y\,y'$, $\dfrac{d}{dx}(e^y) = e^y y'$. You need it in orthogonal trajectories and in Clairaut equations.

**CHAIN RULE AS USED IN WEEK 3, P.4** If $u = y^{1-n}$, find $\dfrac{du}{dx}$.

1. Differentiate the power: $\dfrac{d}{dy}\, y^{1-n} = (1-n)y^{-n}$.
2. Multiply by the inner derivative $\dfrac{dy}{dx}$ ($y$ is a function of $x$).

**ANSWER**

$$\frac{du}{dx} = (1-n)\, y^{-n}\, \frac{dy}{dx}$$

## 2. Integrals that keep coming back

| Integral | Answer (+ C) | Where it appears |
|---|---|---|
| $\int x^n\, dx,\ n \neq -1$ | $\dfrac{x^{n+1}}{n+1}$ | everywhere |
| $\int \dfrac{dx}{x}$ | $\ln\lvert x\rvert$ | every separable equation |
| $\int \dfrac{f'(x)}{f(x)}\, dx$ | $\ln\lvert f(x)\rvert$ | integrating factors |
| $\int e^{ax}\, dx$ | $\dfrac{1}{a}e^{ax}$ | linear equations |
| $\int \tan x\, dx$ | $-\ln\lvert\cos x\rvert$ | Bernoulli Example 16 |
| $\int \cot x\, dx$ | $\ln\lvert\sin x\rvert$ | Bernoulli Exercise 5.4 |
| $\int \sec^2 x\, dx$ | $\tan x$ | substitution $u = x + y$ |
| $\int \cos^2 x\, dx$ | $\dfrac{x}{2} + \dfrac{\sin 2x}{4}$ | substitution $u = x + y$ |
| $\int \dfrac{dx}{1+x^2}$ | $\arctan x$ | special transformation |
| $\int \dfrac{dx}{a^2+x^2}$ | $\dfrac{1}{a}\arctan\dfrac{x}{a}$ | special transformation |

★ **The one integral to rule them all**

$\displaystyle\int \frac{f'(x)}{f(x)}\, dx = \ln\lvert f(x)\rvert$. Whenever the top is (a multiple of) the derivative of the bottom, the answer is a logarithm. $\int \frac{2x}{x^2+1}dx = \ln(x^2+1)$, $\int \frac{1}{x\ln x}dx = \ln\lvert\ln x\rvert$ (Week 4 quiz).

## 3. Substitution (u-substitution)

Pick the "inside" piece $z$, compute $dz$, and rewrite the whole integral in $z$. The lecturer does this in the margin of almost every solution.

**INTEGRAL OF TAN X — WEEK 3, P.5** Compute $\int \tan x \, dx$.

1. Write $\tan x = \dfrac{\sin x}{\cos x}$.
2. Let $z = \cos x$, so $dz = -\sin x \, dx$ (the numerator is almost $dz$).
3. $\displaystyle\int \frac{\sin x}{\cos x} dx = -\int \frac{dz}{z} = -\ln|z| = -\ln|\cos x|$.

**ANSWER**
$\int \tan x \, dx = -\ln|\cos x| + C$, so $e^{-\int \tan x \, dx} = e^{\ln|\cos x|} = \cos x$ (up to sign)

**WEEK 4, P.16 (QUIZ)** Compute $\displaystyle\int \frac{\ln x}{x} dx$.

1. Let $z = \ln x$, $dz = \dfrac{1}{x} dx$.
2. $\displaystyle\int z \, dz = \frac{z^2}{2}$.

**ANSWER**
$$\int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2} + C$$

**WEEK 3, P.23** Compute $\displaystyle\int \frac{u}{u^2 - 1} du$.

1. Let $z = u^2 - 1$, $dz = 2u \, du$, so $u \, du = \frac{1}{2} dz$.
2. $\displaystyle\int \frac{u \, du}{u^2 - 1} = \frac{1}{2}\int \frac{dz}{z} = \frac{1}{2}\ln|u^2 - 1|$.

**ANSWER**
$\frac{1}{2}\ln|u^2 - 1| + C$

## 4. Integration by parts

$$\int f \, g' \, dx = f \, g - \int f' \, g \, dx$$

**INTEGRATION BY PARTS**

Choose $f$ = the part that gets *simpler* when differentiated (a polynomial like $x$, or $\ln x$); $g'$ = the part you can integrate (like $e^{-x}$, $\dfrac{1}{x^2}$).

**TYPICAL IN LINEAR EQUATIONS** Compute $\displaystyle\int x e^{-x} \, dx$.

1. $f = x$ ($f' = 1$), $g' = e^{-x}$ ($g = -e^{-x}$).

2. $\int xe^{-x}dx = -xe^{-x} + \int e^{-x}dx = -xe^{-x} - e^{-x}$.

**ANSWER**

$$\int xe^{-x}dx = -(x+1)e^{-x} + C$$

**NEEDED IN BERNOULLI PROBLEMS WITH LN X** Compute $\int \frac{\ln x}{x^2}\,dx$.

1. $f = \ln x$ ($f' = \frac{1}{x}$), $g' = x^{-2}$ ($g = -\frac{1}{x}$).
2. $-\frac{\ln x}{x} + \int \frac{1}{x^2}dx = -\frac{\ln x}{x} - \frac{1}{x}$.

**ANSWER**

$$\int \frac{\ln x}{x^2}dx = -\frac{\ln x + 1}{x} + C$$

## 5. Partial fractions

When a separable equation gives you $\int \frac{\text{polynomial}}{(\ldots)(\ldots)}$, split the fraction into simple pieces, each of which integrates to a logarithm.

**WEEK 4, P.19 (QUIZ Q2)** Split $\frac{1}{u(u-1)}$ and integrate.

1. Write $\frac{1}{u(u-1)} = \frac{A}{u} + \frac{B}{u-1}$.
2. Cover-up: at $u = 0$: $A = \frac{1}{0-1} = -1$; at $u = 1$: $B = \frac{1}{1} = 1$.
3. $\int \left(\frac{1}{u-1} - \frac{1}{u}\right)du = \ln|u-1| - \ln|u| = \ln\left|\frac{u-1}{u}\right|$.

**ANSWER**

$$\int \frac{du}{u(u-1)} = \ln\left|\frac{u-1}{u}\right| + C$$

**WEEK 3, P.32 (EXAMPLE 20)** Split $\frac{4-3z}{(3z+1)(z-1)}$.

1. $\frac{4-3z}{(3z+1)(z-1)} = \frac{A}{3z+1} + \frac{B}{z-1}$, so $4 - 3z = A(z-1) + B(3z+1)$.
2. Compare coefficients (the lecturer's way): $z$: $A + 3B = -3$; constant: $-A + B = 4$. Adding: $4B = 1$, $B = \frac{1}{4}$, $A = -\frac{15}{4}$.
3. Then $\int \frac{-15/4}{3z+1}dz = -\frac{5}{4}\ln|3z+1|$ (the inner derivative 3 divides out) and $\int \frac{1/4}{z-1}dz = \frac{1}{4}\ln|z-1|$.

**ANSWER**

$-\frac{5}{4}\ln|3z+1| + \frac{1}{4}\ln|z-1| + C$

## 6. Logs, exponentials and the integrating factor

$$e^{\ln f} = f, \qquad e^{-\ln f} = \frac{1}{f}, \qquad e^{a\ln x} = x^a, \qquad a\ln x = \ln x^a, \qquad \ln A + \ln B = \ln(AB)$$

**RULES USED IN EVERY INTEGRATING FACTOR**

- $e^{-\int \frac{3}{x}dx} = e^{-3\ln x} = x^{-3} = \dfrac{1}{x^3}$ (Week 3, p.40).
- $e^{-\int \frac{dx}{x+1}} = e^{-\ln(x+1)} = \dfrac{1}{x+1}$ (Week 3, p.15).
- $e^{\int \frac{dx}{x\ln x}} = e^{\ln(\ln x)} = \ln x$ (Week 4, p.18).
- $4\ln|\cos x| = \ln\cos^4 x$ (Week 3, p.6).

> **Absolute values inside the integrating factor**
> Strictly $e^{\ln|\cos x|} = |\cos x|$. The lecturer simply writes $\cos x$: any nonzero constant multiple of an integrating factor still works, and on an interval $|\cos x| = \pm\cos x$. You may do the same.

### The linear-equation engine

Bernoulli, Riccati and many substitutions all end in a **linear** equation. This is the routine the lecturer writes every time:

$$\frac{du}{dx} + P(x)\,u = Q(x), \qquad \mu(x) = e^{\int P(x)\,dx}, \qquad \int \frac{d}{dx}(\mu\,u)\,dx = \int \mu\,Q\,dx \;\Rightarrow\; u = \frac{1}{\mu}\Big(\int \mu Q\,dx$$

**FIRST-ORDER LINEAR EQUATION**

Why it works: after multiplying by $\mu$, the left side $\mu u' + \mu P u$ is exactly $(\mu u)'$ by the product rule, because $\mu' = P\mu$.

## 7. Handling the constant like the lecturer

The constant of integration may be renamed freely, as long as it stays an arbitrary constant. The lecturer's favourite move: write the constant as a logarithm so the logs combine.

**PATTERN FROM WEEK 3, P.24** Solve $\ln|z| = \ln|x| + C$ for $z$.

1. Write the constant as $\ln|C|$ with $C \neq 0$: $\ln|z| = \ln|x| + \ln|C| = \ln|Cx|$.
2. Remove the logs: $|z| = |Cx|$, so $z = \pm Cx$.
3. Rename: $K = \pm C$, with $K \neq 0$. So $z = Kx$, $K \neq 0$.

4 Check $z = 0$ separately (it was lost when you divided by $z$). If $z = 0$ also solves the equation, allow $K = 0$ too and call the constant $D \in \mathbb{R}$.

**ANSWER**
$\ln|z| = \ln|x| + C \implies z = Dx,\ D \in \mathbb{R}$ (when $z = 0$ is also a solution)

- $e^{x+C} = e^C e^x = Ke^x$ with $K = \pm e^C \neq 0$ (Week 3, p.44).
- $\frac{1}{4}\ln|A| = \ln|Cu|$ gives $|A| = C^4u^4$ — rename $\tilde{C} = C^4 > 0$, then $A = Ku^4$, $K = \pm\tilde{C} \neq 0$ (Week 3, p.33).
- $\frac{1}{m-1} \cdot \frac{1}{z^{m-1}} = x + C$ can be rewritten with $\tilde{C} = (m-1)C$ (Week 3, p.43).

**Dividing means losing**
Every time you divide by something ($y$, $u$, $u^2 - 1$, $z - 1$ ...) to separate variables, write "for ... $\neq 0$" and afterwards **plug those values back into the original equation**. Some are solutions you must list, some are already inside the family, and some are not solutions at all. The lecturer checks this on every example — and it is worth points.

## In one minute

- Every method ends in a **separable** or a **linear** equation, then an integral.
- Chain rule: $\frac{d}{dx}y^k = ky^{k-1}y'$ — the key to every substitution.
- $\int \frac{f'}{f} = \ln|f|$; $\int \tan x = -\ln|\cos x|$; $\int \cot x = \ln|\sin x|$.
- By parts for $xe^{ax}$, $\ln x \cdot x^k$; partial fractions for $\frac{1}{(\ldots)(\ldots)}$.
- $e^{\ln f} = f$, $e^{a\ln x} = x^a$: integrating factors simplify to plain functions.
- Linear: $\mu = e^{\int P}$, $(\mu u)' = \mu Q$.
- $\ln|y| = \ln|x| + \ln|C| \Rightarrow y = Kx$; then check the value you divided by (maybe $K = 0$ is allowed).

CHAPTER 2

# Differential equations and their solutions

After this chapter you can look at any equation and say at once: ODE or PDE, which order, linear or non-linear. You will also know exactly what a solution is, and the difference between general, particular and singular solutions.

From Week+1-3.pdf, pages 6–21

## What is a differential equation?

In ordinary algebra the unknown is a *number* (solve $x^2 = 4$). In a differential equation the unknown is a *function* $y(x)$, and the equation tells you something about its derivatives.

**Definition 1 (Week+1-3.pdf p.6)**

A **differential equation** is an equation involving one or more derivatives of an unknown function.

### ODE or PDE?

Look at what the unknown function depends on.

- If it depends on **one** variable only ($y = y(x)$), you only see ordinary derivatives $\frac{dy}{dx}, y'', \ldots$. That is an **ordinary differential equation (ODE)**.
- If it depends on **several** variables ($u = u(x, y)$), you see partial derivatives $\frac{\partial u}{\partial x}$. That is a **partial differential equation (PDE)**.

**EXAMPLE 1 · WEEK+1-3.PDF P.7** Classify: (1) $\frac{dy}{dx} + y = x^2$; (2) $y'' = -k^2 y$; (3) $\frac{d^4y}{dx^4} + \left(\frac{d^2y}{dx^2}\right)^2 + \cos x = 0$; (4) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

1. In (1), (2), (3) the unknown $y$ depends only on $x$. (only ordinary derivatives appear)
2. In (4) the unknown $u$ depends on $x$ and $y$. (partial derivatives $\partial/\partial x$, $\partial/\partial y$)

**ANSWER**

(1), (2), (3) are ODEs; (4) is a PDE. In this course we only solve ODEs.

## Order

**Definition 2 (p.8)**

The **order** of a differential equation is the order of the highest derivative occurring in it. Any $n$-th order ODE can be written as $F(x, y, y', \ldots, y^{(n)}) = 0$. Here $y$ is the **dependent variable** and $x$ is the **independent variable**.

**Order is not power**

The order is about *how many times* you differentiate, not about exponents. In $\left(\frac{dy}{dx}\right)^3 = y^2$ the highest derivative is $y'$, so the order is **1** (the cube is a power, not a third derivative). In Example 1 (3), $(y'')^2$ has a square, but $y^{(4)}$ is there, so the order is **4**.

## Linear or non-linear?

**Definition 3 (p.9)**

An equation that can be written as

$$a_0(x)y^{(n)}(x) + a_1(x)y^{(n-1)}(x) + \cdots + a_n(x)y(x) = f(x),$$

where $a_0, a_1, \ldots, a_n$ (with $a_0 \neq 0$) and $f$ are functions of $x$ only, is a **linear differential equation of order** $n$. It is linear in $y, y', \ldots, y^{(n)}$. An equation that does not satisfy this definition is **non-linear**.

A practical way to test linearity: look only at $y$ and its derivatives. The equation is linear when

- $y, y', y'', \ldots$ appear only to the **first power**,
- they are **never multiplied with each other** (no $y\,y'$),
- they are **never inside another function** (no $\sin y$, $e^y$, $\sqrt{y}$).

The coefficients may be any functions of $x$: $x^2y'$ or $\frac{1}{1+x^2}y'$ or $\sin(x)\,y$ are all fine.

**EXAMPLE 2 · P.10** Which are linear? (1) $y'' + x^2y' + \sin(x)y = e^x$; (2) $xy'' + \frac{1}{1+x^2}y' = \sin x$; (3) $y'' + x\sin(y') - xy = x^2$; (4) $y' = \sqrt{y/x}$; (5) $y'' + (2y')^2 = e^x$.

1. (1) and (2): every term is (function of $x$) × ($y$, $y'$ or $y''$). ($\sin x$ and $\frac{1}{1+x^2}$ are coefficients; they only contain $x$)
2. (3): $y'$ sits inside $\sin$. (a function of the unknown)
3. (4): $y$ sits inside a square root.
4. (5): $y'$ is squared.

**ANSWER**

(1), (2) linear (both order 2); (3), (4), (5) non-linear.

**EXERCISES 1 · P.11** Classify by type, order and linearity: (1) $y\frac{dy}{dx} = x$; (2) $\frac{d^2y}{dx^2} - 16y = 0$; (3) $\left(\frac{dy}{dx}\right)^3 = y^2$; (4) $\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial u}{\partial y}\right) = xyu$; (5) $\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = 0$.

1. (1) ODE, order 1, **non-linear**. ($y \cdot y'$ is a product of the unknown with its derivative)
2. (2) ODE, order 2, **linear**. ($a_0 = 1, a_1 = 0, a_2 = -16, f = 0$)
3. (3) ODE, order 1, **non-linear**. ($y'$ is cubed and $y$ is squared)
4. (4) PDE, order 1, **non-linear**. (product of two derivatives of $u$)
5. (5) PDE, order 1, **linear**. (each derivative appears alone, to the first power)

**ANSWER**
(1) ODE·1·non-lin. (2) ODE·2·lin. (3) ODE·1·non-lin. (4) PDE·1·non-lin. (5) PDE·1·lin.

**Note**
Definition 3 on the slide is written for ODEs. For the two PDEs we use the same idea (the unknown $u$ and its partial derivatives appear only to the first power and are not multiplied together).

## What is a solution?

**Definition 4 (p.12)**
A **solution** of an $n$-th order differential equation on an interval $I$ is any function $y = y(x)$ that is (at least) $n$-times differentiable on $I$ and that satisfies the differential equation identically for all $x$ in $I$.

Two words matter here. **Identically** means the equation must hold for *every* $x$ in $I$, not just at a few points. **Interval** means a solution always comes with a domain; where the formula or the equation is undefined, the solution stops.

**EXAMPLE 3 · P.13** (1) Show that $y = e^{2x}$ is a solution of $y' = 2y$ for all $x \in \mathbb{R}$. (2) Where can solutions of $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}(y - 1)$ live?

1. Differentiate the candidate: $y' = 2e^{2x}$. (chain rule)
2. Substitute: left side $2e^{2x}$, right side $2y = 2e^{2x}$. They agree for every real $x$.
3. For (2): $\frac{1}{2\sqrt{x}}$ is undefined when $x \leq 0$. (square root of a negative / division by 0)

**ANSWER**
(1) $y = e^{2x}$ is a solution on $\mathbb{R}$. (2) Any solution can only be defined for $x > 0$.

### Explicit and implicit solutions

A solution can come in two shapes (p.14): **explicit** $y = \varphi(x)$ ($y$ is alone on one side) or **implicit** $\Phi(x, y) = 0$ (a relation between $x$ and $y$).

**Explicit solution (Def. 5)** A function $y = \varphi(x)$ that satisfies $F(x, y, y', \dots, y^{(n)}) = 0$ for all $x$ in $I$ when substituted for $y$.

**Implicit solution (Def. 6)** A relation $\Phi(x, y) = 0$ that defines one or more explicit solutions of the equation on $I$.

**EXAMPLE 4 · P.15** Show that $x^2 + y^2 = 4$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$. On which interval?

1. Solve the relation for $y$: $y = \pm\sqrt{4 - x^2}$. (the relation hides two functions)
2. Check one branch: $y = \sqrt{4 - x^2} \Rightarrow y' = \frac{-x}{\sqrt{4-x^2}} = -\frac{x}{y}$. The other branch works the same way.
3. At $x = \pm 2$ we get $y = 0$, but the equation contains $\frac{x}{y}$, so it is only defined for $y \neq 0$. (remove the endpoints)

**ANSWER**
$x^2 + y^2 = 4$ contains two explicit solutions $y = \pm\sqrt{4 - x^2}$, valid for $-2 < x < 2$.

★ **Checking an implicit solution fast**
Differentiate the relation implicitly: $2x + 2yy' = 0 \Rightarrow y' = -x/y$. That is exactly the equation, so it is a solution.

## General, particular and singular solutions

**EXAMPLE 5 · P.16** Find all solutions of $\frac{d^2y}{dx^2} = 5x$.

1. Integrate once: $\frac{dy}{dx} = \frac{5}{2}x^2 + c_1$. ($c_1$ arbitrary constant)
2. Integrate again: $y = \frac{5}{6}x^3 + c_1x + c_2$. (a second constant $c_2$)

**ANSWER**
$y = \frac{5}{6}x^3 + c_1x + c_2$ — the **general solution** (two constants for a 2nd-order equation).

**General solution (Def. 7, p.17)** A solution of an $n$-th order ODE that contains $n$ arbitrary constants.

**Particular solution (Def. 8, p.19)** A solution that does **not** contain any arbitrary constants. Assigning fixed values to the constants of the general solution gives a particular solution (Remark 2).

**Singular solution (Def. 9, p.20)** A solution that **cannot** be obtained from the general solution by any choice of the $n$ arbitrary constants.

**Example 6 (p.17).** $y'' = e^{-x}$. Integrating twice: general solution $y = e^{-x} + c_1 x + c_2$, $x \in \mathbb{R}$.

**Example 7 (p.19).** Choosing $c_1 = 1, c_2 = 0$ gives $y = e^{-x} + x$, a **particular** solution of $y'' = e^{-x}$.

**Remark 1 (p.18): not every equation has a general solution**

$\left(\frac{dy}{dx}\right)^2 + (y-1)^2 = 0$. A sum of two squares is 0 only if both are 0, so $y' = 0$ and $y = 1$. The only solution is $y(x) \equiv 1$. It has no arbitrary constant, so this equation has **no** general solution.

**EXAMPLE 8 · P.20** For $\left(\frac{dy}{dx}\right)^2 - 4y = 0$, show that $y = (x+c)^2$ is the general solution and that $y \equiv 0$ is a singular solution.

1. $y = (x+c)^2 \Rightarrow y' = 2(x+c)$, so $(y')^2 - 4y = 4(x+c)^2 - 4(x+c)^2 = 0$. (one constant, first order $\rightarrow$ general solution)
2. Particular solutions: $c = 2$ gives $y = (x+2)^2$; $c = -5$ gives $y = (x-5)^2$.
3. $y \equiv 0$: $y' = 0$, and $0 - 0 = 0$. So it is a solution.
4. Can $(x+c)^2$ be 0 for *all* $x$? No: whatever $c$ is, $(x+c)^2 > 0$ for $x \neq -c$. (no choice of $c$ works)

**ANSWER**

General solution $y = (x+c)^2$; $y \equiv 0$ is a **singular solution**.

**How to prove something is singular**

Two steps. (1) Plug it into the equation: it must be a solution. (2) Show that no value of the constant(s) in the general solution produces it *for all* $x$.

## Exercises 1.1 (p.21), solved

**EXERCISES 1.1 #1 · P.21** Show that $y'' - y = 0$ has the general solution $y = c_1 \cosh x + c_2 \sinh x$. Find $c_1, c_2$ so that $y(0) = 0$, $y'(0) = 1$.

1. $(\cosh x)' = \sinh x$, $(\sinh x)' = \cosh x$. So $y' = c_1 \sinh x + c_2 \cosh x$ and $y'' = c_1 \cosh x + c_2 \sinh x = y$. Hence $y'' - y = 0$. (2 constants, 2nd order $\rightarrow$ general)
2. $y(0) = c_1 \cosh 0 + c_2 \sinh 0 = c_1 = 0$. ($\cosh 0 = 1,\ \sinh 0 = 0$)
3. $y'(0) = c_1 \sinh 0 + c_2 \cosh 0 = c_2 = 1$.

**ANSWER**

$c_1 = 0,\ c_2 = 1$: the particular solution is $y = \sinh x$.

**EXERCISES 1.1 #2 · P.21** Find $\alpha \in \mathbb{R}$ such that $y'' - 4y' + 4y = 0$ has the general solution $y = e^{\alpha x}(c_1 x + c_2)$.

1. $y' = \alpha e^{\alpha x}(c_1 x + c_2) + c_1 e^{\alpha x}$. (product rule)
2. $y'' = \alpha^2 e^{\alpha x}(c_1 x + c_2) + 2\alpha c_1 e^{\alpha x}$.
3. Substitute and divide by $e^{\alpha x} \neq 0$: $(\alpha^2 - 4\alpha + 4)(c_1 x + c_2) + (2\alpha - 4)c_1 = 0$.
4. This must hold for **all** $c_1, c_2$ and all $x$: $\alpha^2 - 4\alpha + 4 = (\alpha - 2)^2 = 0$ and $2\alpha - 4 = 0$.

**ANSWER**
$\alpha = 2$: $y = e^{2x}(c_1 x + c_2)$.

**EXERCISES 1.1 #3 · P.21** Show that $2y\frac{dy}{dx} = x\left(4 + \left(\frac{dy}{dx}\right)^2\right)$ has the particular solution $y = x^2 + 1$, and find the one-parameter family that forms the general solution.

1. $y = x^2 + 1,\ y' = 2x$: left $2(x^2 + 1)(2x) = 4x^3 + 4x$; right $x(4 + 4x^2) = 4x + 4x^3$. Equal. ✓
2. Try the family $y = cx^2 + h$ (same shape as the particular solution): $y' = 2cx$.
3. Left: $2(cx^2 + h)(2cx) = 4c^2x^3 + 4chx$. Right: $x(4 + 4c^2x^2) = 4x + 4c^2x^3$.
4. Compare the $x$ terms: $4ch = 4 \Rightarrow h = \frac{1}{c}$.

**ANSWER**
General solution $y = cx^2 + \frac{1}{c}$ $(c \neq 0)$; $c = 1$ gives $x^2 + 1$.

**The slide says y = cx + h, but ...**
The slide asks for $h$ so that $y = cx + h$ is the general solution. If you try that, you get $2c^2 = 4 + c^2$ and $2ch = 0$, so only $c = \pm 2,\ h = 0$ works (that is not a one-parameter family). The family that fits the particular solution $x^2 + 1$ is $y = cx^2 + h$ with $h = 1/c$. We think the slide dropped the square.

**EXERCISES 1.1 #4 · P.21** Show that $y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$ has the general solution $y = cx + c^2$ and that $y = -\frac{1}{4}x^2$ is a singular solution.

1. $y = cx + c^2,\ y' = c$: right side $xc + c^2 = y$. ✓ (1 constant, 1st order → general)
2. $y = -\frac{1}{4}x^2,\ y' = -\frac{x}{2}$: right side $x\left(-\frac{x}{2}\right) + \frac{x^2}{4} = -\frac{x^2}{4} = y$. ✓
3. Could $cx + c^2 = -\frac{1}{4}x^2$ for all $x$? The left side is a line, the right side a parabola; no $c$ makes them equal.

**ANSWER**
$y = cx + c^2$ is the general solution; $y = -\frac{1}{4}x^2$ is a singular solution.

## In one minute

- A DE contains derivatives of an unknown function. One independent variable $\rightarrow$ **ODE**; partial derivatives $\rightarrow$ **PDE**.
- **Order** = highest derivative (not the highest power).
- **Linear**: $y, y', \ldots$ only to the 1st power, not multiplied together, not inside $\sin, e^{\cdot}, \sqrt{}$; coefficients depend on $x$ only.
- A **solution** is $n$-times differentiable on an interval $I$ and satisfies the equation for *all* $x \in I$. It can be explicit $y = \varphi(x)$ or implicit $\Phi(x, y) = 0$.
- **General** = $n$ arbitrary constants; **particular** = no constants; **singular** = a solution that no choice of constants produces.
- To show a function is a solution: differentiate and substitute. To show it is singular: also show no constant gives it.
- Some equations have no general solution ($(y')^2 + (y-1)^2 = 0$ has only $y \equiv 1$).

CHAPTER 3

# Initial value problems and the existence–uniqueness theorem

Here you learn to pick the one solution that passes through a given point, and to use Theorem 1 to say whether such a solution exists and is unique, without solving anything.

From Week+1-3.pdf, pages 22–26

## One-parameter families

A first-order equation $\frac{dy}{dx} = f(x, y)$ usually has infinitely many solutions. Solving it gives a formula with one constant, $y = \Phi(x, c)$. The slides call this the **one-parameter family of solutions** (p.22).

Each value of $c$ is one curve. An extra condition such as $y(0) = 1$ chooses one curve from the family.

**EXAMPLE 9 · WEEK+1-3.PDF P.22** $y = \cos x + c$ is a one-parameter family of solutions of $y' = -\sin x$. Which one satisfies $y(0) = 1$?

1. Check the family: $(\cos x + c)' = -\sin x$. ✓
2. Use the condition: $y(0) = \cos 0 + c = 1 + c = 1$. (put $x = 0$, $y = 1$)
3. So $c = 0$.

**ANSWER**

$y = \cos x$ is the only solution with $y(0) = 1$.

## What an IVP looks like

A problem of this kind is an **initial value problem (IVP)**. It always has two parts (p.23):

$$\begin{cases} \dfrac{dy}{dx} = f(x, y) & \text{differential equation} \\ y(x_0) = y_0 & \text{initial condition} \end{cases}$$

**INITIAL VALUE PROBLEM**

The recipe is always the same:

1. Find the general solution (one constant $c$).
2. Put $x = x_0$ and $y = y_0$ into it and solve for $c$.
3. Write the solution with that $c$, and say on which interval it lives.

## Theorem 1: existence and uniqueness

Before solving, we may ask: does the IVP have a solution at all? Is there only one? Theorem 1 answers both questions by looking only at $f$ and $\partial f/\partial y$.

**Theorem 1 (Existence and Uniqueness), p.24**

Let $f(x,y)$ and $\frac{\partial}{\partial y}f(x,y)$ be continuous on the rectangle $R := \{(x,y) : \ |x-x_0| \le a,\ |y-y_0| \le b\}$. Then there is a **unique** solution of the IVP

$$\begin{cases} \frac{dy}{dx} = f(x,y) \\ y(x_0) = y_0 \end{cases}$$

in some interval $I : \ |x-x_0| < h$, where $h > 0$.

How to use it, step by step:

1. Write the equation as $y' = f(x,y)$ and name $f$.
2. Compute $\frac{\partial f}{\partial y}$ (treat $x$ as a constant).
3. Find where $f$ is continuous and where $\frac{\partial f}{\partial y}$ is continuous. Typical trouble spots: division by zero, $\sqrt{}$ of something negative, $\ln$ of something $\le 0$.
4. If the point $(x_0, y_0)$ lies **inside** a region where both are continuous, you can draw a small rectangle $R$ around it: a unique solution exists near $x_0$.
5. If $\frac{\partial f}{\partial y}$ is not continuous at $(x_0, y_0)$, the theorem **says nothing**. There may be no solution, one, or several.

**What the theorem does NOT say**

It only promises a solution on *some* small interval $|x-x_0| < h$. It does not tell you how big $h$ is, and it does not promise the solution lives on all of $\mathbb{R}$. Also, if the hypotheses fail, that does **not** prove non-uniqueness; it just means the theorem can't help.

**EXAMPLE 10 · P.25** Consider $\frac{dy}{dx} = x\sqrt{y},\ y(x_0) = y_0$. What does Theorem 1 say for $(x_0, y_0) = (1,2)$? And for $(0,0)$?

1. $f(x,y) = x\sqrt{y}$ is continuous for $x \in \mathbb{R}$, $y \ge 0$. (square root needs $y \ge 0$)
2. $\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$ is continuous for $x \in \mathbb{R}$, $y > 0$. (now $y = 0$ also breaks it: division by 0)
3. $(1,2)$: $y_0 = 2 > 0$, so both are continuous on a small rectangle around it. A unique solution exists on some interval $1 - h < x < 1 + h$.
4. $(0,0)$: $y_0 = 0$ is on the boundary where $\partial f/\partial y$ fails. The theorem does not apply.
5. Indeed there are two solutions through $(0,0)$: $y \equiv 0$ ($0 = x \cdot 0$) and $y = \frac{x^4}{16}$ ($y' = \frac{x^3}{4}$ and $x\sqrt{x^4/16} = x \cdot \frac{x^2}{4} = \frac{x^3}{4}$).

**ANSWER**
At $(1, 2)$: unique solution near $x = 1$. At $(0, 0)$: two solutions, $y \equiv 0$ and $y = \frac{x^4}{16}$; uniqueness does **not** hold.

**Shortcut for the region**
Write down $f$ and $f_y$. Every place where one of them divides by zero, takes an even root of a negative number, or takes $\ln$ of a non-positive number is a *forbidden* set. The theorem guarantees a unique solution through every point in the open region that avoids all forbidden sets.

## Exercises 2 (p.26), solved

**EXERCISES 2 #1 · P.26** Show that the IVP $\frac{dy}{dx} + p(x)y = q(x),\ y(x_0) = y_0$, where $p, q$ are continuous functions and $x_0, y_0 \in \mathbb{R}$, admits a unique solution in an interval $|x - x_0| < h$ for some $h > 0$.

1. Put it in the form $y' = f(x, y)$: $f(x, y) = q(x) - p(x)y$.
2. $f$ is continuous everywhere: $p, q$ are continuous and $y$ is a polynomial. (sums and products of continuous functions are continuous)
3. $\frac{\partial f}{\partial y} = -p(x)$, which is continuous. (the $q(x)$ term has no $y$)
4. So both are continuous on any rectangle around $(x_0, y_0)$. Theorem 1 applies.

**ANSWER**
A unique solution exists on some interval $|x - x_0| < h$. (Every linear first-order IVP with continuous $p, q$ has this property.)

**EXERCISES 2 #2 · P.26** Show that $\frac{dy}{dx} = 1 + y^2,\ y(0) = 0$ admits a unique solution in some neighbourhood of 0.

1. $f(x, y) = 1 + y^2$ is a polynomial, so continuous on all of $\mathbb{R}^2$.
2. $\frac{\partial f}{\partial y} = 2y$, also continuous everywhere.
3. Any rectangle around $(0, 0)$ works, so Theorem 1 gives a unique solution on some $|x| < h$.

**ANSWER**
Unique solution near $x = 0$. (It is $y = \tan x$, which you can find with Chapter 4's method.)

**Why only some neighbourhood?**
Even though $f = 1 + y^2$ is perfect everywhere, the solution $y = \tan x$ blows up at $x = \pm\frac{\pi}{2}$. So it exists only on $-\frac{\pi}{2} < x < \frac{\pi}{2}$, not on all of $\mathbb{R}$. This is exactly why the theorem only promises an interval $|x - x_0| < h$.

## More practice in the same style

**PRACTICE (MODELED ON P.25)** For $y' = y^{2/3}$, where does Theorem 1 guarantee a unique solution through $(x_0, y_0)$? Check $(0, 0)$.

1. $f = y^{2/3}$ is continuous for all $(x, y)$. (cube root is defined for negative numbers too)
2. $\frac{\partial f}{\partial y} = \frac{2}{3}y^{-1/3} = \frac{2}{3\sqrt[3]{y}}$: not defined at $y = 0$.
3. So uniqueness is guaranteed through every point with $y_0 \neq 0$.
4. At $(0, 0)$: $y \equiv 0$ works, and so does $y = \frac{x^3}{27}$ ($y' = \frac{x^2}{9} = \left(\frac{x^3}{27}\right)^{2/3}$).

**ANSWER**
Unique for every $y_0 \neq 0$. Through $(0, 0)$ there are at least two solutions: $y \equiv 0$ and $y = x^3/27$.

**PRACTICE: INTERVAL OF EXISTENCE** Solve $y' = -2xy^2$, $y(0) = -1$ (the equation of Example 11 (3) in Chapter 4) and give the largest interval containing 0 where the solution exists.

1. The general solution is $y = \frac{1}{x^2 + C}$ (see Chapter 4).
2. $y(0) = \frac{1}{C} = -1 \Rightarrow C = -1$, so $y = \frac{1}{x^2 - 1}$.
3. The formula breaks at $x = \pm 1$. A solution lives on an *interval*, and the one that contains $x_0 = 0$ is $(-1, 1)$.

**ANSWER**
$y = \frac{1}{x^2 - 1}$ on $-1 < x < 1$.

**! Same equation, different interval**
With $y(0) = 1$ the same equation gives $y = \frac{1}{x^2 + 1}$, defined on all of $\mathbb{R}$. The interval depends on the initial condition, not only on the equation.

## In one minute

- IVP = equation $y' = f(x, y)$ + initial condition $y(x_0) = y_0$.
- To solve: general solution, plug in $(x_0, y_0)$, find $c$, state the interval.
- **Theorem 1**: if $f$ and $\partial f/\partial y$ are continuous on a rectangle around $(x_0, y_0)$, there is a **unique** solution on some $|x - x_0| < h$.
- Check $f_y$, not just $f$. Typical failure: $\sqrt{y}$ or $y^{2/3}$ at $y = 0$.
- If the hypotheses fail, the theorem is silent. Example 10: two solutions $0$ and $x^4/16$ through $(0, 0)$.
- The guarantee is local: $y' = 1 + y^2$ is perfect everywhere, but $y = \tan x$ only lives on $(-\pi/2, \pi/2)$.
- Linear IVPs $y' + p(x)y = q(x)$ with continuous $p, q$ always have a unique solution near $x_0$.

CHAPTER 4

# Separable equations

The first real solving method: move all the y's to one side and all the x's to the other, then integrate. You will also learn the step most students forget, which is checking for lost (singular) solutions.

From Week+1-3.pdf, pages 27–38

## Two ways to write a first-order equation

The slides (p.27) use two forms, and you must be able to switch between them:

- **Derivative form:** $\frac{dy}{dx} = f(x, y)$.
- **Differential form:** $M(x, y)\,dx + N(x, y)\,dy = 0$.

Going from one to the other is just algebra with $dx$ and $dy$. For example $\frac{dy}{dx} = -2y^2x$ is the same as $2y^2x\,dx + dy = 0$.

## What makes an equation separable

**Definition 10 (Separable Equations), p.28**

An equation of the form

$$f(y)\frac{dy}{dx} = g(x)$$

is called a **separable equation**.

In words: you can put everything with $y$ (including $dy$) on one side and everything with $x$ (including $dx$) on the other side.

The most common shape on exams is the one in Remark 3 (p.30):

$$\frac{dy}{dx} = g(x)\,h(y) \quad \Longrightarrow \quad \frac{1}{h(y)}\,dy = g(x)\,dx \qquad (h(y) \neq 0)$$

SEPARABLE: RIGHT SIDE = (FUNCTION OF X) × (FUNCTION OF Y)

**Quick separability test**

Try to factor the right side into a product $g(x) \cdot h(y)$. $x^2 + x^2y^2 = x^2(1 + y^2)$ factors, so it is separable. $x + y$ does not factor, so it is not.

## The solution technique

From p.29: first write the equation as $f(y)\,dy = g(x)\,dx$. Then integrate both sides:

$$\underbrace{\int^{y} f(\zeta)\,d\zeta}_{F(y)} = \underbrace{\int^{x} g(\eta)\,d\eta}_{G(x)} + c$$

**GENERAL SOLUTION $F(Y) = G(X) + C$, $C$ ARBITRARY**

One constant on one side is enough: two constants $c_1, c_2$ on both sides combine into one.

### The lost-solution check (Remark 3)

When you divide by $h(y)$, you silently assume $h(y) \neq 0$. But if $h(y_1) = 0$ for some number $y_1$, then the constant function $y \equiv y_1$ gives $y' = 0$ and $g(x)h(y_1) = 0$. So it **is** a solution, and dividing threw it away.

**Remark 3, p.30**

The equation $\frac{dy}{dx} = g(x)h(y)$ is separable because it can be written as $\frac{1}{h(y)}dy = g(x)dx$, provided that $h(y) \neq 0$. What happens if $h(y) = 0$? In that case we get **lost solutions**. If these solutions are **not** included in the general solution for definite values of the arbitrary constants, then they are called **singular solutions**.

**The lecturer's recipe (mirror this on the exam)**

1. Name the form (derivative or differential) and show it is separable by labelling $h(y)$ and $g(x)$.
2. Write **"for** $h(y) \neq 0$**"** and separate: $\frac{1}{h(y)}dy = g(x)dx$.
3. Integrate both sides; add **one** constant $C$; write "general solution".
4. Write **"if** $h(y) = 0$**"**: find those constants $y = y_1$ and check them in the equation.
5. Decide: is $y = y_1$ contained in the general solution for some $C$? If not, it is a **singular solution**.

The same idea works in differential form: if you divide by something containing $x$, such as $x + 1$, check the vertical lines $x = \text{const}$ too (see Exercise 3.1 #1 below).

## Example 11 (p.31), solved

**EXAMPLE 11 (1) · P.31** Solve $e^y \frac{dy}{dx} = x\cos x$.

1. Already in the form $f(y)y' = g(x)$ with $f(y) = e^y$, $g(x) = x\cos x$. Separate: $e^y\,dy = x\cos x\,dx$. (nothing to divide by, so nothing is lost)
2. $\int e^y\,dy = e^y$.

3. $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x$. (by parts: $u = x$, $dv = \cos x\,dx$)
4. $e^y = x\sin x + \cos x + C$.

**ANSWER**
$y = \ln(x\sin x + \cos x + C)$, wherever $x\sin x + \cos x + C > 0$.

**EXAMPLE 11 (2) · P.31** Solve $\frac{dy}{dx} = 2xy$.

1. $g(x) = 2x$, $h(y) = y$. For $y \neq 0$: $\frac{dy}{y} = 2x\,dx$.
2. Integrate: $\ln|y| = x^2 + C$. ($\int \frac{dy}{y} = \ln|y|$)
3. Exponentiate: $|y| = e^C e^{x^2}$, so $y = Ke^{x^2}$ with $K = \pm e^C \neq 0$.
4. If $y = 0$: $y' = 0$ and $2x \cdot 0 = 0$. ✓ So $y \equiv 0$ is a solution.
5. Is it included? Yes: take $K = 0$ in $y = Ke^{x^2}$.

**ANSWER**
$y = Ke^{x^2}$, $K$ arbitrary ($K = 0$ gives $y \equiv 0$). Here $y = 0$ is **not** singular.

**EXAMPLE 11 (3) · P.32 (HANDWRITTEN)** Solve $\frac{dy}{dx} = -2y^2x$.

1. Derivative form. $-2y^2x = (-y^2)\cdot(2x)$: $h(y) = -y^2$, $g(x) = 2x$. Separable.
2. For $y \neq 0$: $-\int \frac{1}{y^2}dy = \int 2x\,dx$.
3. $-\int y^{-2}dy = \frac{1}{y}$, so $\frac{1}{y} = x^2 + C$, $C$ arbitrary constant.
4. General solution: $y = \frac{1}{x^2+C}$.
5. If $y = 0$: $\frac{dy}{dx} = 0$ and $-2x \cdot 0^2 = 0$. ✓ So $y = 0$ is also a solution (lost solution).
6. $\frac{1}{x^2+C}$ is never 0, for any $C$.

**ANSWER**
General solution $y = \frac{1}{x^2+C}$; $y = 0$ is a **singular solution**, since it cannot be included in the general solution by any choice of $C$.

**! Same check, different verdict**

In 11(2) the lost solution $y = 0$ comes back with $K = 0$, so it is **not** singular. In 11(3) it never comes back, so it **is** singular. Always write out both: "$y = 0$ is a solution" and then whether it is contained in the general solution.

# Exercises 3.1 (p.33): "find the solutions and domain of validity"

**EXERCISES 3.1 #1 · P.34 (HANDWRITTEN)** Solve $(y^2+1)\,dx + (x+1)\,dy = 0$.

1. Differential form. Write $y$-terms on the left, $x$-terms on the right. For $x \neq -1$: $\frac{1}{y^2+1}dy = -\frac{1}{x+1}dx$.
2. Integrate both sides: $\int \frac{dy}{1+y^2} = -\int \frac{dx}{x+1}$.
3. $\arctan y = -\ln|x+1| + C$, $C$ arbitrary constant. (general solution)
4. For $x = -1$: $dx = 0$. Substitute into the equation: $(y^2+1)\cdot 0 + (x+1)dy = 0 \cdot dy = 0$. ✓ So $x = -1$ is also a solution.
5. $x = -1$ is not included in the general solution, so it is a singular solution.
6. Domain: explicitly $y = \tan(C - \ln|x+1|)$, valid on intervals that avoid $x = -1$ and where $|C - \ln|x+1|| < \frac{\pi}{2}$. (tan blows up at $\pm\pi/2$)

**ANSWER**
$\arctan y + \ln|x+1| = C$; also the singular solution $x = -1$. ($y^2+1$ is never 0, so no constant $y$ is lost.)

**EXERCISES 3.1 #2 · P.35 (HANDWRITTEN)** Solve $e^{y^2}\ln(x)\,dx - y\,dy = 0$.

1. Differential form. Rewrite: $\frac{y}{e^{y^2}}dy = \ln x\,dx$. ($e^{y^2} > 0$, nothing lost; $\ln x$ needs $x > 0$)
2. Left: $\int ye^{-y^2}dy$. Put $u = y^2$, $du = 2y\,dy$: $\frac{1}{2}\int e^{-u}du = -\frac{1}{2}e^{-u} = -\frac{1}{2}e^{-y^2}$.
3. Right: $\int \ln x\,dx = x\ln x - x$. (by parts)
4. $-\frac{1}{2}e^{-y^2} = x\ln x - x + C$.
5. Multiply by $-2$: $e^{-y^2} = -2x\ln x + 2x + \tilde{C}$, where $\tilde{C} = -2C$ is an arbitrary constant.

**ANSWER**
$e^{-y^2} = 2x - 2x\ln x + \tilde{C}$, for $x > 0$ (and where the right side is in $(0, 1]$, so that $y = \pm\sqrt{-\ln(\dots)}$ is real).

**The handwritten line has a slip**
Multiplying $x\ln x - x + C$ by $-2$ gives $-2x\ln x + 2x - 2C$. The handwritten line has $+x$ and $\tilde{C} = -C$; the correct factors are $+2x$ and $\tilde{C} = -2C$. (Checked with sympy.)

**EXERCISES 3.1 #3 · P.33** Solve $\frac{dy}{dx} - x^2y^2 = x^2$.

1. Move the $x^2y^2$: $\frac{dy}{dx} = x^2 + x^2y^2 = x^2(y^2+1)$. Separable. (this is the hint written on the slide)
2. $y^2 + 1 \neq 0$ always, so $\frac{dy}{1+y^2} = x^2dx$ loses nothing.
3. $\arctan y = \frac{x^3}{3} + C$.

**ANSWER**
$y = \tan\left(\frac{x^3}{3} + C\right)$, valid where $\left|\frac{x^3}{3} + C\right| < \frac{\pi}{2}$. No singular solutions.

**EXERCISES 3.1 #4 · P.36–37 (HANDWRITTEN)** Solve $\frac{dy}{dx} = \cos(y)\sin(x)$.

1. $h(y) = \cos y$, $g(x) = \sin x$: separable.
2. For $\cos y \neq 0$: $\int \frac{1}{\cos y} dy = \int \sin x \, dx$, i.e. $\int \sec y \, dy = \int \sin x \, dx$.
3. $\ln|\sec y + \tan y| = -\cos x + C$, $C$ arbitrary constant. (standard integral of $\sec y$)
4. For $\cos y = 0$: $y = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$. Then $\frac{dy}{dx} = 0$ and $\cos y \sin x = 0 \cdot \sin x = 0$. ✓
5. At $\cos y = 0$, $\sec y$ is undefined, so these lines are not in the general solution.

**ANSWER**
$\ln|\sec y + \tan y| = -\cos x + C$; singular solutions $y(x) = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$.

**EXERCISES 3.1 #5 · P.37–38 (HANDWRITTEN)** Solve $\frac{dy}{dx} = (y^2 - 1)\ln x$. (The slide says: be careful for $y = \mp 1$!)

1. $h(y) = y^2 - 1$, $g(x) = \ln x$ (so $x > 0$). For $y \neq \mp 1$: $\int \frac{1}{y^2-1} dy = \int \ln x \, dx$.
2. Partial fractions: $\frac{1}{y^2-1} = \frac{A}{y-1} + \frac{B}{y+1} \Rightarrow A + B = 0,\ A - B = 1 \Rightarrow A = \frac{1}{2},\ B = -\frac{1}{2}$.
3. Right side by parts ($u = \ln x$, $dv = dx$): $x\ln x - \int \frac{1}{x} \cdot x \, dx = x\ln x - x$.
4. $\frac{1}{2}(\ln|y-1| - \ln|y+1|) = x\ln x - x + C$, i.e. $\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = x\ln x - x + C$.
5. Solve for $y$: $\frac{y-1}{y+1} = Kx^{2x}e^{-2x}$ ($K = \pm e^{2C}$), so $y = \frac{1+Kx^{2x}e^{-2x}}{1-Kx^{2x}e^{-2x}}$.
6. Constant solutions: $y = 1$ and $y = -1$ make $y^2 - 1 = 0$, so $y' = 0 = 0 \cdot \ln x$. ✓ Both are solutions.
7. In the explicit family, $K = 0$ gives $y = 1$. No $K$ gives $y = -1$: with $A = Kx^{2x}e^{-2x}$, $\frac{1+A}{1-A} = -1$ would need $1 + A = -1 + A$, i.e. $1 = -1$.

**ANSWER**
$\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = x\ln x - x + C$ ($x > 0$), plus $y = 1$ and $y = -1$. With the family written as $y = \frac{1+Kx^{2x}e^{-2x}}{1-Kx^{2x}e^{-2x}}$, $K$ arbitrary, $y = 1$ is included ($K = 0$) and only $y = -1$ is singular.

**The handwritten solution ends with ln C x^x, but …**
On p.37–38 the right side $x\ln x - x + \ln C$ is rewritten as $\ln(Cx^x)$. That drops the $-x$: the correct form is $\ln(Cx^xe^{-x})$. The $\ln(Cx^x)$ version does not satisfy the equation.

**Singular depends on how you write the family**

In the $\ln$ form, both $y = 1$ and $y = -1$ are outside the formula. After solving for $y$ and letting $K = 0$, $y = 1$ comes back. So on the exam, list **all** constant solutions explicitly, and say which ones the family you wrote contains.

## In one minute

- Separable: $f(y)y' = g(x)$, or $y' = g(x)h(y)$ (the right side factors into an $x$-part times a $y$-part).
- Separate $\frac{dy}{h(y)} = g(x)dx$, integrate both sides, add **one** constant.
- Always say "for $h(y) \neq 0$" and then check $h(y) = 0$: each root $y_1$ gives a constant solution $y \equiv y_1$.
- Lost solution not obtainable from the general solution for any $C$ = **singular solution** (e.g. $y = 0$ for $y' = -2xy^2$).
- Some lost solutions come back with $K = 0$ ($y' = 2xy$: $y = Ke^{x^2}$), so they are not singular.
- In differential form, dividing by an $x$-expression such as $x + 1$ can lose a vertical line $x = -1$.
- State the domain: $\ln x$ needs $x > 0$, $\tan$ needs its argument in $(-\pi/2, \pi/2)$, and so on.

CHAPTER 5

# Exact equations

Some equations are secretly the total differential of a function F(x,y). Spot them with one test, rebuild F in two ways, and the answer is simply F(x,y) = C.

From Week 2-2.pdf, pages 1–9

## The idea: dF = 0 means F is constant

If $F(x, y)$ is a function of two variables, a small change in $x$ and $y$ changes $F$ by

**Definition 11 (Total differential), Week 2-2.pdf p.1**

Let $F$ be a function of two real variables with continuous first partial derivatives in a domain $D$. The **total differential of** $F$ is

$$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy \quad \text{for all } (x, y) \in D.$$

Now look at an equation in differential form, $M(x, y)\,dx + N(x, y)\,dy = 0$. It has the same shape as $dF$. If we can find $F$ with

$$\frac{\partial F}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y} = N(x, y),$$

then the equation says $dF = 0$. A function whose change is always zero is constant, so

$$dF = 0 \quad \Longrightarrow \quad F(x, y) = C, \qquad C \text{ arbitrary constant}$$

THE GENERAL SOLUTION OF AN EXACT EQUATION (P.2)

**Definition 12, p.3**

The differential equation $M(x, y)dx + N(x, y)dy = 0$ is said to be **exact in a domain** $D$ if there exists a function $F$ such that $\frac{\partial F}{\partial x} = M(x, y)$ and $\frac{\partial F}{\partial y} = N(x, y)$ for all $(x, y) \in D$.

## The test for exactness

**Theorem 2 (Test for Exactness), p.4**

Let $M, N$ and their first-order partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ be continuous in a rectangular domain $D$. Then $M(x, y)dx + N(x, y)dy = 0$ is exact for all $(x, y) \in D$ **if and only if**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

**Why it works** (the lecturer's handwritten arrows on p.4): if the equation is exact, then $\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right) = \frac{\partial^2 F}{\partial y\,\partial x}$ and $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right) = \frac{\partial^2 F}{\partial x\,\partial y}$. Mixed partial derivatives of a nice function are equal, so $M_y = N_x$.

> **Cross the letters**
> Differentiate $M$ (the $dx$ coefficient) with respect to $y$, and $N$ (the $dy$ coefficient) with respect to $x$. Computing $M_x$ and $N_y$ is the classic slip. Also keep the sign: in $y\,dx - x\,dy = 0$, $N = -x$, not $x$.

## Building F: two ways

Example 12 on p.5–6 is worked out both ways side by side. Learn one well, but know both.

**1st way: start from $M$**

1. $\frac{\partial F}{\partial x} = M \Rightarrow F = \int M\,dx + g(y)$. ($y$ is held constant, so the "constant" may depend on $y$)
2. Differentiate in $y$ and set $\frac{\partial F}{\partial y} = N$.
3. Solve for $g'(y)$ (all $x$ must cancel), integrate to get $g(y)$.

**2nd way: start from $N$**

1. $\frac{\partial F}{\partial y} = N \Rightarrow F = \int N\,dy + h(x)$.
2. Differentiate in $x$ and set $\frac{\partial F}{\partial x} = M$.
3. Solve for $h'(x)$ (all $y$ must cancel), integrate to get $h(x)$.

> **Which way is faster?**
> Integrate whichever of $M\,dx$ or $N\,dy$ is easier. In Exercise 3.2 #3 below, $N = x^2$ is trivial to integrate in $y$, so the 2nd way is quicker.

**EXAMPLE 12 · WEEK 2-2.PDF P.5–6** Find the general solution of $2xe^y\,dx + \left(x^2e^y + \cos y\right)dy = 0$.

1. $M = 2xe^y$, $N = x^2e^y + \cos y$.
2. $\frac{\partial M}{\partial y} = 2xe^y$, $\frac{\partial N}{\partial x} = 2xe^y$. Equal, so the given equation is exact.
3. **1st way.** $F = \int 2xe^y\,dx + g(y) = x^2e^y + g(y)$.
4. $\frac{\partial F}{\partial y} = x^2e^y + g'(y) \overset{!}{=} x^2e^y + \cos y \Rightarrow g'(y) = \cos y \Rightarrow g(y) = \sin y + c_1$.
5. $F = x^2e^y + \sin y + c_1 = c_2$, so $x^2e^y + \sin y = c_2 - c_1 = C$.
6. **2nd way.** $F = \int \left(x^2e^y + \cos y\right)dy + h(x) = x^2e^y + \sin y + h(x)$.
7. $\frac{\partial F}{\partial x} = 2xe^y + h'(x) \overset{!}{=} 2xe^y \Rightarrow h'(x) = 0 \Rightarrow h(x) = 0$. (one choice is enough)

**ANSWER**
$x^2e^y + \sin y = C$, $C$ arbitrary constant (both ways give the same answer).

**Constants**

The lecturer sometimes keeps $c_1$ from $g(y)$ and then merges it: $F = c_2 \Rightarrow \ldots = c_2 - c_1 = C$. It is equally correct to take $g$ without a constant, since the final $= C$ absorbs it.

**g(y) must not contain x**

If after setting $F_y = N$ your $g'(y)$ still contains $x$, you made a mistake (or the equation is not exact). Go back and check $M_y = N_x$ and your integral.

## Exercises 3.2 (p.7), solved

**EXERCISES 3.2 #4 · P.8–9 (HANDWRITTEN)** Solve $\frac{1}{x+y^2}\,dx + \left(3 + \frac{2y}{x+y^2}\right) dy = 0$.

1. $M = \frac{1}{x+y^2}$, $N = 3 + \frac{2y}{x+y^2}$.
2. $\frac{\partial M}{\partial y} = \frac{-2y}{(x+y^2)^2}$ and $\frac{\partial N}{\partial x} = \frac{-2y}{(x+y^2)^2}$. Equal, so the equation is exact.
3. $\frac{\partial F}{\partial x} = M$: $F = \int \frac{1}{x+y^2}dx + g(y) = \ln|x+y^2| + g(y)$.
4. $\frac{\partial F}{\partial y} = N$: $\frac{1}{x+y^2} \cdot 2y + g'(y) = \frac{2y}{x+y^2} + 3 \Rightarrow g'(y) = 3 \Rightarrow g(y) = 3y$.
5. Hence $F(x,y) = \ln|x+y^2| + 3y$.

**ANSWER**

$\ln|x+y^2| + 3y = C$, $C$ arbitrary constant.

**EXERCISES 3.2 #1 · P.7** Solve $(\sec(x+y))^2\,dx + \left(2y + (\sec(x+y))^2\right) dy = 0$.

1. $M = \sec^2(x+y)$, $N = 2y + \sec^2(x+y)$.
2. $M_y = 2\sec^2(x+y)\tan(x+y)$ and $N_x = 2\sec^2(x+y)\tan(x+y)$. Equal, so the equation is exact. ($(\sec u)' = \sec u \tan u$, chain rule)
3. $F = \int \sec^2(x+y)\,dx + g(y) = \tan(x+y) + g(y)$.
4. $F_y = \sec^2(x+y) + g'(y) = 2y + \sec^2(x+y) \Rightarrow g'(y) = 2y \Rightarrow g(y) = y^2$.

**ANSWER**

$\tan(x+y) + y^2 = C$.

**EXERCISES 3.2 #2 · P.7** Solve $\left(2(x+y) - 3x^2\right) dx + 2(x+y)\,dy = 0$.

1. $M = 2x + 2y - 3x^2$, $N = 2x + 2y$. $M_y = 2 = N_x$, so the equation is exact.
2. $F = \int (2x + 2y - 3x^2)\,dx + g(y) = x^2 + 2xy - x^3 + g(y)$.
3. $F_y = 2x + g'(y) = 2x + 2y \Rightarrow g(y) = y^2$.

**ANSWER**
$x^2 + 2xy + y^2 - x^3 = C$, i.e. $(x+y)^2 - x^3 = C$.

**EXERCISES 3.2 #3 · P.7** Solve $(2xy + x\cos x + \sin x)\,dx + x^2\,dy = 0$.

1. $M_y = 2x$, $N_x = 2x$. Exact.
2. 2nd way (easier): $F = \int x^2\,dy + h(x) = x^2y + h(x)$.
3. $F_x = 2xy + h'(x) \overset{!}{=} 2xy + x\cos x + \sin x \Rightarrow h'(x) = x\cos x + \sin x$.
4. $h(x) = \int x\cos x\,dx + \int \sin x\,dx = (x\sin x + \cos x) + (-\cos x) = x\sin x$. (by parts for $x\cos x$)

**ANSWER**
$x^2y + x\sin x = C$.

**EXERCISES 3.2 #5 · P.7** Determine a function $M$ such that $M(x,y)\,dx + \left(2x^2y - 1\right)dy = 0$ is exact, and then solve it.

1. Exactness requires $M_y = N_x = \frac{\partial}{\partial x}(2x^2y - 1) = 4xy$.
2. Integrate in $y$: $M = 2xy^2 + k(x)$, where $k(x)$ is any function of $x$ only. (differentiating in $y$ kills $k(x)$)
3. Simplest choice $k = 0$: $M = 2xy^2$.
4. $F = \int 2xy^2\,dx + g(y) = x^2y^2 + g(y)$; $F_y = 2x^2y + g'(y) = 2x^2y - 1 \Rightarrow g(y) = -y$.

**ANSWER**
$M = 2xy^2$ (more generally $2xy^2 + k(x)$); solution $x^2y^2 - y = C$.

## An exact IVP

**PRACTICE (MODELED ON P.5–6)** Solve $\left(3x^2y + e^y\right)dx + \left(x^3 + xe^y - 2y\right)dy = 0,\ y(0) = 1$.

1. $M_y = 3x^2 + e^y$, $N_x = 3x^2 + e^y$. Exact.
2. $F = \int (3x^2y + e^y)\,dx + g(y) = x^3y + xe^y + g(y)$.
3. $F_y = x^3 + xe^y + g'(y) = x^3 + xe^y - 2y \Rightarrow g(y) = -y^2$.
4. General solution $x^3y + xe^y - y^2 = C$. At $(0,1)$: $0 + 0 - 1 = C$.

**ANSWER**
$x^3y + xe^y - y^2 = -1$.

## In one minute

- Exact means $M\,dx + N\,dy = dF$ for some $F$; then the solution is $F(x,y) = C$.
- **Test (Theorem 2):** $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Always write both partial derivatives and the verdict.

- 1st way: $F = \int M\,dx + g(y)$, then $F_y = N$ gives $g'(y)$.
- 2nd way: $F = \int N\,dy + h(x)$, then $F_x = M$ gives $h'(x)$.
- $g'(y)$ may contain only $y$; if an $x$ is left, something is wrong.
- For an IVP, plug $(x_0, y_0)$ into $F(x, y) = C$ to get $C$.

CHAPTER 6

# Integrating factors

When an equation fails the exactness test, you can often repair it by multiplying by a well-chosen $\mu(x)$ or $\mu(y)$. This chapter shows how to decide which one, how to compute it, and how to finish. It is exactly midterm Question 2.

From Week 2-2.pdf, pages 10–21

## The idea

If $M\,dx + N\,dy = 0$ is **not** exact, it is sometimes possible to make it exact by multiplying the whole equation by a non-zero function (p.10).

**Definition 13, Week 2-2.pdf p.10**

A non-zero function $\mu = \mu(x, y)$ is called an **integrating factor** for $M(x,y)dx + N(x,y)dy = 0$ if the differential equation

$$\mu(x,y)M(x,y)\,dx + \mu(x,y)N(x,y)\,dy = 0$$

is exact.

Applying the exactness test to $\mu M$ and $\mu N$ gives a condition on $\mu$:

**Theorem 3, p.11**

$\mu = \mu(x, y)$ is an integrating factor for $M\,dx + N\,dy = 0$ if and only if it is a solution of the partial differential equation

$$N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y} = \mu\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right). \qquad (*)$$

Solving a PDE is usually no easier than the original problem. So we only look for the two special kinds $\mu = \mu(x)$ and $\mu = \mu(y)$.

## Where the formulas come from

**If $\mu$ depends only on $x$** (handwritten, p.11–12): then $\frac{\partial \mu}{\partial x} = \frac{d\mu}{dx}$ and $\frac{\partial \mu}{\partial y} = 0$. From $(*)$:

$$N\frac{d\mu}{dx} = \mu\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) \implies \frac{d\mu}{dx} = \mu \cdot \underbrace{\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}_{\text{must depend only on } x,\ =f(x)}$$

This is a separable equation for $\mu$: $\int \frac{d\mu}{\mu} = \int f(x)dx$, so $\ln|\mu| = \int f(x)dx$ and $\mu = \mp e^{\int f(x)dx}$. Since we need only one integrating factor, take $\mu(x) = e^{\int f(x)dx}$.

**If $\mu$ depends only on $y$**: now $\frac{\partial \mu}{\partial x} = 0$, and $(*)$ becomes $-M\frac{d\mu}{dy} = \mu\,(M_y - N_x)$. Notice the **minus sign** in front of $M$. It carries into the formula:

$$\frac{d\mu}{dy} = -\mu \cdot \underbrace{\frac{1}{M}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}_{=g(y)} \implies \mu(y) = e^{-\int g(y)\,dy}.$$

## Theorem 4: the two cases

**Theorem 4, p.13–14 (learn exactly as written)**

Consider $M(x,y)dx + N(x,y)dy = 0$.

**Case 1.** There exists an integrating factor that depends only on $x$ if and only if

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x).$$

Then an integrating factor is $\mu(x) = \exp\left\{\int^x f(\eta)\,d\eta\right\}$.

**Case 2.** There exists an integrating factor that depends only on $y$ if and only if

$$\frac{1}{M}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = g(y).$$

Then an integrating factor is $\mu(y) = \exp\left\{-\int^y g(\zeta)\,d\zeta\right\}$.

**The minus sign in Case 2**

The lecturer marked it with a red "!" on p.14. With the slide's $g(y) = \frac{1}{M}(M_y - N_x)$, the factor is $e^{-\int g}$. Forgetting the minus gives $1/\mu$ instead of $\mu$ (e.g. $y^{-5}$ instead of $y^5$), and the new equation is **not** exact. Quick self-check: after multiplying, test $\tilde{M}_y = \tilde{N}_x$ again.

**How to decide (the lecturer's order)**

1. Label $M$ and $N$. Compute $M_y$ and $N_x$. If they differ, write "Not exact".
2. Compute the difference $M_y - N_x$ and simplify/factor it.
3. Divide by $N$. Only $x$ left? Then Case 1, $\mu(x) = e^{\int f}$.
4. Otherwise divide by $M$. Only $y$ left? Then Case 2, $\mu(y) = e^{-\int g}$.
5. Multiply the equation by $\mu$, rename $\tilde{M} = \mu M$, $\tilde{N} = \mu N$, and check $\tilde{M}_y = \tilde{N}_x$.
6. Solve the exact equation as in Chapter 5: $F = \int \tilde{M}\,dx + g(y)$, $F_y = \tilde{N}$. Answer $F(x,y) = C$.

**Factor the difference first**

Factor $M_y - N_x$ and compare it with $N$ and $M$. In Example 13, $M_y - N_x = -2(xy-1)$ and $N = x(xy-1)$: the bracket $(xy-1)$ cancels and only $-\frac{2}{x}$ remains.

## Example 13: μ(x)

**EXAMPLE 13 · P.15–18 (HANDWRITTEN)** Solve $(2x^2 + y)\,dx + x(xy - 1)\,dy = 0$.

1. $M = 2x^2 + y$, $N = x(xy-1) = x^2y - x$.
2. $\frac{\partial M}{\partial y} = 1 \neq \frac{\partial N}{\partial x} = 2xy - 1$. Not exact.
3. $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 1 - 2xy + 1 = -2(xy-1)$.
4. $\frac{1}{M}(\ldots) = \frac{-2(xy-1)}{2x^2+y}$: not a function of $y$ alone. $\frac{1}{N}(\ldots) = \frac{-2(xy-1)}{x(xy-1)} = -\frac{2}{x} = f(x)$. ✓
5. $\mu(x) = e^{\int f(x)dx} = e^{-\int \frac{2}{x}dx} = e^{-2\ln|x|} = \frac{1}{x^2}$.
6. Multiply: $\left(2 + \frac{y}{x^2}\right)dx + \left(y - \frac{1}{x}\right)dy = 0$. Now $\tilde{M}_y = \frac{1}{x^2} = \tilde{N}_x$. Exact.
7. $F = \int \left(2 + \frac{y}{x^2}\right)dx + g(y) = 2x - \frac{y}{x} + g(y)$.
8. $F_y = -\frac{1}{x} + g'(y) = y - \frac{1}{x} \Rightarrow g'(y) = y \Rightarrow g(y) = \frac{1}{2}y^2 + c_1$.

**ANSWER**
$2x - \frac{y}{x} + \frac{1}{2}y^2 = C$, $C$ arbitrary constant.

## Example 14: μ = $x^m y^n$

Sometimes neither case works, and the problem gives you the *shape* of $\mu$. Then multiply, apply the exactness test, and compare coefficients of matching powers.

**EXAMPLE 14 · P.15, 18–19 (HANDWRITTEN)** Determine the integrating factor of the form $\mu(x,y) = x^m y^n$ for $y(3x - 2y^2)\,dx + x(3x - 5y^2)\,dy = 0$, and solve.

1. $M = 3xy - 2y^3$, $N = 3x^2 - 5xy^2$. $M_y = 3x - 6y^2 \neq N_x = 6x - 5y^2$. Not exact.
2. Multiply by $x^m y^n$: $\tilde{M} = 3x^{m+1}y^{n+1} - 2x^m y^{n+3}$, $\tilde{N} = 3x^{m+2}y^n - 5x^{m+1}y^{n+2}$.
3. $\tilde{M}_y = 3(n+1)x^{m+1}y^n - 2(n+3)x^m y^{n+2}$.
4. $\tilde{N}_x = 3(m+2)x^{m+1}y^n - 5(m+1)x^m y^{n+2}$.
5. Match the coefficients of $x^{m+1}y^n$ and of $x^m y^{n+2}$: $3(n+1) = 3(m+2)$ and $-2(n+3) = -5(m+1)$.
6. So $n - m = 1$ and $-2n + 5m = 1$. Then $3m = 3$: $m = 1$, $n = 2$. $\mu = xy^2$.
7. Now solve: $\left(3x^2y^3 - 2xy^5\right)dx + \left(3x^3y^2 - 5x^2y^4\right)dy = 0$ (check: $\tilde{M}_y = 9x^2y^2 - 10xy^4 = \tilde{N}_x$).
8. $F = \int (3x^2y^3 - 2xy^5)\,dx + g(y) = x^3y^3 - x^2y^5 + g(y)$; $F_y = 3x^3y^2 - 5x^2y^4 + g'(y) \Rightarrow g' = 0$.

**ANSWER**
$\mu = xy^2$; general solution $x^3y^3 - x^2y^5 = C$.

# Exercises 3.3 (p.20), solved

**EXERCISES 3.3 #1 · P.20** Solve $(x+y-1)y\,dx+(x+2y-2)\,dy=0$.

1. $M=xy+y^2-y$, $N=x+2y-2$. $M_y=x+2y-1$, $N_x=1$. Not exact.
2. $M_y-N_x=x+2y-2=N$. So $\frac{1}{N}(M_y-N_x)=1=f(x)$. (a constant counts as a function of $x$ only)
3. $\mu(x)=e^{\int 1\,dx}=e^x$.
4. $\tilde{M}=e^x(xy+y^2-y)$, $\tilde{N}=e^x(x+2y-2)$; $\tilde{M}_y=e^x(x+2y-1)=\tilde{N}_x$. ✓
5. Easier from $\tilde{N}$: $F=\int e^x(x+2y-2)\,dy+h(x)=e^x(xy+y^2-2y)+h(x)$.
6. $F_x=e^x(xy+y^2-2y)+e^xy+h'(x)=e^x(xy+y^2-y)+h'(x)\overset{!}{=}\tilde{M}\Rightarrow h'=0$.

**ANSWER**
$e^x\left(xy+y^2-2y\right)=C$.

**EXERCISES 3.3 #2 · P.20** Solve $3xy\,dx+(9x^2+4y)\,dy=0$.

1. $M=3xy$, $N=9x^2+4y$. $M_y=3x\neq N_x=18x$. Not exact.
2. $M_y-N_x=-15x$. $\frac{1}{N}(\ldots)=\frac{-15x}{9x^2+4y}$: contains $y$, so no Case 1.
3. $\frac{1}{M}(\ldots)=\frac{-15x}{3xy}=-\frac{5}{y}=g(y)$. Case 2.
4. $\mu(y)=e^{-\int g(y)dy}=e^{\int \frac{5}{y}dy}=e^{5\ln|y|}=y^5$. (minus times minus)
5. $\tilde{M}=3xy^6$, $\tilde{N}=9x^2y^5+4y^6$; $\tilde{M}_y=18xy^5=\tilde{N}_x$. ✓
6. $F=\int 3xy^6\,dx+g(y)=\frac{3}{2}x^2y^6+g(y)$; $F_y=9x^2y^5+g'(y)=9x^2y^5+4y^6\Rightarrow g=\frac{4}{7}y^7$.

**ANSWER**
$\frac{3}{2}x^2y^6+\frac{4}{7}y^7=C$.

**EXERCISES 3.3 #3 · P.20** Find $m,n$ such that $\mu=x^my^n$ is an integrating factor for $y^2\,dx+x(x+y)\,dy=0$, and then solve it.

1. $\tilde{M}=x^my^{n+2}$, $\tilde{N}=x^{m+2}y^n+x^{m+1}y^{n+1}$.
2. $\tilde{M}_y=(n+2)x^my^{n+1}$; $\tilde{N}_x=(m+2)x^{m+1}y^n+(m+1)x^my^{n+1}$.
3. $\tilde{M}_y$ has no $x^{m+1}y^n$ term, so $m+2=0$. Matching $x^my^{n+1}$: $n+2=m+1$.
4. $m=-2$, $n=-3$: $\mu=x^{-2}y^{-3}$.
5. Equation: $\frac{1}{x^2y}dx+\left(\frac{1}{y^3}+\frac{1}{xy^2}\right)dy=0$.
6. $F=\int\frac{1}{x^2y}dx+g(y)=-\frac{1}{xy}+g(y)$; $F_y=\frac{1}{xy^2}+g'(y)=\frac{1}{y^3}+\frac{1}{xy^2}\Rightarrow g=-\frac{1}{2y^2}$.

**ANSWER**
$\mu=x^{-2}y^{-3}$; $-\frac{1}{xy}-\frac{1}{2y^2}=C$.

**EXERCISES 3.3 #4 · P.20–21 (HANDWRITTEN)** Use an integrating factor of the form $\mu = \mu(xy)$ to find the general solution of $y\,dx + x(1 + x^2y^3)\,dy = 0$.

1. $M = y$, $N = x + x^3y^3$. $M_y = 1 \neq N_x = 1 + 3x^2y^3$. Not exact.
2. Try $\mu = (xy)^n$: $\tilde{M} = x^n y^{n+1}$, $\tilde{N} = x^{n+1}y^n + x^{n+3}y^{n+3}$.
3. $\tilde{M}_y = (n+1)x^n y^n$; $\tilde{N}_x = (n+1)x^n y^n + (n+3)x^{n+2}y^{n+3}$.
4. Equal iff $n + 3 = 0$: $n = -3$, $\mu = (xy)^{-3}$. Then the obtained equation is exact.
5. Equation: $x^{-3}y^{-2}dx + \left(x^{-2}y^{-3} + 1\right)dy = 0$.
6. $F = \int x^{-3}y^{-2}dx + g(y) = -\frac{1}{2x^2y^2} + g(y)$; $F_y = \frac{1}{x^2y^3} + g'(y) = \frac{1}{x^2y^3} + 1 \Rightarrow g = y$.

**ANSWER**
$\mu = (xy)^{-3}$; $y - \frac{1}{2x^2y^2} = C$. (The slide stops at "Solve it!"; this is the finish.)

> **Note**
> When you multiply by something like $\frac{1}{x^2}$ or $(xy)^{-3}$, the new equation is not defined on $x = 0$ or $y = 0$. The answers above are valid where $\mu$ is defined and non-zero. The lecturer does not discuss these lines for integrating-factor problems.

## Midterm-style: μ(y) with trig

**PRACTICE (MODELED ON 2023 MIDTERM Q2)** Find the general solution of $(2x - \cot y)\,dx + (x^2 \cot y + x)\,dy = 0$.

1. $M = 2x - \cot y$, $N = x^2 \cot y + x$. $M_y = \csc^2 y$, $N_x = 2x \cot y + 1$. Not exact.
2. $M_y - N_x = \csc^2 y - 1 - 2x \cot y = \cot^2 y - 2x \cot y = -\cot y\,(2x - \cot y)$. $(\csc^2 y - 1 = \cot^2 y)$
3. $\frac{1}{M}(\ldots) = -\cot y = g(y)$. Case 2.
4. $\mu(y) = e^{-\int g} = e^{\int \cot y\,dy} = e^{\ln|\sin y|} = \sin y$.
5. Multiply: $(2x \sin y - \cos y)\,dx + (x^2 \cos y + x \sin y)\,dy = 0$. $\tilde{M}_y = 2x \cos y + \sin y = \tilde{N}_x$. ✓
6. $F = \int (2x \sin y - \cos y)\,dx + g(y) = x^2 \sin y - x \cos y + g(y)$; $F_y = x^2 \cos y + x \sin y + g'(y) \Rightarrow g' = 0$.

**ANSWER**
$x^2 \sin y - x \cos y = C$.

## In one minute

- Integrating factor $\mu \neq 0$: multiplying by it makes $M\,dx + N\,dy = 0$ exact.
- Compute $M_y - N_x$, factor it, then divide by $N$ and by $M$.
- $\frac{1}{N}(M_y - N_x) = f(x)$ gives $\mu(x) = e^{\int f(x)dx}$.

- $\frac{1}{M}(M_y - N_x) = g(y)$ gives $\mu(y) = e^{-\int g(y)dy}$ (note the **minus**).
- Given a shape $x^m y^n$ or $(xy)^n$: multiply, test exactness, match coefficients.
- After multiplying, re-check $\tilde{M}_y = \tilde{N}_x$, then build $F$ and write $F = C$.
- $e^{-2\ln|x|} = 1/x^2$, $e^{5\ln|y|} = y^5$, $e^{\ln|\sin y|} = \sin y$: simplify $e^{\ln}$.

CHAPTER 7

# First-order linear equations

The most reliable recipe of the course: put the equation in standard form, multiply by $\mu = e^{\int p}$, and the left side becomes one derivative. You will also see when to swap the roles of x and y.

From Week 2-2.pdf, pages 22–28

## Standard form

A linear equation of order $n = 1$ (Definition 3 with $n = 1$) is $a_0(x)y'(x) + a_1(x)y(x) = f(x)$. Dividing by $a_0(x)$ (where $a_0(x) \neq 0$) gives the form we always use (handwritten, p.22):

$$y' + \underbrace{\frac{a_1(x)}{a_0(x)}}_{p(x)} y = \underbrace{\frac{f(x)}{a_0(x)}}_{q(x)}.$$

**Definition 14 (Linear Differential Equation), Week 2-2.pdf p.22**

A first-order ordinary differential equation is **linear** in the dependent variable $y$ and the independent variable $x$ if it can be written in the form

$$\frac{dy}{dx} + p(x)y = q(x).$$

**Standard form first**

The coefficient of $y'$ must be exactly **1** before you read off $p(x)$. For $xy' - 3y = x^2$, divide by $x$: $y' - \frac{3}{x}y = x$, so $p(x) = -\frac{3}{x}$ (not $-3$), and the work is valid only on an interval with $x \neq 0$. Also keep the sign: in $y' - \tan(x)y = \sec x$, $p(x) = -\tan x$.

## Theorem 5 and why it works

**Theorem 5, p.23**

The linear differential equation $\frac{dy}{dx} + p(x)y = q(x)$ has an integrating factor

$$\mu(x) := \exp\left\{\int^x p(\eta)\, d\eta\right\}.$$

A **one-parameter family of solutions** for this equation is

$$y := \frac{1}{\mu(x)}\left[\int^x \mu(\eta)q(\eta)\, d\eta + c\right].$$

Furthermore, this one-parameter family of solutions includes **all** solutions.

**Why** (handwritten, p.24): multiply $y' + p(x)y = q(x)$ by $\mu = e^{\int p(x)dx}$:

$$e^{\int p\,dx}\,y' + p(x)e^{\int p\,dx}\,y = e^{\int p\,dx}\,q(x).$$

By the product rule, $\frac{d}{dx}\left(e^{\int p\,dx}\,y\right) = e^{\int p\,dx}\,p(x)\,y + e^{\int p\,dx}\,y'$. That is exactly the left side! So the equation is

$$\frac{d}{dx}\big(\mu(x)\,y\big) = \mu(x)\,q(x).$$

Integrate both sides with respect to $x$:

$$\mu(x)\,y = \int \mu(x)q(x)\,dx + C \qquad \Longrightarrow \qquad y(x) = e^{-\int p(x)dx}\left(\int e^{\int p(x)dx}q(x)\,dx + C\right)$$

GENERAL SOLUTION, C ARBITRARY CONSTANT (P.24–25)

→ **The recipe in 5 lines**

1. Standard form $y' + p(x)y = q(x)$; name $p$ and $q$; say "1st order linear diff. eqn."
2. $\mu(x) = e^{\int p(x)dx}$ (no constant needed; simplify $e^{\ln}$).
3. Write $\frac{d}{dx}(\mu y) = \mu q$.
4. Integrate: $\mu y = \int \mu q\,dx + C$. Do not forget $C$.
5. Divide by $\mu$. For an IVP, now plug in $x_0, y_0$.

! **Where the constant goes**

Add $C$ **before** dividing by $\mu$. Then $C$ gets multiplied by $e^{-\int p}$: $y = \ldots + Ce^{-\int p}$. Writing $y = \frac{1}{\mu}\int \mu q\,dx\ + C$ is wrong.

Note that the linear method is the integrating-factor idea of Chapter 6: written as $(py - q)\,dx + dy = 0$, we have $\frac{1}{N}(M_y - N_x) = p(x)$, so Case 1 gives $\mu = e^{\int p}$.

## Example 15 (an IVP)

EXAMPLE 15 · P.25–27 (HANDWRITTEN) Solve the initial value problem $\frac{dy}{dx} + xy = xe^{x^2/2},\ y(0) = 1.$

1. $p(x) = x$, $q(x) = xe^{x^2/2}$. 1st order linear diff. eqn.
2. $\mu(x) = e^{\int p(x)dx} = e^{\int x\,dx} = e^{x^2/2}$.
3. Multiply both sides: $e^{x^2/2}y' + xe^{x^2/2}y = xe^{x^2}$. The left side is $\frac{d}{dx}\left(e^{x^2/2}y\right)$.
4. Integrate: $e^{x^2/2}y = \int xe^{x^2}dx$. Put $u = x^2$, $du = 2x\,dx$: $= \frac{1}{2}\int e^u\,du = \frac{1}{2}e^{x^2} + C$.
5. Divide: $y(x) = \frac{1}{2}e^{x^2/2} + Ce^{-x^2/2}$.
6. $y(0) = \frac{1}{2}e^0 + Ce^0 = 1 \Rightarrow C = \frac{1}{2}$.

**ANSWER**
$y(x) = \frac{1}{2}e^{x^2/2} + \frac{1}{2}e^{-x^2/2}$, for all $x \in \mathbb{R}$.

## Domain caveats

Theorem 5 (and Exercise 2 #1 in Chapter 3) needs $p$ and $q$ **continuous** on an interval $I$. The solution lives on that interval. So:

- If you divided by $a_0(x)$, exclude its zeros ($xy' + \ldots$ means $x \neq 0$: work on $x > 0$ or $x < 0$).
- $\tan x$, $\sec x$ need $x \neq \frac{\pi}{2} + k\pi$: work on e.g. $-\frac{\pi}{2} < x < \frac{\pi}{2}$.
- $\cot x$, $\csc x$ need $x \neq k\pi$: work on e.g. $0 < x < \pi$.
- $e^{\int p}$ often produces $|x|$ or $|\cos x|$. On an interval where the inside has a fixed sign, you may drop the absolute value (any non-zero constant multiple of $\mu$ also works).

## Exercises 4 (p.28), solved

**EXERCISES 4 #1 · P.28** Find the general solution of $y' + 3y = x$.

1. $p = 3$, $q = x$. $\mu = e^{\int 3\,dx} = e^{3x}$.
2. $\left(e^{3x}y\right)' = xe^{3x}$.
3. $\int xe^{3x}dx = \frac{x}{3}e^{3x} - \frac{1}{9}e^{3x}$. (by parts: $u = x$, $dv = e^{3x}dx$)
4. $e^{3x}y = \frac{x}{3}e^{3x} - \frac{1}{9}e^{3x} + C$.

**ANSWER**
$y = \frac{x}{3} - \frac{1}{9} + Ce^{-3x}$, $x \in \mathbb{R}$.

**EXERCISES 4 #2 · P.28** Find the general solution of $y' - \tan(x)y = \sec(x)$.

1. $p = -\tan x$, $q = \sec x$; both continuous on $-\frac{\pi}{2} < x < \frac{\pi}{2}$.
2. $\mu = e^{-\int \tan x\,dx} = e^{\ln|\cos x|} = \cos x$. ($\int \tan x\,dx = -\ln|\cos x|$; $\cos x > 0$ there)
3. $(y\cos x)' = \cos x \sec x = 1$.
4. $y\cos x = x + C$.

**ANSWER**
$y = (x + C)\sec x$ on $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

**EXERCISES 4 #3 · P.28** Solve $y' + e^x y = e^x$.

1. $p = e^x$, $q = e^x$. $\mu = e^{\int e^x dx} = e^{e^x}$.
2. $\left(e^{e^x}y\right)' = e^x e^{e^x}$.
3. $\int e^x e^{e^x}\,dx = e^{e^x}$. ($u = e^x$, $du = e^x dx$)

4 $e^{e^x} y = e^{e^x} + C$.

**ANSWER**
$y = 1 + Ce^{-e^x}$, $x \in \mathbb{R}$. (The equation is also separable: $y' = e^x(1 - y)$.)

**EXERCISES 4 #4 · P.28** Solve $y' + \cot(x)y = x$.

1 $p = \cot x$, continuous on $0 < x < \pi$. $\mu = e^{\int \cot x\, dx} = e^{\ln|\sin x|} = \sin x$.
2 $(y \sin x)' = x \sin x$.
3 $\int x \sin x\, dx = -x \cos x + \sin x$. (by parts)
4 $y \sin x = \sin x - x \cos x + C$.

**ANSWER**
$y = 1 - x \cot x + C \csc x$ on $0 < x < \pi$.

**EXERCISES 4 #5 · P.28** Depending on the values of $\alpha \in \mathbb{R}$, solve $\frac{dy}{dx} + \frac{\alpha}{x} y = \beta$, where $\beta \in \mathbb{R}$.

1 $p = \frac{\alpha}{x}$, $q = \beta$; work on $x > 0$. $\mu = e^{\alpha \ln x} = x^\alpha$.
2 $(x^\alpha y)' = \beta x^\alpha$.
3 **Case** $\alpha \neq -1$: $\int \beta x^\alpha dx = \frac{\beta x^{\alpha+1}}{\alpha+1}$. So $x^\alpha y = \frac{\beta x^{\alpha+1}}{\alpha+1} + C$.
4 **Case** $\alpha = -1$: $\int \beta x^{-1} dx = \beta \ln x$. So $x^{-1} y = \beta \ln x + C$. (the power rule fails for exponent $-1$)

**ANSWER**
$\alpha \neq -1$: $y = \frac{\beta}{\alpha+1} x + Cx^{-\alpha}$. $\alpha = -1$: $y = \beta x \ln x + Cx$. $(x > 0)$

**EXERCISES 4 #6 · P.28 (HANDWRITTEN START)** Obtain the general solution of $y^2\, dx + (3xy - 1)\, dy = 0$.

1 As $\frac{dy}{dx} = \frac{y^2}{1-3xy}$ it is **not** linear in $y$. Flip it: for $y \neq 0$, $\frac{dx}{dy} = \frac{1-3xy}{y^2} = \frac{1}{y^2} - \frac{3}{y} x$.
2 So $\frac{dx}{dy} + \frac{3}{y} x = \frac{1}{y^2}$: linear in $x$ (the unknown is now $x(y)$), with $p(y) = \frac{3}{y}$, $q(y) = \frac{1}{y^2}$.
3 $\mu(y) = e^{\int \frac{3}{y} dy} = e^{3 \ln|y|} = y^3$ (on $y > 0$; $-y^3$ works the same on $y < 0$).
4 $\frac{d}{dy}\left(y^3 x\right) = y^3 \cdot \frac{1}{y^2} = y \Rightarrow y^3 x = \frac{y^2}{2} + C$.

**ANSWER**
$x = \frac{1}{2y} + \frac{C}{y^3}$, i.e. $xy^3 - \frac{y^2}{2} = C$. (Also $y \equiv 0$ satisfies the differential form: $dy = 0$ and $y^2 = 0$.)

**When to flip x and y**
If the equation is not linear in $y$ but $x$ appears only to the first power (as in $3xy$ above), solve for $\frac{dx}{dy}$ instead and treat $x$ as the unknown.

## In one minute

- Linear 1st order: $y' + p(x)y = q(x)$. Get the coefficient of $y'$ to be 1 first.
- $\mu(x) = e^{\int p(x)dx}$; then $\frac{d}{dx}(\mu y) = \mu q$.
- $y = \frac{1}{\mu}\left(\int \mu q \, dx + C\right)$; the $C$ goes inside the bracket.
- Theorem 5: this family contains **all** solutions, so there are no singular solutions.
- Domain: $p, q$ continuous on $I$; avoid zeros of the leading coefficient, $\tan$ / $\sec$ at $\frac{\pi}{2} + k\pi$, $\cot$ / $\csc$ at $k\pi$.
- Typical $\mu$: $p = \frac{a}{x} \Rightarrow x^a$; $p = -\tan x \Rightarrow \cos x$; $p = \cot x \Rightarrow \sin x$; $p = \text{const } k \Rightarrow e^{kx}$.
- Not linear in $y$? Try linear in $x$: $\frac{dx}{dy} + p(y)x = q(y)$.

CHAPTER 8

# Bernoulli equations

A Bernoulli equation is a linear equation spoiled by one power of y. After this chapter you can spot it, kill the power with u = y^(1−n), solve the linear equation, and remember the solution y = 0 that the substitution hides.

From Week+3.pdf, pages 1–15

## The idea in one picture

A linear equation $y' + p(x)y = q(x)$ is easy (integrating factor). A Bernoulli equation looks almost the same, but the right side has an extra factor $y^n$:

$$\frac{dy}{dx} + p(x)\,y = q(x)\,y^n$$

DEFINITION 15 — BERNOULLI EQUATION (WEEK+3.PDF P.1)

That single $y^n$ makes it nonlinear. The trick: divide by $y^n$, and notice that what is left is the derivative of a new unknown $u = y^{1-n}$. In terms of $u$ the equation is linear again.

### The two easy cases (p.2)

- $n = 0$: $y^0 = 1$, so $y' + p(x)y = q(x)$ — already **linear**.
- $n = 1$: $y' + p(x)y = q(x)y$, i.e. $y' + \big(p(x) - q(x)\big)y = 0$ — linear (and also separable: $\frac{dy}{dx} = -\tilde{p}(x)\,y$).

So the real Bernoulli work is for $n \neq 0$ and $n \neq 1$.

## The substitution $u = y^{1-n}$

$$u := y^{1-n} \quad\Longrightarrow\quad \frac{du}{dx} = (1-n)\,y^{-n}\,\frac{dy}{dx} \quad\Longrightarrow\quad \frac{1}{y^n}\frac{dy}{dx} = \frac{1}{1-n}\frac{du}{dx}$$

THEOREM 6 (P.3): FOR N OTHER THAN 0 AND 1 THIS SUBSTITUTION MAKES BERNOULLI LINEAR

Here is the lecturer's derivation (p.4), step by step.

1. First check $y = 0$: then $y' = 0$ and the equation reads $0 + p \cdot 0 = q \cdot 0$. — true. So $y = 0$ **is a solution** (when $n > 0$).
2. For $y \neq 0$, divide everything by $y^n$: $\frac{1}{y^n}\frac{dy}{dx} + p(x)\,y^{1-n} = q(x)$ ($\frac{1}{y^n}\,p\,y = p\,y^{1-n}$).
3. Replace $\frac{1}{y^n}\frac{dy}{dx}$ by $\frac{1}{1-n}\frac{du}{dx}$ and $y^{1-n}$ by $u$: $\frac{1}{1-n}\frac{du}{dx} + p(x)\,u = q(x)$.
4. Multiply by $(1-n)$:

$$\frac{du}{dx} + \underbrace{(1-n)p(x)}_{\tilde{p}(x)}\, u = \underbrace{(1-n)q(x)}_{\tilde{q}(x)}$$

THE LINEAR EQUATION YOU ALWAYS GET

**Letters**

The slides call the new unknown $u$ (Theorem 6: $u := y^{1-n}$). Some books and teachers write $v = y^{1-n}$ — it is the same thing. This guide follows the slides and uses $u$.

**Practical version: multiply instead of divide**

In the worked examples the lecturer does not divide and then multiply; he multiplies the whole equation once by $(1-n)y^{-n}$ (for $n=2$: by $-\frac{1}{y^2}$; for $n=3$: by $-\frac{2}{y^3}$). Then the first term is exactly $\frac{du}{dx}$ and the second is a multiple of $u$. Fewer steps, fewer sign errors.

### The recipe

1. Bring the equation to the form $y' + p(x)y = q(x)y^n$. Read off $p$, $q$, $n$.
2. Note that $y = 0$ is a solution if $n > 0$.
3. Set $u = y^{1-n}$, $u' = (1-n)y^{-n}y'$. Multiply the equation by $(1-n)y^{-n}$.
4. Solve the linear equation $u' + (1-n)p\,u = (1-n)q$ with $\mu = e^{\int (1-n)p\,dx}$.
5. Back-substitute $u = y^{1-n}$ and state the general solution (plus $y = 0$).

## Example 16 — an initial value problem

**EXAMPLE 16 · WEEK+3.PDF P.3, SOLVED P.5–7** Solve $\dfrac{dy}{dx} + \tan(x)\,y = \dfrac{4\sin x}{\cos^2 x}\,y^2$, $y(0) = 1$.

1. Identify: $p(x) = \tan x$, $q(x) = \dfrac{4\sin x}{\cos^2 x}$, $n = 2$.
2. $u = y^{1-2} = \dfrac{1}{y}$, so $\dfrac{du}{dx} = -\dfrac{1}{y^2}\dfrac{dy}{dx}$.
3. For $y \neq 0$, multiply the equation by $-\dfrac{1}{y^2}$: $-\dfrac{1}{y^2}\dfrac{dy}{dx} - \tan x \cdot \dfrac{1}{y} = -\dfrac{4\sin x}{\cos^2 x}$.
4. Recognise $u$: $\dfrac{du}{dx} - \tan x\, u = -\dfrac{4\sin x}{\cos^2 x}$ (linear in $u$).
5. Integrating factor: $\mu(x) = e^{-\int \tan x\,dx} = e^{-\int \frac{\sin x}{\cos x}dx}$. With $z = \cos x$, $dz = -\sin x\,dx$: $\mu = e^{\int \frac{dz}{z}} = e^{\ln z} = \cos x$.
6. Multiply: $\cos x\,\dfrac{du}{dx} - \sin x\, u = -4\tan x$, and the left side is $\dfrac{d}{dx}(u\cos x)$.
7. Integrate: $u\cos x = -\int 4\tan x\,dx = 4\ln|\cos x| + C$.
8. So $\dfrac{1}{y} = u = \dfrac{4\ln|\cos x| + C}{\cos x}$, i.e. $y = \dfrac{\cos x}{\ln\cos^4 x + C}$ $(4\ln|\cos x| = \ln\cos^4 x)$.
9. Initial condition: $y(0) = \dfrac{\cos 0}{\ln 1 + C} = \dfrac{1}{0 + C} = 1 \Rightarrow C = 1$.

**ANSWER**
General solution $y(x) = \dfrac{\cos x}{\ln \cos^4 x + C}$ (and $y = 0$); solution of the IVP: $y(x) = \dfrac{\cos x}{\ln \cos^4 x + 1}$.

**The slide and the handwriting differ**
The printed slide (p.3) says the right side is $4\sin(x)\,y^2$; the handwritten solution (p.5) uses $\dfrac{4\sin x}{\cos^2 x}\,y^2$. The solution above follows the handwriting. With the printed version the same steps give $(u\cos x)' = -4\sin x\cos x$, $u\cos x = 2\cos^2 x + C$, $y = \dfrac{\cos x}{2\cos^2 x + C}$ and, with $y(0) = 1$, $C = -1$: $y = \dfrac{\cos x}{2\cos^2 x - 1} = \dfrac{\cos x}{\cos 2x}$. Both are checked with sympy.

## Exercise 5.2 — $n = 3$

**EXERCISE 5.2 · WEEK+3.PDF P.11, SOLVED P.14–15** Find the general solution of $y' + \dfrac{1}{2(x+1)}\,y = (1-x^2)\,y^3$.

1. $p(x) = \dfrac{1}{2(x+1)}$, $q(x) = 1 - x^2$, $n = 3$.
2. $u = y^{1-3} = y^{-2}$, $\dfrac{du}{dx} = -2y^{-3}\dfrac{dy}{dx}$.
3. Multiply by $-\dfrac{2}{y^3}$: $-\dfrac{2}{y^3}\dfrac{dy}{dx} - \dfrac{1}{x+1}\cdot\dfrac{1}{y^2} = -2(1-x^2)$.
4. So $\dfrac{du}{dx} - \dfrac{1}{x+1}\,u = -2(1-x)(1+x)$ (linear).
5. $\mu(x) = e^{-\int \frac{dx}{x+1}} = e^{-\ln(x+1)} = \dfrac{1}{x+1}$.
6. $\dfrac{1}{x+1}\dfrac{du}{dx} - \dfrac{1}{(x+1)^2}u = -2(1-x)$, i.e. $\dfrac{d}{dx}\left(\dfrac{u}{x+1}\right) = 2(x-1)$ (the factor $1+x$ cancels).
7. Integrate: $\dfrac{u}{x+1} = x^2 - 2x + C$, so $u = (x+1)(x^2 - 2x + C)$.

**ANSWER**
$\dfrac{1}{y^2} = (x+1)(x^2 - 2x + C)$, $C$ arbitrary; plus the solution $y = 0$.

## Exercise 5.5 — when $x$ is the Bernoulli variable

Sometimes the equation is not Bernoulli in $y(x)$, but it *is* Bernoulli if you treat $x$ as a function of $y$. Flip the derivative: $\dfrac{dx}{dy} = 1 \Big/ \dfrac{dy}{dx}$.

**EXERCISE 5.5 · WEEK+3.PDF P.11–13** Obtain the general solution of $x\,y' = \dfrac{e^{y^2}}{y\,e^{y^2} - x^2}$.

1. Flip (for $x \neq 0$; note $x = 0$ is not a solution): $\dfrac{1}{x}\dfrac{dx}{dy} = \dfrac{ye^{y^2} - x^2}{e^{y^2}} = y - e^{-y^2}x^2$.

2. Multiply by $x$: $\dfrac{dx}{dy} - y\,x = -e^{-y^2}x^3$. This is Bernoulli in $x(y)$ with $p(y) = -y$, $q(y) = -e^{-y^2}$, $n = 3$.
3. $u = x^{1-3} = x^{-2}$, $\dfrac{du}{dy} = -2x^{-3}\dfrac{dx}{dy}$. Multiply by $-2x^{-3}$: $-2x^{-3}\dfrac{dx}{dy} + 2y\,x^{-2} = 2e^{-y^2}$.
4. So $\dfrac{du}{dy} + 2y\,u = 2e^{-y^2}$ (linear in $u(y)$).
5. $\mu(y) = e^{\int 2y\,dy} = e^{y^2}$; then $\dfrac{d}{dy}\left(e^{y^2}u\right) = 2$.
6. $e^{y^2}u = 2y + C$, $u = 2ye^{-y^2} + Ce^{-y^2}$.

**ANSWER**
$\dfrac{1}{x^2} = 2y\,e^{-y^2} + Ce^{-y^2}$, $C$ arbitrary.

## Try it yourself (homework Exercises 5, p.11)

Same recipe each time; answers checked with sympy.

| Equation | $n$, $u$ | Linear equation | General solution |
|---|---|---|---|
| 5.1 $y' + \frac{1}{x}y = (1-x^2)y^2$ | $2,\ y^{-1}$ | $u' - \frac{1}{x}u = x^2 - 1$ | $\frac{1}{y} = x\left(\frac{x^2}{2} - \ln\lvert x\rvert + C\right)$ |
| 5.3 $y' + \tan(x)\,y = y^2$ | $2,\ y^{-1}$ | $u' - \tan x\,u = -1$ | $y = \dfrac{\cos x}{C - \sin x}$ |
| 5.4 $y' + \cot(x)\,y = \cos(x)\,y^3$ | $3,\ y^{-2}$ | $u' - 2\cot x\,u = -2\cos x$ | $\frac{1}{y^2} = 2\sin x + C\sin^2 x$ |

In every case $y = 0$ is an extra solution.

## A substitution that *creates* a Bernoulli equation

Exam question 3 in 2023 was not written as a Bernoulli equation. A substitution turned it into one.

**Midterm 2023, Q3 pattern**
$e^y y' + e^y = e^{3y+2x}$. Notice $e^{3y} = (e^y)^3$ and $e^y y' = (e^y)'$. With $z = e^y$, $z' = e^y y'$, the equation becomes $z' + z = e^{2x}z^3$: **Bernoulli with** $n = 3$. Then $u = z^{-2} = e^{-2y}$, $u' - 2u = -2e^{2x}$, $\mu = e^{-2x}$, and the answer is $e^{-2y} = -2xe^{2x} + Ce^{2x}$. Since $z = e^y$ is never 0, there is no lost solution here.

**How to spot such a substitution**
Look for a function of $y$ whose derivative (times $y'$) is sitting in the equation: $e^y y' \Rightarrow z = e^y$; $\cos y\,y' \Rightarrow z = \sin y$; $2y\,y' \Rightarrow z = y^2$. After the substitution, the leftover powers of $z$ tell you $n$.

## Remark 4 and Example 17 — a jump in $q$

The last part of the Bernoulli section is about a **linear** equation ($n = 0$) whose right side jumps.

**Remark 4 (p.8)**

Sometimes the function $q$ in the equation may not be continuous but may have a jump discontinuity. In this case the existence theorem does **not** necessarily guarantee the existence of a unique solution of the IVP.

What we do instead: solve on each side of the jump separately, then choose the constants so that the pieces **join continuously**.

**EXAMPLE 17 · WEEK+3.PDF P.8, SOLVED P.9–10** Find a continuous solution to $\frac{dy}{dx} + y = q(x)$, $y(0) = 0$, where $q(x) = 1$ for $x \geq 0$ and $q(x) = -1$ for $x < 0$.

1. Linear equation: $\mu(x) = e^{\int 1\,dx} = e^x$. Then $\frac{d}{dx}(e^x y) = e^x q(x)$, so $y = e^{-x} \int e^x q(x)\,dx + Ce^{-x}$.
2. For $x \geq 0$, $q = 1$: $y = e^{-x}e^x + Ce^{-x} = 1 + Ce^{-x}$. From $y(0) = 1 + C = 0$: $C = -1$, so $y = 1 - e^{-x}$.
3. For $x < 0$, $q = -1$: $y = -1 + \tilde{C}e^{-x}$ with a *new* constant $\tilde{C}$ (different interval, different constant).
4. Continuity at 0: $\lim_{x\to 0^+}(1 - e^{-x}) = 0$ must equal $\lim_{x\to 0^-}(-1 + \tilde{C}e^{-x}) = -1 + \tilde{C}$. So $\tilde{C} = 1$.

**ANSWER**

$$y(x) = \begin{cases} 1 - e^{-x}, & x \geq 0 \\ -1 + e^{-x}, & x < 0 \end{cases}$$

## Traps

**Forgetting y = 0**

For $n > 0$, $y \equiv 0$ solves every Bernoulli equation, but the formula for $u = y^{1-n}$ can never produce it (you divided by $y^n$). In a "find all solutions" question you must write it down. (For $n < 0$, $y^n$ is undefined at 0, so $y = 0$ is not a solution.)

**Multiplying only one side by (1–n)**

The linear equation is $u' + (1-n)p\,u = (1-n)q$. Both $p$ and $q$ get the factor $(1-n)$. For $n = 2$ that is a factor $-1$: $y' + \tan x\, y = \ldots y^2$ becomes $u' - \tan x\, u = -\ldots$. Losing this sign changes the integrating factor completely.

**Stopping at u**

The question asks for $y$. Always finish with $u = y^{1-n}$ put back, e.g. $\frac{1}{y^2} = \ldots$ or $y = \ldots$.

**Wrong n**

Read $n$ after putting the equation in the form $y' + py = qy^n$. In Exercise 5.5 the "$y$" is actually $x$ and $n = 3$ appears only after flipping and multiplying by $x$.

## In one minute

- Bernoulli: $y' + p(x)y = q(x)y^n$; $n = 0, 1$ are already linear.
- Substitute $u = y^{1-n}$ (slides' letter $u$); $u' = (1-n)y^{-n}y'$.
- Result: $u' + (1-n)p\,u = (1-n)q$ — linear; solve with $\mu = e^{\int (1-n)p\,dx}$.
- Shortcut: multiply the original equation by $(1-n)y^{-n}$.
- $y = 0$ is an extra solution when $n > 0$ — never forget it.
- If $x$ appears with a power instead of $y$, flip to $\frac{dx}{dy}$ and treat $x$ as the unknown.
- Hidden Bernoulli: $z = e^y$, $z = \sin y$, ... (midterm Q3).
- Jump in $q$: solve on each side, match the pieces by continuity.

CHAPTER 9

# Homogeneous equations (y = ux)

Some equations only care about the ratio y/x. After this chapter you can test for that, substitute y = ux to get a separable equation, and collect every solution — including the ones lost when you divide.

From Week+3.pdf, pages 16–28, and the quiz in Week+4.pdf, pages 15–19

## The idea

Look at $y' = \frac{y}{x} + \frac{x}{y}$. The right side depends on $x$ and $y$ only through the ratio $\frac{y}{x}$ (because $\frac{x}{y} = 1/\frac{y}{x}$). Such equations are called **homogeneous**. If we give the ratio its own name, $u = \frac{y}{x}$, the equation becomes separable.

$$M(x,y)\,dx + N(x,y)\,dy = 0 \text{ is homogeneous} \iff y' = f(x,y) = g\left(\frac{y}{x}\right)$$

DEFINITION 16 (WEEK+3.PDF P.16)

**Not the same word as later**

"Homogeneous" here means "depends only on $y/x$". It has nothing to do with "homogeneous linear equation" (right side zero) that you will meet with second-order equations.

## Testing for homogeneity

### Way 1: write $f$ as a function of $y/x$

Divide numerator and denominator by a suitable power of $x$ and see whether only $\frac{y}{x}$ remains.

**EXAMPLE 18.1 · WEEK+3.PDF P.17–18** Is $(x^2 - y^2)\,dx + 2xy\,dy = 0$ homogeneous?

1. For $x, y \neq 0$: $\dfrac{dy}{dx} = -\dfrac{x^2 - y^2}{2xy} = -\dfrac{1}{2}\left(\dfrac{x^2}{xy} - \dfrac{y^2}{xy}\right) = \dfrac{1}{2}\left(\dfrac{y}{x} - \dfrac{x}{y}\right)$.
2. $\dfrac{x}{y} = \dfrac{1}{y/x}$, so $\dfrac{dy}{dx} = \dfrac{1}{2}\left(\dfrac{y}{x} - \dfrac{1}{y/x}\right) = g\left(\frac{y}{x}\right)$.

ANSWER

Yes — it is a homogeneous differential equation.

**EXAMPLE 18.2 · WEEK+3.PDF P.17, 19 (HOMEWORK)** Is $\left(x+\sqrt{x^2+y^2}\right)dx - x\,dy = 0$ homogeneous?

1. For $x \neq 0$: $\dfrac{dy}{dx} = \dfrac{x+\sqrt{x^2+y^2}}{x} = 1+\sqrt{\dfrac{x^2+y^2}{x^2}} = 1+\sqrt{1+\left(\frac{y}{x}\right)^2}$ (for $x>0$; the slide writes it this way).

**ANSWER**
Yes — $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$, homogeneous.

### Way 2: homogeneous functions of the same degree

$$F(\lambda x, \lambda y) = \lambda^n F(x,y)$$

**DEFINITION 17 (P.17): HOMOGENEOUS FUNCTION OF DEGREE N**

Why this helps (p.20): put $\lambda = \frac{1}{x}$. Then $M\left(1, \frac{y}{x}\right) = \left(\frac{1}{x}\right)^n M(x,y)$, i.e.

$$M(x,y) = x^n M\left(1, \tfrac{y}{x}\right), \qquad N(x,y) = x^n N\left(1, \tfrac{y}{x}\right)$$

**SO IN DY/DX = −M/N THE COMMON POWER OF X CANCELS (P.20)**

**Conclusion 1 (p.21)**
If $M$ and $N$ in $M(x,y)dx + N(x,y)dy = 0$ are both homogeneous functions **of the same degree** $n$, then the differential equation is a homogeneous differential equation.

**Quick test**
Equivalent and fastest in practice: replace $x \to \lambda x$, $y \to \lambda y$ in $f(x,y) = y'$. If all the $\lambda$'s cancel ($f(\lambda x, \lambda y) = f(x,y)$), it is homogeneous. Eyeball version: every term in the numerator and denominator has the **same total degree**. $\frac{x^2+y^2}{xy}$: all degree 2 — yes. $\frac{x+y+1}{x}$: the "1" has degree 0 — no.

## The substitution $y = ux$

$$y = ux \quad \left(u = \frac{y}{x}\right) \quad \Longrightarrow \quad \frac{dy}{dx} = u + x\frac{du}{dx}$$

**THEOREM 7 (P.21): HOMOGENEOUS BECOMES SEPARABLE**

The derivative comes from the product rule: $\frac{d}{dx}(u \cdot x) = \frac{du}{dx}\,x + u \cdot 1$. Then $u + xu' = g(u)$, i.e. $x\,\frac{du}{dx} = g(u) - u$: the variables separate as $\dfrac{du}{g(u)-u} = \dfrac{dx}{x}$.

### The recipe

1. Show it is homogeneous (same degree for $M$ and $N$, or write $y' = g(y/x)$).
2. Put $y = ux$, $y' = u + xu'$. Simplify to $x\,u' = g(u) - u$.

3. For $g(u) - u \neq 0$: separate and integrate. Combine the logs and the constant.
4. Check the values with $g(u) - u = 0$: $u = \text{const}$ gives the straight lines $y = u_0 x$. Are they in the family? If not, list them.
5. Back-substitute $u = y/x$. In differential form, also check $x = 0$.

## Example 19 — the full lecture solution

**EXAMPLE 19 · WEEK+3.PDF P.21–25** Solve $(x^2 - 3y^2)\,dx + 2xy\,dy = 0$.

1. **Test (1st way):** $M = x^2 - 3y^2$: $M(\lambda x, \lambda y) = \lambda^2 x^2 - 3\lambda^2 y^2 = \lambda^2 M$, degree 2. $N = 2xy$: $N(\lambda x, \lambda y) = 2\lambda x \lambda y = \lambda^2 N$, degree 2. Same degree ⇒ homogeneous.
2. **Test (2nd way):** $\dfrac{dy}{dx} = \dfrac{3y^2 - x^2}{2xy} = \dfrac{3}{2}\dfrac{y}{x} - \dfrac{1}{2}\dfrac{1}{y/x} = g\left(\frac{y}{x}\right)$.
3. $y = ux$ (for $x \neq 0$): $x\dfrac{du}{dx} + u = \dfrac{3}{2}u - \dfrac{1}{2}\dfrac{1}{u}$, so $x\dfrac{du}{dx} = \dfrac{1}{2}\left(u - \dfrac{1}{u}\right) = \dfrac{1}{2}\,\dfrac{u^2 - 1}{u}$ (separable).
4. For $u \neq \pm 1$: $\displaystyle\int \frac{u}{u^2 - 1}du = \int \frac{dx}{2x}$. With $z = u^2 - 1$, $u\,du = \frac{1}{2}dz$: $\frac{1}{2}\ln|z| = \frac{1}{2}\ln|x| + \frac{1}{2}\ln|C|$, $C \neq 0$.
5. $\ln|z| = \ln|Cx| \Rightarrow z = \pm Cx = Kx$, $K \neq 0$: $u^2 - 1 = Kx$.
6. **Lost solutions:** if $u = \pm 1$, then $\frac{du}{dx} = 0$ and $\frac{u^2-1}{u} = 0$: the $u$-equation holds. So $u = \pm 1$ are solutions too — they are exactly $K = 0$. Hence $u^2 - 1 = Dx$, $D \in \mathbb{R}$.
7. Back to $y$: $\dfrac{y^2}{x^2} - 1 = Dx$, i.e. $y^2 = x^2 + Dx^3$.
8. $x = 0$: then $dx = 0$ and the term $2xy\,dy$ is 0 too, so the differential equation is satisfied. $x = 0$ is also a solution (the lecturer calls it a singular solution).

**ANSWER**
General solution $y^2 = x^2 + Dx^3$, $D \in \mathbb{R}$ (contains $y = \pm x$ for $D = 0$); also $x = 0$.

**Correction to the handwritten slide**
On p.25 the final line reads $y^2(x) = x^2 + Dx$. That is a slip: from $\frac{y^2}{x^2} - 1 = Dx$ you get $y^2 = x^2 + Dx^3$. Sympy confirms that $y^2 = x^2 + Dx^3$ solves the equation and $y^2 = x^2 + Dx$ does not.

**x = 0 only counts in differential form**
$x = 0$ is a "solution" here because the equation is written as $M\,dx + N\,dy = 0$ and both terms vanish on the line $x = 0$. If the equation is given as $y' = f(x, y)$ with $x$ in a denominator (like the quiz below), $x = 0$ is not even in the domain — do not list it.

## Exercise 6.3 — a logarithm inside

**EXERCISE 6.3 · WEEK+3.PDF P.26–28** Find the general solution of $x\,y' = \big(1 + \alpha \ln y - \alpha \ln x\big)\,y$, $\alpha > 0$.

1. For $x \neq 0$: $y' = \left(1 + \alpha \ln \dfrac{y}{x}\right)\dfrac{y}{x}$ — a function of $\frac{y}{x}$, so homogeneous.
2. $y = ux$: $u + x\dfrac{du}{dx} = u + \alpha\, u \ln u$, so $x\dfrac{du}{dx} = \alpha\, u \ln u$ (separable).
3. For $u \neq 1$, $u \neq 0$ (note $u = 1$ is a solution, and $u$ cannot be 0 because of $\ln u$): $\displaystyle\int \frac{du}{u \ln u} = \int \frac{\alpha}{x} dx$.
4. With $z = \ln u$, $dz = \frac{1}{u}du$: $\ln|z| = \alpha \ln|x| + \ln C = \ln\big(C|x|^{\alpha}\big)$, $C \neq 0$.
5. $z = \pm C x^{\alpha} = K x^{\alpha}$, $K \neq 0$; so $\ln u = K x^{\alpha}$. Including $u = 1$ ($\ln u = 0$): $\ln u = D x^{\alpha}$, $D$ arbitrary.

**ANSWER**
$\ln \dfrac{y}{x} = D\,x^{\alpha}$, $D$ arbitrary constant (i.e. $y = x\,e^{Dx^{\alpha}}$).

## The Week 4 quiz — "find all solutions" (same as midterm Q4)

**This exact equation was midterm 2023 Q4**
$y' = \dfrac{y(y+2x)}{3x^2}$ is homework Exercise 6.2 (p.26), the Week 4 quiz (Week+4.pdf p.18–19) **and** Question 4 of the 2023 midterm ("Find all solutions"). Learn both ways below.

**QUIZ Q2, 1ST WAY · WEEK+4.PDF P.18–19 (FINISHED HERE)** Find all solutions of $y' = \dfrac{y(y+2x)}{3x^2}$.

1. $y' = \dfrac{1}{3}\left(\dfrac{y}{x}\right)^2 + \dfrac{2}{3}\left(\dfrac{y}{x}\right)$ — homogeneous.
2. $y = ux$: $u + x\dfrac{du}{dx} = \dfrac{1}{3}u^2 + \dfrac{2}{3}u \Rightarrow x\dfrac{du}{dx} = \dfrac{1}{3}(u-1)u$ (separable).
3. For $u \neq 0, u \neq 1$: $\displaystyle\int \frac{du}{u(u-1)} = \frac{1}{3}\int \frac{dx}{x}$. Partial fractions: $\frac{1}{u(u-1)} = \frac{1}{u-1} - \frac{1}{u}$.
4. $\ln\left|\dfrac{u-1}{u}\right| = \dfrac{1}{3}\ln|x| + c \Rightarrow \dfrac{u-1}{u} = Kx^{1/3}$, $K \neq 0$.
5. Solve for $u$: $u - 1 = Kx^{1/3}u \Rightarrow u = \dfrac{1}{1 - Kx^{1/3}}$, so $y = \dfrac{x}{1 - Kx^{1/3}}$.
6. **Lost values.** $u = 1$ ($y = x$): it solves the equation and it is $K = 0$ in the formula — already included. $u = 0$ ($y = 0$): it solves the equation ($0 = 0$) but no value of $K$ gives it — list it separately.

**ANSWER**
All solutions: $y = \dfrac{x}{1 + Cx^{1/3}}$, $C \in \mathbb{R}$ (renaming $C = -K$), and $y = 0$.

**QUIZ Q2, 2ND WAY · WEEK+4.PDF P.19** Same equation, as a Bernoulli equation.

1. $\frac{dy}{dx} = \frac{1}{3x^2}y^2 + \frac{2}{3x}y \Rightarrow \frac{dy}{dx} - \frac{2}{3x}y = \frac{1}{3x^2}y^2$: Bernoulli, $n = 2$.
2. $u = y^{-1}$, $u' = -\frac{1}{y^2}y'$. Multiply by $-\frac{1}{y^2}$: $\frac{du}{dx} + \frac{2}{3x}u = -\frac{1}{3x^2}$ (linear).
3. $\mu = e^{\int \frac{2}{3x}dx} = x^{2/3}$; $\left(x^{2/3}u\right)' = -\frac{1}{3}x^{-4/3} \Rightarrow x^{2/3}u = x^{-1/3} + C$.
4. $u = \frac{1}{x} + Cx^{-2/3}$, i.e. $\frac{1}{y} = \frac{1 + Cx^{1/3}}{x}$.

**ANSWER**
$y = \frac{x}{1 + Cx^{1/3}}$, plus $y = 0$ (Bernoulli with $n = 2 > 0$). Same answer as the first way.

## More homework, with answers

| Exercise (p.26) | Separable equation in $u$ | Answer |
|---|---|---|
| 6.1 $y' = \frac{3y - x}{3x - y}$ | $xu' = \frac{u^2 - 1}{3 - u}$ | $y - x = K(x + y)^2$, $K \in \mathbb{R}$; and $y = -x$ |
| 6.4 $y' = \frac{y^2 + (\alpha + 1)xy - \alpha x^2}{x\left((\alpha + 1)x - (\alpha - 1)y\right)}$ | $xu' = \frac{\alpha(u^2 - 1)}{(\alpha + 1) - (\alpha - 1)u}$ | $y - x = K\,x\,(x + y)^{\alpha}$; and $y = -x$ |

In both, $u = 1$ ($y = x$) is inside the family ($K = 0$) but $u = -1$ ($y = -x$) is an extra solution. For 6.1 the partial fractions are $\frac{3-u}{(u-1)(u+1)} = \frac{1}{u-1} - \frac{2}{u+1}$.

## The other quiz question (Week 4, p.15–18)

The same quiz had an equation you can solve in two ways — with an integrating factor (exact equations) or as a linear equation. It shows that one equation may fit several methods.

**QUIZ Q1 · WEEK+4.PDF P.15–18** Solve $(\ln x - y)\,dx - x \ln x\,dy = 0$.

1. 1st way: $M_y = -1 \neq N_x = -\ln x - 1$ — not exact. $\frac{M_y - N_x}{N} = \frac{\ln x}{-x \ln x} = -\frac{1}{x}$ depends on $x$ only, so $\mu = e^{-\int \frac{1}{x}dx} = \frac{1}{x}$.
2. $\frac{1}{x}(\ln x - y)dx - \ln x\,dy = 0$ is exact; $F = -y \ln x + g(x)$, $F_x = -\frac{y}{x} + g'(x) = \frac{\ln x}{x} - \frac{y}{x} \Rightarrow g = \frac{(\ln x)^2}{2}$.
3. 2nd way: $\frac{dy}{dx} + \frac{1}{x \ln x}y = \frac{1}{x}$ is linear; $\mu = e^{\int \frac{dx}{x \ln x}} = \ln x$ and $\frac{d}{dx}(y \ln x) = \frac{\ln x}{x}$.

**ANSWER**
$-y \ln x + \frac{(\ln x)^2}{2} = C$ (both ways give the same).

## Traps

**The derivative of y = ux**

$y' = u + xu'$, not $y' = u'$ and not $y' = xu'$. Forgetting the "$+u$" is the most common way to lose the whole question.

**Lost straight-line solutions**

Separating needs $g(u) - u \neq 0$. Each root $u_0$ gives a line $y = u_0 x$ that solves the equation. Check whether a value of the constant reproduces it (then it is already in the general solution) or not (then list it). Example 19: $y = \pm x$ included ($D = 0$); quiz: $y = x$ included, $y = 0$ **not**; Exercise 6.1: $y = -x$ not.

**Same degree means the same n**

$x^2 + xy$ and $x^3$ are both homogeneous functions, but of degrees 2 and 3, so $(x^2 + xy)dx + x^3 dy = 0$ is **not** a homogeneous equation.

## In one minute

- Homogeneous: $y' = g(y/x)$; test with $f(\lambda x, \lambda y) = f(x, y)$, or $M, N$ homogeneous of the **same** degree.
- Substitute $y = ux$, $y' = u + xu' \Rightarrow xu' = g(u) - u$ (separable).
- Integrate, write the constant as $\ln|C|$, drop the logs, rename $K = \pm C$.
- Roots of $g(u) = u$ give lines $y = u_0 x$: include them (maybe as $K = 0$).
- Differential form: check $x = 0$ too.
- $y' = \frac{y(y+2x)}{3x^2}$ (midterm Q4): $y = \frac{x}{1+Cx^{1/3}}$ and $y = 0$; also solvable as Bernoulli $n = 2$.

CHAPTER 10

# Shifting and clever substitutions

Equations with linear expressions like (x − 2y + 1) are almost homogeneous — a shift of the origin (or the substitution z = ax + by) fixes them. After this chapter you can do that, and you can use a hint like u = sin y to reduce an equation to a linear one.

From Week+3.pdf, pages 29–44

## Equations of the form $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

Without the constants $c_1, c_2$ this would be homogeneous (every term has degree 1). The constants spoil it. Two cases, depending on whether the two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ intersect.

$$\Delta := a_1b_2 - a_2b_1$$

THEOREM 8 (WEEK+3.PDF P.29) — COMPUTE THE DETERMINANT FIRST

| Case | What to do | You get |
|---|---|---|
| **1.** $\Delta \neq 0$ (lines intersect) | Solve $a_1h + b_1k + c_1 = 0,\ a_2h + b_2k + c_2 = 0$. Put $x = u + h$, $y = v + k$ (so $dx = du$, $dy = dv$). | Homogeneous: $(a_1u + b_1v)du + (a_2u + b_2v)dv = 0$ |
| **2.** $\Delta = 0$ (lines parallel) | Put $z = a_1x + b_1y$ (the second bracket is then a multiple of $z$ plus a constant). | Separable in $z$ and $x$ |

→ **Why the shift works**

$(h, k)$ is the point where both lines meet. Moving the origin there ($u = x - h$, $v = y - k$) makes both brackets vanish at the new origin, so the constants disappear: $a_1x + b_1y + c_1 = a_1u + b_1v$. What remains is homogeneous, and you already know how to solve that ( $v = zu$).

## Example 20.1 — Case 1

EXAMPLE 20.1 · WEEK+3.PDF P.30–35 Solve $(x - 2y + 1)\,dx + (4x - 3y - 6)\,dy = 0$.

1. $a_1 = 1, b_1 = -2, a_2 = 4, b_2 = -3$: $\Delta = 1(-3) - 4(-2) = 5 \neq 0$ — Case 1.
2. Solve $h - 2k + 1 = 0$, $4h - 3k - 6 = 0$: multiply the first by 4 and subtract: $-5k = -10$, $k = 2$, $h = 3$.
3. $x = u + 3$, $y = v + 2$, $dx = du$, $dy = dv$: $(u - 2v)du + (4u - 3v)dv = 0$.

4. For $4u-3v\neq 0$, $u\neq 0$: $\dfrac{dv}{du}=-\dfrac{u-2v}{4u-3v}=-\dfrac{1-2\frac{v}{u}}{4-3\frac{v}{u}}=g\left(\frac{v}{u}\right)$ — homogeneous. (Note $4u-3v=0$ and $u=0$ are not solutions.)
5. $z=\frac{v}{u}$, $v=zu$, $\frac{dv}{du}=u\frac{dz}{du}+z$: $u\dfrac{dz}{du}=\dfrac{2z-1}{4-3z}-z=\dfrac{3z^2-2z-1}{4-3z}=\dfrac{(3z+1)(z-1)}{4-3z}$ (separable).
6. Notice $z=1$ and $z=-\frac{1}{3}$ are solutions of this $z$-equation. For $z\neq 1,-\frac{1}{3}$: $\displaystyle\int\frac{4-3z}{(3z+1)(z-1)}dz=\int\frac{du}{u}$.
7. Partial fractions: $A=-\frac{15}{4}$, $B=\frac{1}{4}$. So $-\frac{5}{4}\ln|3z+1|+\frac{1}{4}\ln|z-1|=\ln|u|+\ln|C|$, $C\neq 0$.
8. $\frac{1}{4}\ln\dfrac{|z-1|}{|3z+1|^5}=\ln|Cu|\Rightarrow\left|\dfrac{z-1}{(3z+1)^5}\right|=\tilde{C}u^4$ ($\tilde{C}=C^4>0$) $\Rightarrow z-1=Ku^4(3z+1)^5$, $K=\pm\tilde{C}\neq 0$.
9. $z=\frac{v}{u}$: $\dfrac{v}{u}-1=Ku^4\left(\dfrac{3v+u}{u}\right)^5\Rightarrow v-u=K(3v+u)^5$.
10. Back to $x,y$ ($u=x-3$, $v=y-2$): $y-x+1=K(3y+x-9)^5$, $K\neq 0$. Since $z=1$ is also a solution (that is $K=0$), write $D\in\mathbb{R}$.
11. $z=-\frac{1}{3}$: $v=-\frac{1}{3}u\Rightarrow y-2=-\frac{1}{3}(x-3)\Rightarrow y=-\frac{1}{3}x+3$. No value of $D$ gives this line, so it is a **singular solution**.

**ANSWER**
General solution $y-x+1=D(3y+x-9)^5$, $D\in\mathbb{R}$; singular solution $y=-\frac{1}{3}x+3$.

## Example 20.2 — Case 2

**EXAMPLE 20.2 · WEEK+3.PDF P.35–36** Solve $(x+2y+3)\,dx+(2x+4y-1)\,dy=0$.

1. $a_1=1,b_1=2,a_2=2,b_2=4$: $\Delta=1\cdot 4-2\cdot 2=0$ — Case 2. Note $2x+4y=2(x+2y)$.
2. $z=x+2y$: $\dfrac{dz}{dx}=1+2\dfrac{dy}{dx}$, so $\dfrac{dy}{dx}=\dfrac{1}{2}\dfrac{dz}{dx}-\dfrac{1}{2}$.
3. The equation says $\dfrac{dy}{dx}=-\dfrac{x+2y+3}{2x+4y-1}=-\dfrac{z+3}{2z-1}$. So $\dfrac{1}{2}\dfrac{dz}{dx}-\dfrac{1}{2}=-\dfrac{z+3}{2z-1}$.
4. $\dfrac{1}{2}\dfrac{dz}{dx}=\dfrac{-2z-6+2z-1}{2(2z-1)}\Rightarrow\dfrac{dz}{dx}=-\dfrac{7}{2z-1}$ (separable).
5. For $2z-1\neq 0$: $\displaystyle\int(2z-1)dz=-\int 7\,dx\Rightarrow z^2-z=-7x+C$.
6. Check $z=\frac{1}{2}$, i.e. $x+2y=\frac{1}{2}$: then $dx+2dy=0$; substituting into the given equation gives $(\frac{1}{2}+3)dx+0\cdot dy=0$, false. So $x+2y=\frac{1}{2}$ is **not** a solution.

**ANSWER**
$(x+2y)^2-x-2y=-7x+C$, or $4y^2+4xy+x^2-2y=-6x+C$, $C$ arbitrary.

**Substitute the excluded value back — every time**

In Example 20.1 the excluded $z = 1$ was inside the family, $z = -\frac{1}{3}$ was a genuinely new (singular) solution, and in Example 20.2 the excluded $z = \frac{1}{2}$ was **not a solution at all**. You only know which by plugging it into the original equation.

## Homework Exercises 6.1 (p.37)

**EXERCISE 6.1.3 · WEEK+3.PDF P.37–38 (LAST STEPS COMPLETED IN THIS GUIDE)** Solve $2(y+\alpha)^2\,dx - (y+x+\alpha+\beta)^2\,dy = 0$.

1. Here $a_1 = 0, b_1 = 1$ (inside the square), $a_2 = 1, b_2 = 1$: $\Delta = 0\cdot 1 - 1\cdot 1 = -1 \neq 0$ — Case 1.
2. $k+\alpha = 0 \Rightarrow k = -\alpha$; $k+h+\alpha+\beta = 0 \Rightarrow h = -\beta$. So $x = u-\beta$, $y = v-\alpha$.
3. $2v^2\,du - (v+u)^2\,dv = 0 \Rightarrow \dfrac{du}{dv} = \dfrac{1}{2}\left(1 + 2\dfrac{u}{v} + \dfrac{u^2}{v^2}\right)$.
4. Let $z = \frac{u}{v}$ (here $u$ is written as a function of $v$): $u = zv$, $\dfrac{du}{dv} = z + v\dfrac{dz}{dv}$. Then $z + v\dfrac{dz}{dv} = \dfrac{1}{2}(1 + 2z + z^2) \Rightarrow v\dfrac{dz}{dv} = \dfrac{1}{2}(1+z^2)$. (The slide stops here.)
5. Separate: $\displaystyle\int \frac{dz}{1+z^2} = \frac{1}{2}\int \frac{dv}{v} \Rightarrow \arctan z = \tfrac{1}{2}\ln|v| + C$.
6. Back: $z = \dfrac{u}{v} = \dfrac{x+\beta}{y+\alpha}$, $v = y+\alpha$.

**ANSWER**

$$\arctan\frac{x+\beta}{y+\alpha} - \tfrac{1}{2}\ln|y+\alpha| = C$$

**Which ratio?**

In Example 20 the lecturer uses $z = v/u$; in Exercise 6.1.3 he uses $z = u/v$ because the equation is easier to write as $\frac{du}{dv}$. Both are fine — just stay consistent within one problem.

Answers for the other two homework problems (checked with sympy):

- **6.1.1** $(x - y + \alpha - \beta)dx + (x + y - \alpha - \beta)dy = 0$: $\Delta = 2$, $(h,k) = (\beta,\alpha)$, homogeneous $(u-v)du + (u+v)dv = 0$. Answer: $\arctan\dfrac{y-\alpha}{x-\beta} + \frac{1}{2}\ln\left((x-\beta)^2 + (y-\alpha)^2\right) = C$.
- **6.1.2** $(2x + 2\alpha y - 1)dx + (2\alpha x + 2\alpha^2 y - \beta)dy = 0$: $\Delta = 4\alpha^2 - 4\alpha^2 = 0$, Case 2 with $z = x + \alpha y$: $\dfrac{dz}{dx} = \dfrac{\alpha - \beta}{2\alpha z - \beta}$. Answer: $\alpha z^2 - \beta z - (\alpha - \beta)x = C$ with $z = x + \alpha y$.

## Substitution: no general rule

The slides (p.39) say it honestly: some equations look different from all the standard types, and a **special substitution** (usually given as a hint) reduces them to something you know. There is no general rule — look for a piece of the equation whose derivative also appears.

**EXAMPLE 21 · WEEK+3.PDF P.39–40** Solve $(1+3x\sin y)\,dx - x^2\cos y\,dy = 0$ by letting $u = \sin y$.

1. $u = \sin y \Rightarrow du = \cos y\,dy$ (exactly the $\cos y\,dy$ in the equation).
2. Then $(1+3xu)\,dx - x^2\,du = 0 \Rightarrow \dfrac{du}{dx} = \dfrac{1+3xu}{x^2} \Rightarrow \dfrac{du}{dx} - \dfrac{3}{x}u = \dfrac{1}{x^2}$ (linear).
3. $\mu(x) = e^{-\int \frac{3}{x}dx} = \dfrac{1}{x^3}$: $\dfrac{d}{dx}\left(\dfrac{u}{x^3}\right) = \dfrac{1}{x^5}$.
4. $\dfrac{u}{x^3} = -\dfrac{1}{4x^4} + C \Rightarrow u = -\dfrac{1}{4x} + Cx^3$.

**ANSWER**
$\sin y = -\dfrac{1}{4x} + Cx^3$, $C$ arbitrary.

## Exercises 6.2 (p.41) — which substitution?

| Equation | Substitution | Result |
|---|---|---|
| 1. $y' = \tan^2(x+y)$ | $u = x+y$ | $u' = 1+\tan^2 u = \sec^2 u$: separable |
| 2. $y' = (\alpha x+\beta y+\gamma)^2$ | $u = \alpha x+\beta y+\gamma$ | $u' = \alpha+\beta u^2$: separable |
| 3. $y' = \dfrac{(\alpha x+\beta y)^2+\gamma}{\alpha x+\beta y} - \dfrac{\alpha}{\beta}$ | $u = \alpha x+\beta y$ | $u' = \beta\dfrac{u^2+\gamma}{u}$: separable; $u^2+\gamma = Ke^{2\beta x}$ |
| 4. $y' = \dfrac{(x+y^n)^m - 1}{ny^{n-1}}$ | $u = y^n$, then $z = x+u$ | $z' = z^m$: separable |
| 5. $ny^{n-1}y' = y^n + x - 1$ | $u = y^n$ | $u' - u = x - 1$: linear; $y^n = Ce^x - x$ |

**EXERCISE 6.2.1 · WEEK+3.PDF P.41 (SOLUTION WRITTEN FOR THIS GUIDE)** Solve $y' = \tan^2(x+y)$.

1. $u = x+y \Rightarrow u' = 1+y' = 1+\tan^2 u = \sec^2 u$.
2. Separate: $\cos^2 u\,du = dx$. Using $\int \cos^2 u\,du = \frac{u}{2} + \frac{\sin 2u}{4}$: $\dfrac{u}{2} + \dfrac{\sin 2u}{4} = x + C$.
3. Multiply by 4 and put $u = x+y$ back: $2(x+y) + \sin(2x+2y) = 4x + C$.

**ANSWER**
$2y - 2x + \sin(2x+2y) = C$

**EXERCISE 6.2.4 · WEEK+3.PDF P.41–44** Solve $y' = \dfrac{(x+y^n)^m - 1}{n\,y^{n-1}}$, $n, m \in \mathbb{N}$.

1. $u = y^n \Rightarrow \dfrac{du}{dx} = ny^{n-1}\dfrac{dy}{dx}$. For $y \neq 0$ (note $y = 0$ is not a solution): $ny^{n-1}\dfrac{dy}{dx} = (x+y^n)^m - 1$, i.e. $\dfrac{du}{dx} = (x+u)^m - 1$.
2. Let $z = x+u$: $\dfrac{dz}{dx} = 1 + \dfrac{du}{dx} = z^m$ (separable).

3. $z = 0$ satisfies $z' = z^m$ (for $m \geq 1$), so $z = 0$ is a solution. For $z \neq 0$ and $m \neq 1$, $m \neq 0$: $\int z^{-m} dz = \int dx \Rightarrow \frac{z^{1-m}}{1-m} = -\frac{1}{m-1}\frac{1}{z^{m-1}} = x + C$.
4. Solve for $z^{m-1}$: $(x+y^n)^{m-1} = \frac{-1}{(m-1)(x+C)} = \frac{1}{(1-m)x + \tilde{C}}$, with $\tilde{C} = (1-m)C$.
5. $m = 1$: $\int \frac{dz}{z} = \int dx \Rightarrow \ln|z| = x + C \Rightarrow z = Ke^x$, $K = \pm e^C \neq 0$; with $z = 0$: $x + y^n = De^x$, $D$ arbitrary.
6. $m = 0$: $z' = 1 \Rightarrow z = x + L \Rightarrow y^n = L$.

**ANSWER**

$m \geq 2$: $(x+y^n)^{m-1} = \frac{1}{(1-m)x + \tilde{C}}$ (and $x + y^n = 0$); $m = 1$: $x + y^n = De^x$; $m = 0$: $y^n = L$.

**A sign slip on the slide**

Integrating $z^{-m}$ gives $\frac{z^{1-m}}{1-m} = -\frac{1}{(m-1)z^{m-1}}$. The handwritten line on p.42 drops this minus sign and ends (p.43) with $(x+y^n)^{m-1} = \frac{1}{(m-1)x+\tilde{C}}$. That changes the sign of the $x$-term, which a renamed constant cannot fix. The correct family is $(x+y^n)^{m-1} = \frac{1}{(1-m)x + \tilde{C}}$; sympy confirms it (e.g. $m = 3, n = 2$) and rejects the slide's version. Quick sanity check with $m = 2$: $z' = z^2$ gives $z = \frac{1}{-x+\tilde{C}}$.

**Recognising the hint**

$\cos y\, dy$ next to $\sin y \Rightarrow u = \sin y$. $ny^{n-1}y'$ next to $y^n \Rightarrow u = y^n$. A single combination like $x + y$ or $\alpha x + \beta y + \gamma$ inside a function $\Rightarrow u$ = that combination. $e^y y'$ with powers of $e^y \Rightarrow z = e^y$ (midterm Q3, see the Bernoulli chapter).

## In one minute

- $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$: compute $\Delta = a_1b_2 - a_2b_1$.
- $\Delta \neq 0$: solve for the intersection $(h, k)$; $x = u + h$, $y = v + k \Rightarrow$ homogeneous; then $v = zu$.
- $\Delta = 0$: $z = a_1x + b_1y \Rightarrow$ separable in $z, x$.
- Excluded values ($z = z_0$): substitute back — included, singular, or not a solution.
- Special substitutions ($u = \sin y$, $u = y^n$, $u = x + y$) have no general rule; look for "a function and its derivative".
- Usually the result is linear (use $\mu$) or separable.

CHAPTER 11

# Riccati and Clairaut equations

Two named equations with their own tricks. Riccati: if someone hands you one particular solution, a single substitution gives all the others. Clairaut: y = xp + f(p) is solved by straight lines y = cx + f(c), plus one special curve, the singular solution.

From Week+4.pdf, pages 1–14

## Riccati equations

$$\frac{dy}{dx} = p(x)\,y^2 + q(x)\,y + r(x)$$

DEFINITION 18 (WEEK+4.PDF P.1) — RICCATI EQUATION

It is "quadratic in $y$". Two quick observations from the lecturer:

- If $r(x) = 0$, it is $\frac{dy}{dx} - q(x)y = p(x)y^2$: a **Bernoulli** equation with $n = 2$ (p.1).
- In general (p.2) its solutions **cannot** be expressed in terms of elementary functions. We need help: one known solution.

### Theorem 9: one solution gives all the others

$$y = y_1 + \frac{1}{u} \quad\Longrightarrow\quad \frac{du}{dx} + \big(2p(x)\,y_1 + q(x)\big)\,u = -p(x)$$

THEOREM 9 (P.2–4): A PARTICULAR SOLUTION TURNS RICCATI INTO A LINEAR EQUATION

**Letters, exactly as on the slides**
Theorem 9 writes the substitution as $y := y_1 + \frac{1}{u}$ (new unknown $u$); Exercise 7.1 uses $y := u + y_1$ without the fraction. Some books call the new unknown $v$ — the same thing. This guide follows the slides.

Where it comes from (p.3–4):

1. $y_1$ is a solution: $y_1' = p\,y_1^2 + q\,y_1 + r$.
2. $y = y_1 + \frac{1}{u} \Rightarrow y' = y_1' - \frac{1}{u^2}u'$.
3. Substitute: $y_1' - \frac{1}{u^2}u' = p\big(y_1 + \frac{1}{u}\big)^2 + q\big(y_1 + \frac{1}{u}\big) + r = \underbrace{p y_1^2 + q y_1 + r}_{= y_1'} + 2p y_1 \frac{1}{u} + p\frac{1}{u^2} + q\frac{1}{u}$.
4. Cancel $y_1'$ on both sides: $-\frac{1}{u^2}u' = (2py_1 + q)\frac{1}{u} + p\frac{1}{u^2}$.
5. Multiply by $-u^2$: $u' + (2py_1 + q)u = -p$. Linear!

**What to write in the exam**

"Since $y_1$ is a particular solution, let $y = y_1 + \frac{1}{u}$." Then either substitute directly (1st way) or use the ready formula $u' + (2py_1 + q)u = -p$ (2nd way). The lecturer shows both and both get full credit.

**No particular solution, no method**

Theorem 9 needs a particular solution $y_1$. If none is given, look for an easy one (a constant, $y = ax + b$, $y = x^\beta$) by plugging in — or the problem tells you its form (Exercises 7.3, 7.4). Without $y_1$ you cannot start.

**EXAMPLE 22 · WEEK+4.PDF P.4–6** Solve $\frac{dy}{dx} = y^2 - 2xy + 2$, which has the particular solution $y_1(x) = 2x$.

1. $p = 1$, $q = -2x$, $r = 2$, $y_1 = 2x$. (Check: $y_1' = 2$ and $4x^2 - 4x^2 + 2 = 2$.)
2. $y = 2x + \frac{1}{u}$, $y' = 2 - \frac{1}{u^2}u'$.
3. **1st way** — substitute: $2 - \frac{1}{u^2}u' = \left(2x + \frac{1}{u}\right)^2 - 2x\left(2x + \frac{1}{u}\right) + 2$. Expand and multiply by $-u^2$: $u' = -1 - 4xu + 2xu$, i.e. $u' = -1 - 2xu$.
4. **2nd way** — formula: $2py_1 + q = 2 \cdot 1 \cdot 2x - 2x = 2x$, so $u' + 2xu = -1$. Same equation.
5. Linear: $\mu = e^{\int 2x\,dx} = e^{x^2}$; $\frac{d}{dx}\left(e^{x^2}u\right) = -e^{x^2}$.
6. $e^{x^2}u = -\int e^{x^2}dx + C$, so $u = e^{-x^2}\left(-\int e^{x^2}dx + C\right)$.

**ANSWER**

$y(x) = 2x + \dfrac{e^{x^2}}{-\int e^{x^2}dx + C}$, $C \in \mathbb{R}$ (the integral $\int e^{x^2}dx$ has no elementary form — leave it like this).

**The particular solution itself**

The particular solution $y_1$ itself corresponds to "$u = \infty$" and is not produced by any $C$. If a question says "find all solutions", mention $y = y_1$ separately.

## Homework Exercises 7 (p.7–8)

**EXERCISE 7.3 · WEEK+4.PDF P.7 (SOLUTION WRITTEN FOR THIS GUIDE)** Find $\alpha$ and $\beta$ such that $y' = y^2 - \frac{y}{x} + \frac{\alpha}{x^2}$ has a particular solution $y_1 = x^\beta$. Then find the general solution.

1. Plug in $y_1 = x^\beta$: $\beta x^{\beta-1} = x^{2\beta} - x^{\beta-1} + \alpha x^{-2}$.
2. The $x^{2\beta}$ term must cancel against the $x^{\beta-1}$ terms, so $2\beta = \beta - 1$: $\beta = -1$ (and then $\beta - 1 = -2$ also matches the $\frac{\alpha}{x^2}$ term). Then $-x^{-2} = x^{-2} - x^{-2} + \alpha x^{-2} \Rightarrow \alpha = -1$.

3. So $y_1 = \frac{1}{x}$, $p = 1$, $q = -\frac{1}{x}$: $2py_1 + q = \frac{2}{x} - \frac{1}{x} = \frac{1}{x}$. Linear: $u' + \frac{1}{x}u = -1$.
4. $\mu = x$: $(xu)' = -x \Rightarrow xu = -\frac{x^2}{2} + C \Rightarrow u = \dfrac{2C - x^2}{2x}$.

**ANSWER**

$\alpha = -1$, $\beta = -1$; $y = \dfrac{1}{x} + \dfrac{2x}{K - x^2}$ (with $K = 2C$), and $y = \frac{1}{x}$.

**EXERCISE 7.4 · WEEK+4.PDF P.7 (SOLUTION WRITTEN FOR THIS GUIDE)** Find the general solution of $y' = (y - \alpha x + \beta)^2 + \alpha$ if it has a particular solution which is a straight line.

1. Try $y_1 = ax + b$: $a = (ax + b - \alpha x + \beta)^2 + \alpha$ for all $x$ forces $a = \alpha$ and $b = -\beta$. So $y_1 = \alpha x - \beta$.
2. Expand: $y' = y^2 - 2(\alpha x - \beta)y + (\alpha x - \beta)^2 + \alpha$, so $p = 1$, $q = -2(\alpha x - \beta)$.
3. $2py_1 + q = 2(\alpha x - \beta) - 2(\alpha x - \beta) = 0$: the linear equation is just $u' = -1$, $u = C - x$.

**ANSWER**

$y = \alpha x - \beta + \dfrac{1}{C - x}$, and $y = \alpha x - \beta$.

- **Exercise 7.1** (p.7): with $y = y_1 + u$ (no fraction) the same computation gives $u' - (2py_1 + q)u = p\,u^2$ — a **Bernoulli** equation with $n = 2$. (Then $\frac{1}{u}$ makes it linear — which is exactly why Theorem 9 uses $y_1 + \frac{1}{u}$ directly.)
- **Exercise 7.2** (p.7–8): $y = -\dfrac{u'}{p\,u}$ turns Riccati into a second-order linear equation. If $p$ is a constant the result is $u'' - q\,u' + p\,r\,u = 0$; in general $u'' - \left(q + \frac{p'}{p}\right)u' + p\,r\,u = 0$. (The handwritten solution writes $u'' + qu' + rpu = 0$: it loses the sign of the $q$-term, because $q\,y = -q\frac{u'}{pu}$, and it ignores $p'$.)

## Clairaut equations

$$y = x\,p + f(p), \qquad p := \frac{dy}{dx}$$

**DEFINITION 19 (P.9) — CLAIRAUT EQUATION**

The unknown slope $p$ appears inside a function $f$, so we cannot solve for $y'$ directly. The trick is to **differentiate the whole equation**.

### Solution technique (p.9–11)

1. Differentiate with respect to $x$ (product rule on $xp$, chain rule on $f(p)$): $\dfrac{dy}{dx} = p + x\dfrac{dp}{dx} + f'(p)\dfrac{dp}{dx}$.
2. The left side is $p$, so $p$ cancels: $0 = \dfrac{dp}{dx}\big(x + f'(p)\big)$.
3. **Either** $\dfrac{dp}{dx} = 0 \Rightarrow p = c$. Substitute $c$ for $p$ in the equation: the **general solution** is the family of lines

$$y = cx + f(c), \qquad c \text{ an arbitrary constant}$$

**GENERAL SOLUTION: REPLACE P BY A CONSTANT C**

4. **Or** $x + f'(p) = 0$, i.e. $f'(p) = -x$. Solve it to find the **singular solution**: an extra solution which is **not** a member of the one-parameter family.

$$x = -f'(p), \qquad y = x\,p + f(p) \quad \Rightarrow \quad \text{eliminate } p$$

**SINGULAR SOLUTION — PARAMETRIC FORM, THEN ELIMINATE THE PARAMETER**

**EXAMPLE 23 · WEEK+4.PDF P.12–13** Solve $y = xp + p^2$.

1. Differentiate: $\dfrac{dy}{dx} = p + x\dfrac{dp}{dx} + 2p\dfrac{dp}{dx} \Rightarrow 0 = \dfrac{dp}{dx}(x + 2p)$.
2. $\dfrac{dp}{dx} = 0 \Rightarrow p = c$: general solution $y = cx + c^2$.
3. $x + 2p = 0 \Rightarrow p = -\dfrac{x}{2}$, i.e. $\dfrac{dy}{dx} = -\dfrac{x}{2} \Rightarrow y = -\dfrac{x^2}{4} + k$ (lecturer's way: integrate the slope).
4. Which $k$? Plug into the original equation with $p = y' = -\frac{x}{2}$: $-\frac{x^2}{4} + k = x\left(-\frac{x}{2}\right) + \frac{x^2}{4} = -\frac{x^2}{4} \Rightarrow k = 0$.
5. Shortcut (same result): put $p = -\frac{x}{2}$ straight into $y = xp + p^2$: $y = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4}$.

**ANSWER**

General solution $y = cx + c^2$; singular solution $y = -\dfrac{x^2}{4}$.

**Check the singular solution in 10 seconds**

For $y = -\frac{x^2}{4}$: $p = -\frac{x}{2}$, and $xp + p^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4} = y$. It works, and it is a parabola, not a line — so it cannot be one of the $y = cx + c^2$.

## Homework Exercises 8 (p.14)

**EXERCISE 8.1 · WEEK+4.PDF P.14 (SOLUTION WRITTEN FOR THIS GUIDE)** Solve the Clairaut equation $y = xp + p^2 + 1$.

1. $f(p) = p^2 + 1$, $f'(p) = 2p$. Differentiating: $0 = \frac{dp}{dx}(x + 2p)$.
2. $p = c$: $y = cx + c^2 + 1$.
3. $x + 2p = 0 \Rightarrow p = -\frac{x}{2} \Rightarrow y = -\frac{x^2}{2} + \frac{x^2}{4} + 1 = 1 - \frac{x^2}{4}$.

**ANSWER**

General: $y = cx + c^2 + 1$; singular: $y = 1 - \dfrac{x^2}{4}$.

**EXERCISE 8.2 · WEEK+4.PDF P.14 (SOLUTION WRITTEN FOR THIS GUIDE)** Solve the Clairaut equation $y = xp + e^{-p} + 1$.

1. $f(p) = e^{-p} + 1$, $f'(p) = -e^{-p}$. Differentiating: $0 = \frac{dp}{dx}\left(x - e^{-p}\right)$.
2. $p = c$: $y = cx + e^{-c} + 1$.
3. $x = e^{-p} \Rightarrow p = -\ln x$ $(x > 0)$. Then $y = x(-\ln x) + x + 1$.

**ANSWER**
General: $y = cx + e^{-c} + 1$; singular: $y = x - x\ln x + 1$, $x > 0$.

**Lagrange equations (Exercises 8.3–8.4)**
The slide also names equations of the form $y = x\,f(p) + g(p)$ "**Lagrange differential equations**" and says they are solved "by using the same steps" (differentiate with respect to $x$). Clairaut is the special case $f(p) = p$. No worked solution is given in the slides.

## Traps

**Forgetting the singular solution**
"Solve the Clairaut equation" means **both** the family $y = cx + f(c)$ **and** the singular solution from $x + f'(p) = 0$. The singular one is where most points are lost.

**Riccati sign**
The linear equation is $u' + (2py_1 + q)u = -p$ — note the minus on the right. Check it by quickly redoing the substitution if unsure.

**Clairaut vs. Riccati**
Clairaut has $y$ *alone* on the left and $p = y'$ inside a function: $y = xp + f(p)$. Riccati has $y'$ alone on the left and is quadratic in $y$: $y' = py^2 + qy + r$.

## In one minute

- Riccati: $y' = py^2 + qy + r$; if $r = 0$ it is Bernoulli $(n = 2)$.
- Needs a particular solution $y_1$; then $y = y_1 + \frac{1}{u}$.
- Linear equation: $u' + (2py_1 + q)u = -p$; solve with $\mu$; and $y = y_1$ is also a solution.
- Find $y_1$ by trying a guessed form (constant, line, $x^\beta$).
- Clairaut: $y = xp + f(p)$; differentiate $\Rightarrow \frac{dp}{dx}(x + f'(p)) = 0$.
- General solution: $y = cx + f(c)$ (replace $p$ by $c$).
- Singular solution: $x = -f'(p)$, $y = xp + f(p)$, eliminate $p$ (e.g. $y = xp + p^2 \Rightarrow y = -\frac{x^2}{4}$).

CHAPTER 12

# Orthogonal trajectories

Given a family of curves, find the curves that cross every one of them at a right angle. It is a three-step recipe: get the family's slope, flip it to −1/slope, solve the new (usually separable) equation.

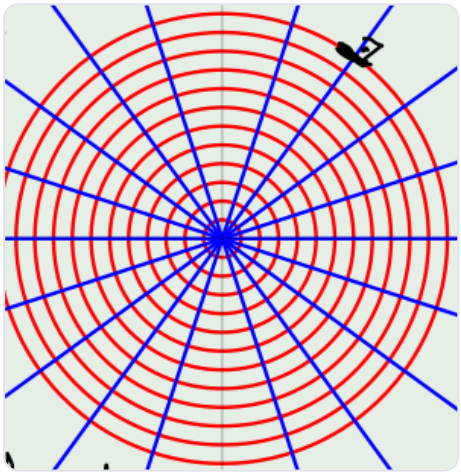
From Week+4.pdf, pages 20–29

## What we are looking for

This is the first **application** of first-order equations in the course (p.20: applied problems formulated in terms of first-order differential equations).

**Definition 20 (p.21)**

Let $\Phi(x, y, c) = 0$ be a given one-parameter family of curves in the $xy$-plane. A curve which intersects the curves of the given family at right angles is called an **orthogonal trajectory** of the given family.



Example 24: the circles $x^2 + y^2 = c^2$ (red) and the lines $y = kx$ through the origin (blue) cross at right angles everywhere (Week+4.pdf p.22)

Example 24 (p.22): every straight line $y = kx$ through the origin is an orthogonal trajectory of the circles $x^2 + y^2 = c^2$, and conversely each circle is an orthogonal trajectory of the lines. "Actually, they are orthogonal trajectories of each other."

**→ Right angle = slopes multiply to −1**

Two lines with slopes $m_1$ and $m_2$ are perpendicular exactly when $m_1 m_2 = -1$, i.e. $m_2 = -\dfrac{1}{m_1}$. Two curves cross at right angles when their **tangent lines** at the crossing point do. The lecturer checks this on Example 24: circles have slope $m_1 = -\frac{x}{y}$, lines have $m_2 = \frac{y}{x}$, and $m_1 m_2 = -\frac{x}{y} \cdot \frac{y}{x} = -1$.

## The recipe (Solution technique, p.23–24)

1. **Slope of the given family.** Differentiate $\Phi(x, y, c) = 0$ implicitly with respect to $x$.
2. **Eliminate** $c$ using the original equation and the derived one. Result: $\dfrac{dy}{dx} = f(x, y)$ — this is $m_1$, the slope of the family member through $(x, y)$. (In general $f = -\Phi_x / \Phi_y$.)
3. **Flip:** the trajectories have slope $m_2 = -\dfrac{1}{f(x, y)}$. Their differential equation is

$$\frac{dy}{dx} = -\frac{1}{f(x, y)} = \frac{\partial \Phi / \partial y}{\partial \Phi / \partial x}$$

DIFFERENTIAL EQUATION OF THE ORTHOGONAL TRAJECTORIES (P.24)

4. **Solve it.** A one-parameter family of solutions $\Psi(x, y, d) = 0$ is the family of orthogonal trajectories.

**! Eliminate c BEFORE flipping**

The slope $f(x, y)$ must not contain $c$. For $y = ce^x$: $y' = ce^x$ still has $c$; replace $ce^x$ by $y$ to get $y' = y$. Only then flip to $y' = -\frac{1}{y}$. If you flip $ce^x$ you get a "family" that depends on $c$ — wrong.

## Example 25 — circles

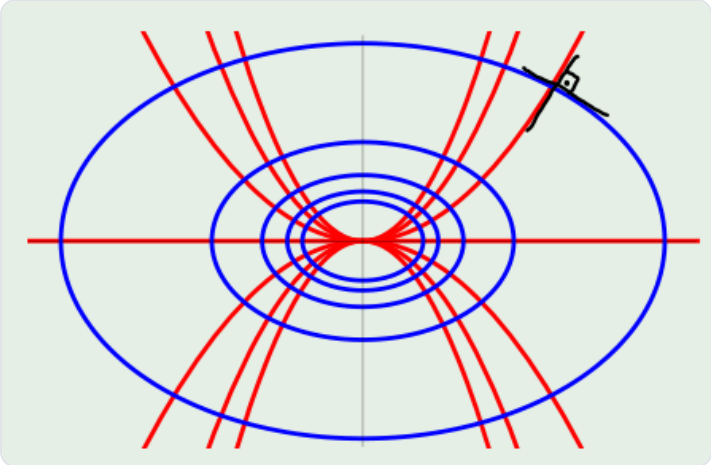
**EXAMPLES 24–25 · WEEK+4.PDF P.22, 25** Find the orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$.

1. Differentiate: $2x + 2y\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = -\dfrac{x}{y}$. ($c$ disappeared by itself.) So $m_1 = -\frac{x}{y}$.
2. Trajectories: $\dfrac{dy}{dx} = -\dfrac{1}{m_1} = \dfrac{y}{x}$ (separable).
3. $\displaystyle\int \frac{dy}{y} = \int \frac{dx}{x} \Rightarrow \ln|y| = \ln|x| + \ln|k| \Rightarrow y = kx.$

**ANSWER**

The orthogonal trajectories are the lines through the origin, $y = kx$.

## Example 26 — parabolas



Example 26: the parabolas y = kx² (red) are crossed perpendicularly by ellipses (blue) (Week+4.pdf p.25)

**EXAMPLE 26 · WEEK+4.PDF P.25–26** Find the orthogonal trajectories of the family of parabolas $y = kx^2$.

1. **1st family.** $y = kx^2 \Rightarrow k = \dfrac{y}{x^2}$.
2. $\dfrac{dy}{dx} = 2kx = 2\dfrac{y}{x^2}x = \dfrac{2y}{x}$ ($k$ eliminated). So $m_1 = \dfrac{2y}{x}$.
3. **2nd family.** $m_2 = -\dfrac{1}{m_1} = -\dfrac{x}{2y}$: $\dfrac{dy}{dx} = -\dfrac{x}{2y}$ (separable).
4. $\displaystyle\int 2y\,dy = -\int x\,dx \Rightarrow y^2 = -\frac{x^2}{2} + C.$

**ANSWER**

$y^2 + \dfrac{x^2}{2} = C$, $C$ arbitrary — a family of ellipses.

## Homework Exercises 9.1 (p.27)

**EXERCISE 9.1.5 · WEEK+4.PDF P.27–28** Obtain the orthogonal trajectories of the family of curves $y = \dfrac{e^x}{e^x + c}$.

1. $e^x y + yc = e^x \Rightarrow c = \dfrac{e^x(1-y)}{y}$.
2. Quotient rule: $\dfrac{dy}{dx} = \dfrac{e^x(e^x + c) - e^x \cdot e^x}{(e^x + c)^2} = \dfrac{ce^x}{(e^x + c)^2}$.
3. Eliminate $c$: from the family, $e^x + c = \dfrac{e^x}{y}$, so $\dfrac{dy}{dx} = \dfrac{c\,y^2}{e^x} = \dfrac{e^x(1-y)}{y} \cdot \dfrac{y^2}{e^x} = (1-y)y$. So $m_1 = (1-y)y$.
4. $m_2 = \dfrac{1}{(y-1)y}$: $\dfrac{dy}{dx} = \dfrac{1}{(y-1)y}$ (separable).

5 $\int y(y-1)\,dy = \int dx \Rightarrow \frac{y^3}{3} - \frac{y^2}{2} = x + C.$

**ANSWER**
$\frac{y^3}{3} - \frac{y^2}{2} = x + C$, $C$ arbitrary.

**EXERCISE 9.1.6 · WEEK+4.PDF P.29 (LAST STEP COMPLETED IN THIS GUIDE)** Show that the orthogonal trajectories of the family $y^2 = 2cx + c^2$ is the family itself.

1 Differentiate: $2y\frac{dy}{dx} = 2c \Rightarrow \frac{dy}{dx} = \frac{c}{y}$.

2 Eliminate $c$: $y^2 + x^2 = x^2 + 2cx + c^2 = (x+c)^2 \Rightarrow x + c = \pm\sqrt{x^2+y^2}$. Taking $c = -x + \sqrt{x^2+y^2}$: $m_1 = \frac{-x + \sqrt{x^2+y^2}}{y}$.

3 $m_2 = -\frac{1}{m_1} = \frac{-y}{-x + \sqrt{x^2+y^2}}$ — a homogeneous equation (the lecturer stops here: "Solve it!").

4 Instead of solving, rationalise: multiply top and bottom by $x + \sqrt{x^2+y^2}$. Since $(-x + \sqrt{x^2+y^2})(x + \sqrt{x^2+y^2}) = y^2$, we get $m_2 = \frac{-x - \sqrt{x^2+y^2}}{y}$.

5 That is exactly $\frac{c}{y}$ for the *other* root $c = -x - \sqrt{x^2+y^2}$. So the trajectory through $(x, y)$ satisfies the same equation $y' = \frac{c}{y}$, $y^2 = 2cx + c^2$, with the other value of $c$.

**ANSWER**
Through every point pass two curves of the family (one for each root $c$), and they meet at a right angle: the family $y^2 = 2cx + c^2$ is its own family of orthogonal trajectories.

Answers for the other homework problems (all separable after flipping, checked with sympy):

| Family | $m_1$ ($c$ eliminated) | Trajectory equation | Orthogonal trajectories |
|---|---|---|---|
| 9.1.1 $3x^2 - y^2 = c$ | $\frac{3x}{y}$ | $y' = -\frac{y}{3x}$ | $xy^3 = K$ |
| 9.1.2 $y = ce^x$ | $y$ | $y' = -\frac{1}{y}$ | $y^2 = -2x + C$ |
| 9.1.3 $y = cx^2$ | $\frac{2y}{x}$ | $y' = -\frac{x}{2y}$ | $x^2 + 2y^2 = C$ (as Example 26) |
| 9.1.4 $y^3 = cx^2$ | $\frac{2y}{3x}$ | $y' = -\frac{3x}{2y}$ | $3x^2 + 2y^2 = C$ |

★ **Shortcut for the slope**
From $\Phi(x, y, c) = 0$ you can also solve for $c$ first ($c = G(x, y)$), then differentiate $G(x, y) = c$: $G_x + G_y\,y' = 0 \Rightarrow y' = -\frac{G_x}{G_y}$. The constant is gone automatically. E.g. $c = \frac{y}{x^2}$: $-\frac{2y}{x^3} + \frac{1}{x^2}y' = 0 \Rightarrow y' = \frac{2y}{x}$.

## In one minute

- Orthogonal trajectory: crosses every curve of the family at a right angle.
- Step 1: differentiate the family implicitly.
- Step 2: eliminate $c \Rightarrow y' = f(x, y)$ (no $c$ left!).
- Step 3: replace $y'$ by $-\frac{1}{y'}$: solve $y' = -\frac{1}{f(x,y)}$.
- The new equation is usually separable (sometimes homogeneous).
- Circles $x^2 + y^2 = c^2$ are orthogonal to lines $y = kx$; parabolas $y = kx^2$ are orthogonal to ellipses $y^2 + \frac{x^2}{2} = C$.
- $y^2 = 2cx + c^2$ is self-orthogonal.

CHAPTER 13

# Applications: cooling, falling bodies, decay and mixtures

Turn a word problem into a first-order differential equation, solve it with methods you already know, and use the numbers in the story to pin down the constants. Question 1 of the midterm is always one of these.

From Week+5.pdf, pages 1–29

## The big picture

Every problem in this chapter is the same story told in four costumes: **cooling**, **falling**, **growth/decay** and **mixing**. In each one, a sentence tells you how fast something changes. You write that sentence as an equation, solve it (it is always separable or linear — methods you already know), and then use the given measurements to find the unknown constants.

→ **The 5-step recipe for every word problem**

1. **Name the unknown** and its units: "Let $A(t)$ be the grams of A present at time $t$ hours."
2. **Translate the rate sentence** into a differential equation (table below).
3. **Write every number in the story as a condition**: "initially 50 g" means $A(0) = 50$; "after 2 hours 10 g" means $A(2) = 10$.
4. **Solve** the equation (separable or linear), then use the condition at $t = 0$ to find $C$, and the second condition to find $k$.
5. **Answer the actual question** (plug in a time, or solve for a time) and write the units.

## Translating words into an equation

"The rate of change of A" always means the derivative $\frac{dA}{dt}$. "Proportional to" always means "equals a constant $k$ times". The rest is just reading carefully what it is proportional to.

| The sentence says... | You write... |
|---|---|
| A changes at a rate proportional to the amount present | $\frac{dA}{dt} = kA$ |
| A **decays / is converted / decreases** at a rate proportional to the amount | $\frac{dA}{dt} = -kA$, $k > 0$ |
| ... proportional to the **square** of the amount | $\frac{dA}{dt} = -kA^2$ |
| ... proportional to the **square root** of the amount | $\frac{dA}{dt} = -k\sqrt{A}$ |

| The sentence says... | You write... |
|---|---|
| bacteria **grow** at a rate proportional to the number present | $\frac{dN}{dt} = kN,\ k > 0$ |
| temperature changes at a rate proportional to the **difference** between the body and the surroundings | $\frac{dT}{dt} = -k(T - T_m)$ |
| force = gravity minus air resistance proportional to speed | $m\frac{dv}{dt} = mg - kv$ |
| salt enters at one rate and leaves at another | $\frac{dx}{dt} = r_{\text{in}} - r_{\text{out}}$ |

| Data sentence | Condition |
|---|---|
| "Initially 50 g is present" | $A(0) = 50$ |
| "After 2 hours only 10 g remain" | $A(2) = 10$ |
| "Half ... disintegrates in 138 years" (half-life) | $N(138) = \frac{1}{2}N_0$ |
| "The number doubles every 10 hours" | $N(10) = 2N_0$ |
| "It is cooling at 5° per minute (at the initial time)" | $T'(0) = -5$ |
| "Dropped without any velocity" | $v(0) = 0$ |

**Where does the minus sign go?**

If the quantity **decreases**, its derivative is negative. The slides use two conventions, and both are fine as long as you stay consistent:

- Write $\frac{dA}{dt} = -kA$ with $k > 0$ (printed slides, Week+5.pdf p.20).
- Write $\frac{dN}{dt} = kN$ and let the data tell you that $k$ is negative (handwritten solution of Example 29: $k = -\frac{\ln 2}{138}$).

What you must **not** do is write $-k$ and then also get a negative $k$: that would make the amount grow.

**Two conditions, two unknowns**

The general solution always contains **two** unknowns: the integration constant $C$ and the proportionality constant $k$. That is why every problem gives you **two** measurements (plus maybe a third time where you have to predict). Use the $t = 0$ data first — it usually gives $C$ immediately.

## 1. Newton's law of cooling

**Intuition.** A hot cup of tea in a cold room cools fast at first and then more and more slowly, because the cooling speed depends on how much hotter it is than the room. When the tea reaches room temperature, it stops changing.

$$\frac{dT}{dt} = -k\,(T - T_m)$$

**NEWTON'S LAW OF COOLING (WEEK+5.PDF P.2)**

$T(t)$ temperature of the body at time $t$ (the unknown)

$T_m$ temperature of the surrounding medium (room, outside air) — a constant

$k$ a **positive** constant of proportionality

$\frac{dT}{dt}$ rate of change of the temperature of the body

**Why the minus sign?** If the body is hotter than the room ($T - T_m > 0$), it must cool, so $\frac{dT}{dt} < 0$. With $k > 0$, only $-k(T - T_m)$ has the right sign. The same formula also works for a cold body warming up: then $T - T_m < 0$ and $\frac{dT}{dt} > 0$.

**Solving it once and for all.** The equation is separable. For $T \neq T_m$:

$$\int \frac{dT}{T - T_m} = -\int k\,dt \;\Rightarrow\; \ln|T - T_m| = -kt + \ln C \;\Rightarrow\; T(t) = T_m + \tilde{C}e^{-kt}.$$

($T \equiv T_m$ is also a solution; it is the case $\tilde{C} = 0$.) Putting $t = 0$ gives $\tilde{C} = T_0 - T_m$, so

$$T(t) = T_m + (T_0 - T_m)\,e^{-kt}$$

**SOLUTION OF THE COOLING LAW (T0 = STARTING TEMPERATURE)**

**EXAMPLE 28 · WEEK+5.PDF P.3–6** A thermometer, which has been showing 26 °C inside a house, is placed outside where the temperature is 10 °C. Three minutes later the thermometer reads 12 °C. Predict the thermometer reading at any time.

1. Unknown: $T(t)$ = reading after $t$ minutes. Data: $T(0) = 26$, $T(3) = 12$, $T_m = 10$.
2. Model: $\frac{dT}{dt} = -k(T - 10)$ (temperature is decreasing, so we use $-k$ with $k > 0$)
3. Separable. For $T \neq 10$: $\frac{dT}{T-10} = -k\,dt \Rightarrow \ln|T - 10| = -kt + \ln C \Rightarrow T(t) = \tilde{C}e^{-kt} + 10$. ($T = 10$ is also a solution, covered by $\tilde{C} = 0$)
4. $T(0) = 26$: $\tilde{C}e^0 + 10 = 26 \Rightarrow \tilde{C} = 16$.
5. $T(3) = 12$: $16e^{-3k} + 10 = 12 \Rightarrow e^{-3k} = \frac{2}{16} = 2^{-3} \Rightarrow -3k = -3\ln 2 \Rightarrow k = \ln 2$.
6. So $T(t) = 16e^{-(\ln 2)t} + 10 = 16 \cdot 2^{-t} + 10$. (because $e^{-(\ln 2)t} = (e^{\ln 2})^{-t} = 2^{-t}$)

**ANSWER**

$T(t) = 16\left(\frac{1}{2}\right)^t + 10 = 2^{4-t} + 10$ °C

★ **Shortcut for k: take a ratio**

From $T - T_m = (T_0 - T_m)e^{-kt}$: $e^{-kt} = \dfrac{T(t) - T_m}{T_0 - T_m}$. So at the measured time $t_1$, $e^{-kt_1}$ is just a ratio of differences. In Example 28: $e^{-3k} = \frac{12-10}{26-10} = \frac{1}{8}$. You often don't even need $k$ itself — keep $e^{-k}$ as a number.

## Exercises 10.1 (homework on Week+5.pdf p.7), solved

**EXERCISES 10.1 #1 · WEEK+5.PDF P.7** An object initially at 20 °C is placed in a room kept at 80 °C. One hour later it is 35 °C. What will its temperature be after three hours?

1. $T(0) = 20$, $T_m = 80$, $T(1) = 35$. Model $\frac{dT}{dt} = -k(T - 80)$, solution $T = 80 + (20 - 80)e^{-kt} = 80 - 60e^{-kt}$. (here the body warms up; the same law works)
2. $T(1) = 35$: $80 - 60e^{-k} = 35 \Rightarrow e^{-k} = \frac{45}{60} = \frac{3}{4}$.
3. $T(3) = 80 - 60\,(e^{-k})^3 = 80 - 60 \cdot \frac{27}{64} = \frac{875}{16}$.

**ANSWER**
$T(3) = \frac{875}{16} \approx 54.69$ °C

**EXERCISES 10.1 #2 · WEEK+5.PDF P.7** A 200 °F cup of tea is left in a 65 °F room. At the initial time the tea is cooling at 5 °F per minute. Write an initial-value problem for the temperature and find the temperature after 5 minutes.

1. IVP: $\frac{dT}{dt} = -k(T - 65)$, $T(0) = 200$. The extra information is about the **rate**: $T'(0) = -5$.
2. Put $t = 0$ into the equation: $-5 = -k(200 - 65) = -135k \Rightarrow k = \frac{5}{135} = \frac{1}{27}$.
3. $T(t) = 65 + 135e^{-t/27}$, so $T(5) = 65 + 135e^{-5/27} \approx 177.18$.

**ANSWER**
$T(5) \approx 177.2$ °F

**The slide's handwritten note says T(1)=195**
The handwriting on the slide reads the data as $T(1) = 195$. That is an approximation of "cooling at 5° per minute". With $T(1) = 195$ you get $e^{-k} = \frac{130}{135}$ and $T(5) \approx 176.8$ °F — almost the same. The exact reading of the sentence is $T'(0) = -5$.

**EXERCISES 10.1 #3 · WEEK+5.PDF P.7** A room is kept at 68 °F. A pie taken out of a 350 °F oven has cooled to 150 °F after 45 minutes. When will it reach 80 °F?

1. $T(t) = 68 + 282e^{-kt}$ $(T_0 - T_m = 350 - 68 = 282)$
2. $T(45) = 150$: $e^{-45k} = \frac{82}{282}$.
3. Want $T(\bar{t}) = 80$: $e^{-k\bar{t}} = \frac{12}{282}$. Take logs and divide: $\dfrac{\bar{t}}{45} = \dfrac{\ln(12/282)}{\ln(82/282)}$.

**ANSWER**
$\bar{t} = 45\,\dfrac{\ln(12/282)}{\ln(82/282)} \approx 115$ minutes

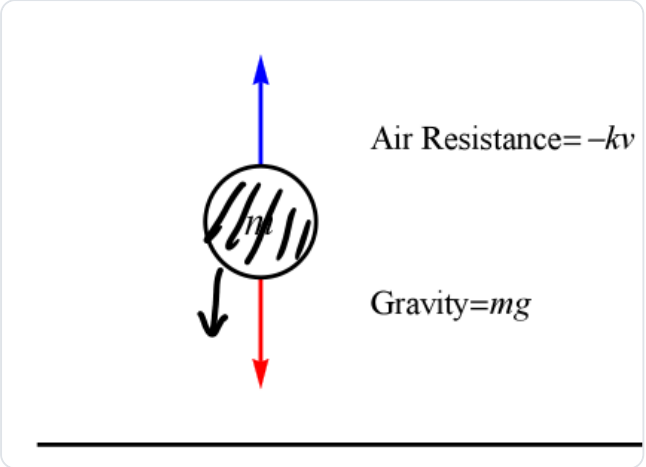
## 2. Problems in mechanics: falling objects

**Newton's second law:** the rate of change of momentum equals the force. For constant mass:

$$\frac{d}{dt}(mv) = F \quad \Longleftrightarrow \quad F = ma = m\frac{dv}{dt}$$

NEWTON'S SECOND LAW (WEEK+5.PDF P.8)

$m$ = mass, $v$ = velocity, $a = \frac{dv}{dt}$ = acceleration, $F$ = total force. Below, $k$ is the drag coefficient and $g$ the gravitational constant.




Forces on a falling object: gravity mg down, air resistance –kv up (Week+5.pdf p.9)

**Falling through air.** Two forces act: gravity $mg$ (down, the positive direction) and air resistance $-kv$ (opposite to the motion, proportional to speed). So

$$m\frac{dv}{dt} = mg - kv \quad \Longleftrightarrow \quad \frac{dv}{dt} + \frac{k}{m}v = g$$

FALLING BODY WITH AIR RESISTANCE (WEEK+5.PDF P.9)

It is both separable and linear. Solving with $v(0) = v_0$ (Week+5.pdf p.10):

$$v(t) = \left(v_0 - \frac{mg}{k}\right) e^{-\frac{k}{m}t} + \frac{mg}{k}$$

VELOCITY OF A FALLING OBJECT

→ **Terminal (limiting) velocity**

As $t \to \infty$, $e^{-\frac{k}{m}t} \to 0$, so $v(t) \to \frac{mg}{k}$. This is the speed at which air resistance exactly cancels gravity: $mg - kv = 0$. You can read it straight from the equation without solving it.

**Distance.** Velocity is the derivative of position, so the distance fallen is $x(t) = \int v(t)\,dt$ with $x(0) = 0$.

**EXERCISES 10.2 #1 · WEEK+5.PDF P.11–16 (HANDWRITTEN SOLUTION)** A parachutist bails out at 10 000 ft, falls freely for 20 seconds, then opens her parachute. Air resistance is proportional to speed with $k_1 = 0.15$ without the parachute and $k_2 = 1.5$ with it. (a) IVP for the free fall; (b) IVP after the parachute opens; (c) total time to land. *The lecturer takes* $m = 60$, $g = 10$.

1. **(a)** $\frac{dv}{dt} = \frac{mg - k_1 v}{m} \Rightarrow \frac{dv}{dt} + \frac{0.15}{60} v = 10$, $v(0) = 0$, for $0 \le t \le 20$.
2. Linear, $\mu = e^{\int \frac{k}{m} dt} = e^{\frac{k}{m} t}$: $\frac{d}{dt}\left(v e^{\frac{k}{m} t}\right) = g e^{\frac{k}{m} t} \Rightarrow v = \frac{mg}{k} + C e^{-\frac{k}{m} t}$. With $v(0) = 0$: $C = -\frac{mg}{k}$.
3. $\frac{mg}{k_1} = \frac{600}{0.15} = 4000$, $\frac{k_1}{m} = \frac{1}{400}$: $v_1(t) = 4000\left(1 - e^{-t/400}\right)$.
4. Distance: $x_1(t) = \int v_1\, dt = 4000t + 1\,600\,000\, e^{-t/400} + C$; $x_1(0) = 0 \Rightarrow C = -1\,600\,000$, so $x_1(t) = 4000t + 1\,600\,000\left(e^{-t/400} - 1\right)$.
5. $x_1(20) \approx 1967 \approx 2000$ and $v_1(20) = 4000(1 - e^{-0.05}) \approx 195$. (the lecturer rounds to 2000)
6. **(b)** Restart the clock when the chute opens: $\frac{dv}{dt} = \frac{600 - 1.5v}{60}$, $v_2(0) = v_1(20) \approx 195$. Now $\frac{mg}{k_2} = 400$, $\frac{k_2}{m} = \frac{1}{40}$: $v_2 = 400 + C e^{-t/40}$, and $400 + C = 195 \Rightarrow C = -205$.
7. $v_2(t) = 400 - 205 e^{-t/40}$; $x_2(t) = 400t + 8200\left(e^{-t/40} - 1\right)$ ($\int -205 e^{-t/40} dt = 8200 e^{-t/40}$, then $x_2(0) = 0$)
8. **(c)** Remaining height $\approx 10\,000 - 2000 = 8000$: solve $400\bar{t} + 8200\left(e^{-\bar{t}/40} - 1\right) = 8000$. This cannot be solved by algebra; trying values gives $\bar{t} \approx 31$.

**ANSWER**
$v_1 = 4000(1 - e^{-t/400})$, $v_2 = 400 - 205 e^{-t/40}$; total time $\approx 20 + 31 = 51$ seconds

**Units in this exercise**
The exercise does not give the mass; the handwritten solution assumes $m = 60$ and $g = 10$ and then treats the 10 000 ft as if it were in the same unit as these numbers. On the exam, use exactly the data you are given; the **method** (two IVPs, the second starting with the final velocity of the first) is the point.

**EXERCISES 10.2 #2 · WEEK+5.PDF P.11** A 1 kg steel ball is dropped from 793 m with no initial velocity. Air resistance is $\frac{1}{8}v$. (a) Find the limiting velocity. (b) Find approximately how long it takes to hit the ground ($g = 9.8$).

1. $1 \cdot \frac{dv}{dt} = 9.8 - \frac{1}{8}v$, $v(0) = 0$. So $v(t) = 78.4\left(1 - e^{-t/8}\right)$. ($\frac{mg}{k} = \frac{9.8}{1/8} = 78.4$, $\frac{k}{m} = \frac{1}{8}$)
2. (a) $\lim_{t \to \infty} v(t) = 78.4$ m/s.
3. (b) $x(t) = \int_0^t v = 78.4t - 627.2\left(1 - e^{-t/8}\right)$. Set $x(t) = 793$ and solve numerically.

**ANSWER**
limiting velocity 78.4 m/s; hits the ground after about 17.2 s

## 3. Rate problems: growth and decay

**Intuition.** The more nuclei (or bacteria) there are, the more decay (or divide) per second. So the rate is proportional to the amount itself.

$$\frac{dN}{dt} = kN \;\Rightarrow\; N(t) = N_0 e^{kt}$$

GROWTH (K POSITIVE) OR DECAY (K NEGATIVE) — WEEK+5.PDF P.17–19

It is separable (the lecturer also solved it as a linear equation with $\mu = e^{-kt}$). With the printed convention $\frac{dx}{dt} = -kx$, $k > 0$, the answer is $x(t) = x_0 e^{-kt}$ — the same thing.

**EXAMPLE 29 · WEEK+5.PDF P.17–21** Half of the original number of Polonium-210 nuclei disintegrate in 138 years. (1) What percentage remains after 414 years? (2) When will only one-tenth remain?

1. $N(t)$ = nuclei at time $t$ years; $N(0) = N_0$, $N(138) = \frac{N_0}{2}$.
2. $\frac{dN}{dt} = kN \Rightarrow N(t) = Ce^{kt}$; $N(0) = N_0 \Rightarrow C = N_0$.
3. $\frac{N_0}{2} = N_0 e^{138k} \Rightarrow -\ln 2 = 138k \Rightarrow k = -\frac{\ln 2}{138}$.
4. So $N(t) = N_0 e^{-\frac{\ln 2}{138} t} = N_0\, 2^{-t/138} = N_0 \left(\frac{1}{2}\right)^{t/138}$.
5. (1) $N(414) = N_0\, 2^{-414/138} = N_0\, 2^{-3} = \frac{N_0}{8}$, i.e. 12.5%. (414 = 3 half-lives)
6. (2) $N_0\, 2^{-T/138} = \frac{N_0}{10} \Rightarrow -\frac{T}{138} \ln 2 = -\ln 10 \Rightarrow T = \frac{138 \ln 10}{\ln 2}$.

**ANSWER**
(1) 12.5% remains; (2) $T = \frac{138 \ln 10}{\ln 2} \approx 458.4$ years (about 458 years and 5 months)

**Half-life form**
If the half-life is $h$, then directly $N(t) = N_0 \left(\frac{1}{2}\right)^{t/h}$. After 1, 2, 3 half-lives: 50%, 25%, 12.5%. Similarly, "doubles every $d$ hours" gives $N(t) = N_0\, 2^{t/d}$.

### Exercises 10.3 (homework on Week+5.pdf p.22–23), solved

**EXERCISES 10.3 #1 · WEEK+5.PDF P.22 — THIS EXACT PROBLEM WAS QUESTION 1 OF THE 2023 MIDTERM**
During a chemical reaction, substance A is converted into substance B at a rate proportional to the **square** of the amount of A. Initially 50 g of A is present, and after 2 hours only 10 g remain. How much A is present after 4 hours?

1. $\frac{dA}{dt} = -kA^2$, $A(0) = 50$, $A(2) = 10$. ("converted" = decreasing, so minus; "square" gives $A^2$)
2. Separable. For $A \neq 0$: $\int \frac{dA}{A^2} = -\int k\, dt \Rightarrow -\frac{1}{A} = -kt - C \Rightarrow A = \frac{1}{kt + C}$.
3. $A(0) = 50 \Rightarrow C = \frac{1}{50} \Rightarrow A(t) = \frac{50}{50kt + 1}$.
4. $A(2) = 10 \Rightarrow \frac{50}{100k + 1} = 10 \Rightarrow 100k = 4 \Rightarrow k = \frac{1}{25}$. So $A(t) = \frac{50}{2t + 1}$.

**ANSWER**
$A(4) = \frac{50}{9} \approx 5.56$ g

**Not exponential!**
With $A^2$ the answer is **not** $Ce^{kt}$. Students who memorise "rate problems give exponentials" write $A = 50e^{-kt}$ and get 2 g — wrong. Always solve the equation you actually wrote.

**EXERCISES 10.3 #2 · WEEK+5.PDF P.22** A substance A transforms into another substance. After 1 hour 50 g of A remain, after 3 hours 25 g remain. (a) How much was present at the beginning? (b) How much after 5 hours? (c) When will there be 2 g?

1. No "square" is mentioned, so use the basic rate law $\frac{dA}{dt} = -kA \Rightarrow A = A_0 e^{-kt}$.
2. Divide the two data: $\frac{A(3)}{A(1)} = e^{-2k} = \frac{25}{50} = \frac{1}{2}$. So the half-life is 2 hours and $A(t) = A_0 \left(\frac{1}{2}\right)^{t/2}$.
3. (a) $A(1) = A_0 \cdot 2^{-1/2} = 50 \Rightarrow A_0 = 50\sqrt{2} \approx 70.7$ g.
4. (b) $A(5)$ is one half-life after $A(3)$: 12.5 g.
5. (c) $50\left(\frac{1}{2}\right)^{(t-1)/2} = 2 \Rightarrow \frac{t-1}{2} = \log_2 25 \Rightarrow t = 1 + \frac{2\ln 25}{\ln 2}$.

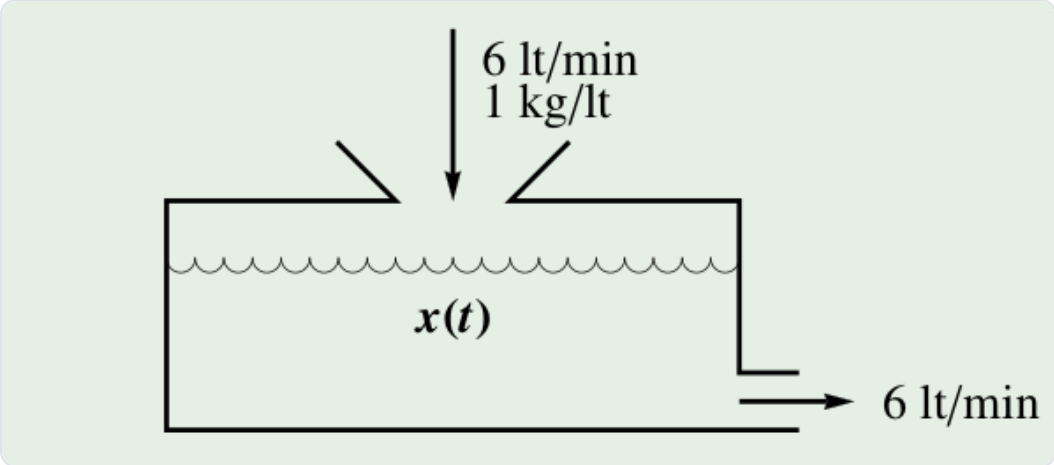
**ANSWER**
(a) $50\sqrt{2} \approx 70.7$ g; (b) 12.5 g; (c) $t \approx 10.3$ hours

**EXERCISES 10.3 #3 · WEEK+5.PDF P.23** Bacteria grow at a rate proportional to the number present. (a) If the number doubles every 10 hours, how long until it triples? (b) If there are 5 million after 6 hours and 8 million after 9 hours, how many were there initially?

1. $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$.
2. (a) $e^{10k} = 2 \Rightarrow k = \frac{\ln 2}{10}$. Triple: $e^{kt} = 3 \Rightarrow t = \frac{10\ln 3}{\ln 2} \approx 15.85$ h.
3. (b) $\frac{N(9)}{N(6)} = e^{3k} = \frac{8}{5}$. Then $N_0 = N(6)e^{-6k} = 5\left(\frac{5}{8}\right)^2 = \frac{125}{64}$.

**ANSWER**
(a) about 15.85 hours; (b) $\frac{125}{64} \approx 1.95$ million

## 4. Mixture problems




Brine flows in, the well-stirred mixture flows out (Week+5.pdf p.26)

**Intuition.** Think of the salt in the tank as money in a bank account: the change per minute is what comes in minus what goes out.

$$\frac{dx}{dt} = r_{\text{in}} - r_{\text{out}}$$

**MIXTURE LAW: X = AMOUNT IN THE TANK (WEEK+5.PDF P.25)**

- $r_{\text{in}} = (\text{inflow rate}) \times (\text{concentration coming in})$ — usually a number.
- $r_{\text{out}} = (\text{outflow rate}) \times \dfrac{x(t)}{V(t)}$ — the concentration in the tank is "salt ÷ volume".
- Volume: $V(t) = V_0 + (\text{inflow} - \text{outflow})\,t$. If the two flows are equal, the volume is constant.

Check the units: (lt/min)·(kg/lt) = kg/min. Both rates must be in "amount per time".

**EXAMPLE 30 · WEEK+5.PDF P.25–29** A large tank holds 1000 lt of water. Brine with 1 kg/lt of salt flows in at 6 lt/min; the well-stirred solution flows out at 6 lt/min. When will the concentration of salt in the tank reach $\frac{1}{2}$ kg/lt?

1. $x(t)$ = kg of salt at time $t$ min; pure water at first, so $x(0) = 0$.
2. $r_{\text{in}} = (6\ \text{lt/min})(1\ \text{kg/lt}) = 6$ kg/min; volume stays 1000 lt, so $r_{\text{out}} = 6 \cdot \frac{x}{1000} = \frac{3}{500}x$ kg/min.
3. IVP: $\frac{dx}{dt} = 6 - \frac{3}{500}x$, $x(0) = 0$. Linear: $\frac{dx}{dt} + \frac{3}{500}x = 6$, $\mu = e^{\frac{3}{500}t}$.
4. $x(t) = e^{-\frac{3}{500}t}\left(\int 6e^{\frac{3}{500}t}dt + C\right) = 1000 + Ce^{-\frac{3}{500}t}$; $x(0) = 0 \Rightarrow C = -1000$.
5. $x(t) = 1000\left(1 - e^{-\frac{3t}{500}}\right)$, concentration $\frac{x}{1000} = 1 - e^{-\frac{3t}{500}}$.
6. Set concentration $= \frac{1}{2}$ (i.e. $x(T) = 500$): $e^{-\frac{3T}{500}} = \frac{1}{2} \Rightarrow T = \frac{500}{3}\ln 2$.

**ANSWER**
$T = \frac{500}{3}\ln 2 \approx 116$ min; in the long run $x(t) \to 1000$ kg

**EXERCISES 10.4 #2 · WEEK+5.PDF P.29 (VOLUME CHANGES!)** A tank holds 200 lt of salt water containing 30 g of salt. Brine of 1 g/lt pours in at 4 lt/min; the well-mixed solution pours out at 5 lt/min. Find the amount of salt at time $t$.

1. Volume: $V(t) = 200 + (4-5)t = 200 - t$ (it loses 1 lt per minute)
2. $r_{\text{in}} = 4 \cdot 1 = 4$ g/min; $r_{\text{out}} = 5 \cdot \frac{x}{200-t}$.
3. $\frac{dx}{dt} + \frac{5}{200-t}x = 4$, $x(0) = 30$. Linear with $\mu = e^{\int \frac{5}{200-t}dt} = e^{-5\ln(200-t)} = (200-t)^{-5}$.
4. $\frac{d}{dt}\left[(200-t)^{-5}x\right] = 4(200-t)^{-5} \Rightarrow (200-t)^{-5}x = (200-t)^{-4} + C$.
5. $x = (200-t) + C(200-t)^5$; $x(0) = 30 \Rightarrow C = -\frac{170}{200^5}$.

**ANSWER**
$x(t) = (200-t) - 170\left(\frac{200-t}{200}\right)^5$ grams, $0 \le t < 200$

**EXERCISES 10.4 #3 · WEEK+5.PDF P.29** A room of 1000 m³ has air with 0.15% $CO_2$. From $t = 0$, outside air with 0.05% $CO_2$ is let in at 500 m³/min (and the mixed air leaves at the same rate). (a) Percentage of $CO_2$ after 3 min? (b) When is it 0.1%?

1. $x(t)$ = m³ of $CO_2$ in the room; $x(0) = 0.0015 \cdot 1000 = 1.5$.
2. $r_{\text{in}} = 500 \cdot 0.0005 = 0.25$, $r_{\text{out}} = 500 \cdot \frac{x}{1000} = \frac{x}{2}$: $\frac{dx}{dt} = 0.25 - \frac{x}{2}$.
3. Solution: $x(t) = 0.5 + e^{-t/2}$ (steady value $0.5$; $x(0) = 1.5$ gives the coefficient 1). Percentage $= \frac{x}{1000} \cdot 100 = 0.05 + 0.1e^{-t/2}\%$.
4. (a) $t = 3$: $0.05 + 0.1e^{-1.5} \approx 0.0723\%$. (b) $0.1e^{-t/2} = 0.05 \Rightarrow t = 2\ln 2$.

**ANSWER**
(a) about 0.072%; (b) $t = 2\ln 2 \approx 1.39$ min

**Exercises 10.4 #1**
On the slide, the concentration of the incoming water is missing ("a salt concentration of lbs/gal"). With an unknown concentration $c$ lb/gal the set-up would be $V(t) = 600 + 3t$, $\frac{dx}{dt} = 9c - \frac{6x}{600+3t}$, $x(0) = 5$ — practise the set-up; the number cannot be computed without $c$.

**Where students lose points in word problems**

- Forgetting the minus sign for decrease, or putting it twice.
- Writing $r_{\text{out}} = (\text{rate}) \cdot x$ instead of $(\text{rate}) \cdot \frac{x}{V}$.
- Forgetting that the volume changes when inflow ≠ outflow.
- Answering with $k$ or $C$ instead of the number the question asks for (and without units).
- Using the exponential formula when the rate is proportional to $A^2$ or $\sqrt{A}$.

## In one minute

- "Rate of change" = derivative; "proportional to" = $k\times$. Decrease ⇒ minus sign with $k > 0$.
- Cooling: $\frac{dT}{dt} = -k(T - T_m) \Rightarrow T = T_m + (T_0 - T_m)e^{-kt}$.
- Falling: $m\frac{dv}{dt} = mg - kv \Rightarrow v = \left(v_0 - \frac{mg}{k}\right) e^{-\frac{k}{m}t} + \frac{mg}{k}$; limiting velocity $\frac{mg}{k}$; distance = $\int v\,dt$.
- Decay/growth: $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$; half-life $h$: $N = N_0(\frac{1}{2})^{t/h}$.
- Rate $\propto A^2$: $\frac{dA}{dt} = -kA^2 \Rightarrow A = \frac{1}{kt+C}$ (midterm 2023 Q1).
- Mixture: $\frac{dx}{dt} = r_{\text{in}} - r_{\text{out}}$, $r_{\text{out}} = \text{flow} \cdot \frac{x}{V(t)}$, $V(t) = V_0 + (\text{in} - \text{out})t$.
- Two data points ⇒ find $C$ (from $t = 0$) and $k$; then answer the question with units.

CHAPTER 14

# Higher-order linear equations and reduction of order

Learn the language of n-th order linear equations (operator, IVP, linear independence, Wronskian, fundamental set, general solution) and the reduction-of-order trick that turns one known solution into a second one — that trick is midterm Question 5.

From Week+6.pdf, pages 1–38

## The big picture

Until now every equation had only $y'$. Now we allow $y''$, $y'''$, ..., $y^{(n)}$. We will **not** yet learn how to solve these in general. This week is about the **rules of the game**: when does a solution exist, how many solutions do we need, how do we check they are "really different", and how do we build all solutions from a few. At the end comes one real solving technique: **reduction of order**.

## 1. Linear equations of order n

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)\,y = f(x)$$

LINEAR ODE OF ORDER N (WEEK+6.PDF P.1)

"Linear" means: $y$ and its derivatives appear only to the **first power**, are **not multiplied together**, and the coefficients $a_0, \ldots, a_n$ depend only on $x$. We assume the $a_i$ and $f$ are continuous on an interval and $a_0(x) \neq 0$ there.

For $n = 2$ this is the case you will see most: $a_0(x)y'' + a_1(x)y' + a_2(x)y = f(x)$.

### The operator notation

Write $D$ for "take the derivative": $D^0 y = y$, $Dy = y'$, $D^2 y = y''$, ..., $D^n y = y^{(n)}$. Then the left side is one machine $L$ that eats a function and returns a function:

$$L[y](x) := a_0(x)y^{(n)}(x) + a_1(x)y^{(n-1)}(x) + \cdots + a_n(x)y(x).$$

$L = L(D)$ is called the **n-th order linear differential operator** (Definition 22). The equation is simply $L(D)y = f$.

| | |
|---|---|
| **non-homogeneous term** | the right-hand side $f$ |
| **homogeneous equation** | $f \equiv 0$: $L(D)y = 0$ |
| **linear mapping** | $L(c_1y_1 + c_2y_2) = c_1L(y_1) + c_2L(y_2)$ — this one property gives almost every theorem of this chapter |

## 2. The IVP and existence–uniqueness

An **n-th order** IVP needs the equation **plus n conditions**, all at the same point $x_0$:

$$L(D)y = f(x), \quad y(x_0) = y_0,\ y'(x_0) = y_1,\ \ldots,\ y^{(n-1)}(x_0) = y_{n-1}.$$

**Tip**

Second order ⇒ two conditions: a value $y(x_0)$ and a slope $y'(x_0)$. Think of a ball: you need where it starts *and* how fast it starts.

**Theorem 10 (Existence and Uniqueness) — Week+6.pdf p.5**

Let $a_0, a_1, \ldots, a_n$ and $f$ be continuous on an interval $I$ with $a_0(x) \neq 0$ for all $x \in I$. Then for any $x_0 \in I$ the IVP above has a **unique solution**, which exists on the **entire interval** $I$.

**EXAMPLE 31 · WEEK+6.PDF P.6–7** Show that $y'' + 3xy' + x^3 y = e^x$, $y(1) = 2$, $y'(1) = 5$ has a unique solution on $(-\infty, \infty)$.

1. Read off: $a_0(x) = 1$, $a_1(x) = 3x$, $a_2(x) = x^3$, $f(x) = e^x$.
2. All are continuous for every $x \in \mathbb{R}$, and $a_0 = 1 \neq 0$ everywhere.
3. By the existence–uniqueness theorem the IVP has a unique solution on $\mathbb{R}$.

**ANSWER**

Unique solution on the whole real line.

**Corollary 1 (the trivial solution)**

If $\varphi$ solves the **homogeneous** equation $L(D)y = 0$ and $\varphi(x_0) = \varphi'(x_0) = \cdots = \varphi^{(n-1)}(x_0) = 0$, then $\varphi \equiv 0$. Reason: $y \equiv 0$ already solves that IVP, and the solution is unique.

## 3. Superposition

A **linear combination** of $f_1, \ldots, f_n$ is $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n$ with constants $c_i$ (Definition 23).

**Theorem 11 (Superposition principle) — Week+6.pdf p.7**

If $\varphi_1, \ldots, \varphi_n$ are solutions of the **homogeneous linear** equation $L(D)y = 0$, then every linear combination $c_1\varphi_1 + \cdots + c_n\varphi_n$ is also a solution.

**Why (the lecturer's proof):** $L(D)(c_1\varphi_1 + \cdots + c_n\varphi_n) = c_1 \underbrace{L(D)\varphi_1}_{0} + \cdots + c_n \underbrace{L(D)\varphi_n}_{0} = 0.$

**Two conditions for superposition**

It needs **(1) linear** and **(2) homogeneous**. If either fails, a combination of solutions is usually not a solution (see Example 36 below).

## 4. Linear dependence and independence

**Intuition.** Functions are **dependent** if one of them is "made of" the others (like $3\sin x$ is made of $\sin x$). Then it brings nothing new. They are **independent** if none of them can be built from the others.

**Definition 24 — Week+6.pdf p.9**

$f_1, \dots, f_n$ are **linearly dependent on** $I$ if there are constants $c_1, \dots, c_n$, **not all zero**, with $c_1 f_1(x) + \cdots + c_n f_n(x) = 0$ for all $x \in I$.
They are **linearly independent on** $I$ if $c_1 f_1(x) + \cdots + c_n f_n(x) = 0$ for all $x \in I$ implies $c_1 = c_2 = \cdots = c_n = 0$.

**EXAMPLE 32 · WEEK+6.PDF P.10–11** (1) Show $f_1 = x$, $f_2 = x^2$ are linearly independent on $[0, 1]$. (2) Show $f_1 = \sin x$, $f_2 = 3\sin x$ are linearly dependent on $[0, \pi]$.

1. (1) Suppose $c_1 x + c_2 x^2 = 0$ for all $x \in [0, 1]$. A polynomial that is zero at infinitely many points is the zero polynomial: $c_1 x + c_2 x^2 = 0x + 0x^2$, so $c_1 = c_2 = 0$. (compare coefficients)
2. (2) $c_1 \sin x + 3c_2 \sin x = (c_1 + 3c_2)\sin x = 0$. Choose $c_1 = -3$, $c_2 = 1$: $(-3 + 3)\sin x = 0$ ✓, and the constants are not both zero.

**ANSWER**
$x, x^2$ independent; $\sin x, 3\sin x$ dependent

## 5. The Wronskian

$$W[f_1, \dots, f_n](x) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

**WRONSKIAN, DEFINITION 25 (WEEK+6.PDF P.12)**

Row 1: the functions. Row 2: their first derivatives. ... The last ($n$-th) row: the $(n-1)$-th derivatives. For two functions:

$$W[f_1, f_2] = \begin{vmatrix} f_1 & f_2 \\ f_1' & f_2' \end{vmatrix} = f_1 f_2' - f_2 f_1'.$$

**Theorems 12, 13 and Corollary 2 — Week+6.pdf p.13–14**

Let $\varphi_1, \ldots, \varphi_n$ be **solutions** of the n-th order homogeneous linear equation $L(D)y = 0$ on $I$.

- (Th. 12) They are linearly **dependent** $\Leftrightarrow W(x) \equiv 0$ on $I$.
- (Th. 13) Either $W(x) \equiv 0$ on $I$, or $W(x) \neq 0$ for **all** $x \in I$ — nothing in between.
- (Cor. 2) They are linearly **independent** $\Leftrightarrow W(x) \neq 0$ for all $x \in I$.

**Practical use**

Because of Theorem 13, for solutions you only need to check $W$ at **one** convenient point. If it is not zero there, it is never zero, and the solutions are independent.

**EXAMPLE 33 · WEEK+6.PDF P.14–15** Show that $\cos x$ and $\sin x$ are linearly independent on $(-\infty, \infty)$.

1. First check they solve one equation: $\varphi_1 = \cos x$, $\varphi_1'' = -\cos x$, so $\varphi_1'' + \varphi_1 = 0$. Likewise $\sin x$. Both solve $y'' + y = 0$.
2. $W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$ for all $x$.
3. So they are independent, $\{\cos x, \sin x\}$ is a fundamental set, and the general solution of $y'' + y = 0$ is $y = c_1 \cos x + c_2 \sin x$.

**ANSWER**

$W = 1 \neq 0 \Rightarrow$ independent; $y = c_1 \cos x + c_2 \sin x$

**EXAMPLE 34 AND REMARK 5 · WEEK+6.PDF P.16–18** Let $f_1(x) = x|x|$, $f_2(x) = x^2$ on $[-1, 1]$. Show $W \equiv 0$ but the functions are independent. Why is that not a contradiction?

1. $x|x| = x^2$ for $x \geq 0$ and $-x^2$ for $x < 0$, so $\frac{d}{dx}(x|x|) = 2|x|$.
2. $W = \begin{vmatrix} x|x| & x^2 \\ 2|x| & 2x \end{vmatrix} = 2x^2|x| - 2x^2|x| = 0$ for every $x \in [-1, 1]$.
3. Independence: suppose $c_1 x|x| + c_2 x^2 = 0$ for all $x$. At $x = 1$: $c_1 + c_2 = 0$. At $x = -1$: $-c_1 + c_2 = 0$. So $c_1 = c_2 = 0$.
4. No contradiction with Corollary 2: these two functions **cannot** both be solutions of one second-order homogeneous linear equation on $[-1, 1]$. The Wronskian test is only guaranteed for **solutions**.

**ANSWER**

$W \equiv 0$, yet independent — the Wronskian criterion needs solutions of $L(D)y = 0$.

**Watch out**

"$W = 0 \Rightarrow$ dependent" is **only** true when the functions are solutions of the same homogeneous linear equation. For arbitrary functions, check the definition directly (plug in points, as in Example 34).

## 6. Fundamental set and general solution

**Theorem 14, Definitions 26–27 — Week+6.pdf p.18–19**

An n-th order homogeneous linear equation $L(D)y = 0$ has **n linearly independent solutions** $\varphi_1, \ldots, \varphi_n$, and every solution can be written as $c_1\varphi_1 + \cdots + c_n\varphi_n$.
Such a set $\{\varphi_1, \ldots, \varphi_n\}$ of n independent solutions is a **fundamental set of solutions** (F.S.S.), and $\varphi = c_1\varphi_1 + \cdots + c_n\varphi_n$ is the **general solution** of $L(D)y = 0$.

**Recipe: proving a general solution**

To show $y = c_1\varphi_1 + c_2\varphi_2$ is the general solution of a 2nd order homogeneous equation: (1) check each $\varphi_i$ solves it; (2) compute $W[\varphi_1, \varphi_2] \neq 0$; (3) conclude F.S.S. and write the combination. Order $n \Rightarrow$ you need exactly $n$ independent solutions.

### Exercises 11 (homework on Week+6.pdf p.20), solved

**EXERCISES 11 #1 · WEEK+6.PDF P.20** Check linear independence: (a) $1, x, x^2, (x+1)^2$; (b) $\cos x, \sin x, \sin 2x$; (c) $\pi, \tan^2 x, \sec^2 x$.

1. (a) $(x+1)^2 = 1 \cdot 1 + 2 \cdot x + 1 \cdot x^2$. So $1 \cdot 1 + 2x + x^2 - (x+1)^2 = 0$ with constants $(1, 2, 1, -1)$, not all zero: **dependent**.
2. (b) Suppose $c_1 \cos x + c_2 \sin x + c_3 \sin 2x = 0$ for all $x$. $x = 0$: $c_1 = 0$. $x = \frac{\pi}{2}$: $c_2 = 0$. $x = \frac{\pi}{4}$: $c_3 = 0$. **Independent.** (their Wronskian is $-3 \sin 2x$, not identically zero, which agrees)
3. (c) $\sec^2 x - \tan^2 x = 1$, so $1 \cdot \pi + \pi \tan^2 x - \pi \sec^2 x = \pi(1 + \tan^2 x - \sec^2 x) = 0$: **dependent**.

**ANSWER**
(a) dependent, (b) independent, (c) dependent

**EXERCISES 11 #2 · WEEK+6.PDF P.20** Show that $e^{\alpha x}$ and $e^{\beta x}$ solve $y'' - (\alpha + \beta)y' + \alpha\beta y = 0$ and are independent exactly when $\alpha \neq \beta$.

1. $y = e^{\alpha x}$: $\alpha^2 e^{\alpha x} - (\alpha + \beta)\alpha e^{\alpha x} + \alpha\beta e^{\alpha x} = e^{\alpha x}(\alpha^2 - \alpha^2 - \alpha\beta + \alpha\beta) = 0$. Same for $e^{\beta x}$ by symmetry.
2. $W = \begin{vmatrix} e^{\alpha x} & e^{\beta x} \\ \alpha e^{\alpha x} & \beta e^{\beta x} \end{vmatrix} = (\beta - \alpha)e^{(\alpha+\beta)x}$.
3. The exponential is never 0, so $W \neq 0 \Leftrightarrow \alpha \neq \beta$.

**ANSWER**
$W = (\beta - \alpha)e^{(\alpha+\beta)x}$: independent $\Leftrightarrow \alpha \neq \beta$

**EXERCISES 11 #3 · WEEK+6.PDF P.20** For $x > 0$, find $\alpha, \beta, \gamma$ so that $1, \frac{1}{x}, x^2$ solve $x^3y''' + \alpha x^2 y'' + \beta x y' + \gamma y = 0$; show they are independent; write the general solution.

1. $y = 1$: all derivatives 0, leaving $\gamma = 0$.
2. $y = x^2$: $y' = 2x$, $y'' = 2$, $y''' = 0$: $2\alpha x^2 + 2\beta x^2 = 0 \Rightarrow \alpha = -\beta$.
3. $y = \frac{1}{x}$: $y' = -x^{-2}$, $y'' = 2x^{-3}$, $y''' = -6x^{-4}$: $\frac{-6+2\alpha-\beta}{x} = 0 \Rightarrow -6 + 3\alpha = 0 \Rightarrow \alpha = 2,\ \beta = -2$.
4. $W = \begin{vmatrix} 1 & x^{-1} & x^2 \\ 0 & -x^{-2} & 2x \\ 0 & 2x^{-3} & 2 \end{vmatrix} = 1 \cdot \left(-2x^{-2} - 4x^{-2}\right) = -\frac{6}{x^2} \neq 0.$

**ANSWER**
$\alpha = 2,\ \beta = -2,\ \gamma = 0$; independent; $y = c_1 + \frac{c_2}{x} + c_3 x^2$

## 7. Non-homogeneous equations

**Theorem 15 — Week+6.pdf p.21**
If $\varphi$ solves $L(D)y = f$ and $\psi$ solves $L(D)y = 0$, then $\varphi + \psi$ solves $L(D)y = f$. (Proof: $L(\psi + \varphi) = L\psi + L\varphi = 0 + f$.)

| | |
|---|---|
| **complementary solution** $\varphi_c$ | the general solution of the homogeneous equation $L(D)y = 0$ |
| **particular solution** $\varphi_p$ | any one solution of $L(D)y = f$ **without arbitrary constants** |
| **general solution of** $L(D)y = f$ | $\varphi_c + \varphi_p$ (Definition 28) |

**EXAMPLE 35 · WEEK+6.PDF P.24–25** Find the general solution of $y'' + y = x$.

1. Homogeneous $y'' + y = 0$: $\varphi_c = c_1 \cos x + c_2 \sin x$ (Example 33).
2. Try $\varphi_p = x$: $\varphi_p' = 1$, $\varphi_p'' = 0$, so $\varphi_p'' + \varphi_p = 0 + x = x$ ✓.

**ANSWER**
$y = c_1 \cos x + c_2 \sin x + x$

**Watch out**
Don't forget to add $\varphi_p$ at the end (the lecturer circles "+x" in red). The complementary part alone is not the general solution of the non-homogeneous equation.

**EXAMPLE 36 · WEEK+6.PDF P.26 (WHY LINEARITY MATTERS)** $yy'' + (y')^2 = 0$ on $[0, 1]$ has the solutions $\varphi_1 \equiv 1$ and $\varphi_2 = \sqrt{x}$. Is $c_1 + c_2\sqrt{x}$ a solution?

1. The equation is **not linear** ($yy''$ and $(y')^2$ are products), so superposition is not promised.
2. Indeed, substituting gives $yy'' + (y')^2 = -\frac{c_1 c_2}{4x^{3/2}}$, which is not zero when $c_1 c_2 \neq 0$.

**ANSWER**
No: superposition fails for non-linear equations (like $Ax = b$ vs $Ax = 0$ in linear algebra).

**Theorem 16 (superposition for the right-hand side) — Week+6.pdf p.27**
If $L(D)\varphi_1 = f_1, \ldots, L(D)\varphi_m = f_m$ and $f = c_1 f_1 + \cdots + c_m f_m$, then $\varphi = c_1\varphi_1 + \cdots + c_m\varphi_m$ solves $L(D)y = f$.

**EXAMPLE 37 · WEEK+6.PDF P.28** The equations $y'' - 5y' + 6y = 1, = x, = e^x$ have particular solutions $\varphi_1 = \frac{1}{6}$, $\varphi_2 = \frac{x}{6} + \frac{5}{36}$, $\varphi_3 = \frac{1}{2}e^x$. Find a solution of $y'' - 5y' + 6y = 2e^x + 36(x + 1)$.

1. Write the right side as a combination: $f = 36 \cdot 1 + 36 \cdot x + 2 \cdot e^x$.
2. Use the same coefficients: $\varphi = 36 \cdot \frac{1}{6} + 36\left(\frac{x}{6} + \frac{5}{36}\right) + 2 \cdot \frac{1}{2}e^x = 6 + 6x + 5 + e^x$.

**ANSWER**
$\varphi = e^x + 6x + 11$

### Exercises 11.1 (homework on Week+6.pdf p.29), solved

**EXERCISES 11.1 #1 · WEEK+6.PDF P.29** $(x + 1)y'' + xy' - y = (x + 1)^2$, $x > -1$. Show $\varphi_1 = x$, $\varphi_2 = e^{-x}$ solve the homogeneous equation and $\varphi_p = x^2 - x + 1$ solves the full one. Write the general solution.

1. $x$: $0 + x \cdot 1 - x = 0$ ✓. $e^{-x}$: $(x + 1)e^{-x} - xe^{-x} - e^{-x} = 0$ ✓.
2. $W[x, e^{-x}] = x(-e^{-x}) - e^{-x} = -(x + 1)e^{-x} \neq 0$ for $x > -1$: independent.
3. $\varphi_p$: $\varphi_p' = 2x - 1$, $\varphi_p'' = 2$: $2(x + 1) + x(2x - 1) - (x^2 - x + 1) = x^2 + 2x + 1 = (x + 1)^2$ ✓.

**ANSWER**
$y = c_1 x + c_2 e^{-x} + x^2 - x + 1$

**EXERCISES 11.1 #2 · WEEK+6.PDF P.29** $y'' - \tan x\, y' - \sec^2 x\, y = \sec x \tan x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Verify $\sec x$, $\tan x$ solve the homogeneous equation and $x \sec x$ the full one; give the general solution.

1. Substituting (with $\frac{d}{dx}\sec x = \sec x \tan x$, $\frac{d}{dx}\tan x = \sec^2 x$) gives 0 for $\sec x$ and $\tan x$, and $\sec x \tan x$ for $x \sec x$.

2 $W[\sec x, \tan x] = \sec x \cdot \sec^2 x - \tan x \cdot \sec x \tan x = \sec x(\sec^2 x - \tan^2 x) = \sec x \neq 0.$

**ANSWER**
$y = c_1 \sec x + c_2 \tan x + x \sec x$

## 8. Reduction of order

**Intuition.** If you already know one solution $\varphi_p$ of a homogeneous linear equation, you look for the next one in the form "the known one times an unknown function". Plugging that in kills the $z$-term (because $\varphi_p$ is a solution), and what remains is an equation for $z'$ of one order lower.

**Theorem 17 — Week+6.pdf p.31**
Let $\varphi_p$ be a non-trivial solution of the n-th order homogeneous linear equation $a_0(x)y^{(n)} + \cdots + a_n(x)y = 0$. Then the transformation $y = \varphi_p z$ reduces it to an $(n-1)$-st order homogeneous linear equation in the dependent variable $w := \frac{dz}{dx}$.

**Notation**
On these slides $\varphi_p$ just means "the known solution" of the **homogeneous** equation — here the "p" is not "particular solution of the non-homogeneous equation". Remark 6: for a second-order equation you must already know one solution.

$$y = \varphi_p z, \qquad y' = \varphi_p' z + \varphi_p z', \qquad y'' = \varphi_p'' z + 2\varphi_p' z' + \varphi_p z''$$

THE SUBSTITUTION AND ITS DERIVATIVES (HANDWRITTEN ON WEEK+6.PDF P.31)

**What always happens for second order.** Put these into $a_0 y'' + a_1 y' + a_2 y = 0$ and group by $z''$, $z'$, $z$:

$$a_0\varphi_p\, z'' + \big(2a_0\varphi_p' + a_1\varphi_p\big)\, z' + \underbrace{\big(a_0\varphi_p'' + a_1\varphi_p' + a_2\varphi_p\big)}_{=0 \text{ since } \varphi_p \text{ is a solution}} z = 0.$$

So with $w = z'$:

$$a_0\varphi_p\, w' + \big(2a_0\varphi_p' + a_1\varphi_p\big)\, w = 0$$

REDUCED EQUATION: FIRST ORDER, LINEAR AND SEPARABLE

**Use it as a check**
Always write the substitution steps on the exam (that is what is graded). But you can use the boxed reduced equation to check your $z'$ coefficient — the place where most mistakes happen (including in the 2023 midterm's handwritten solution, see Chapter 16). If the $z$-coefficient does not come out 0, you made an algebra mistake or $\varphi_p$ is not a solution.

### → The 6 steps of reduction of order (2nd order)

1. Check that the given $y_1$ really is a solution (quick).
2. Put $y = y_1 z$; write $y'$ and $y''$.
3. Substitute and collect: the $z$-term must vanish.
4. Set $w = z'$: solve the first-order equation for $w$ (separate, partial fractions); choose the constant $= 1$.
5. $z = \int w\,dx$ (choose the constant $= 0$); $y_2 = y_1 z$.
6. Write the F.S.S. $\{y_1, y_2\}$ and the general solution $y = c_1 y_1 + c_2 y_2$.

**EXAMPLE 38 · WEEK+6.PDF P.32–36** Given that $\varphi_p(x) = x$ is a solution of $(x^2+1)y'' - 2xy' + 2y = 0$, find a linearly independent solution by reduction of order.

1. $y = xz \Rightarrow y' = z + xz',\ y'' = z' + z' + xz'' = 2z' + xz''$.
2. Substitute: $(x^2+1)(2z' + xz'') - 2x(z + xz') + 2xz = 0$.
3. Collect: $x(x^2+1)z'' + z'\left(2x^2 + 2 - 2x^2\right) + z(-2x + 2x) = 0 \Rightarrow x(x^2+1)z'' + 2z' = 0$. (the $z$-term is 0 — good sign)
4. $w = z'$: $\frac{dw}{dx} + \frac{2}{x(x^2+1)}w = 0$ (linear, separable).
5. Partial fractions: $\frac{2}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} \Rightarrow A = 2,\ B = -2,\ C = 0$.
6. $\mu(x) = e^{\int\left(\frac{2}{x} - \frac{2x}{x^2+1}\right)dx} = e^{2\ln x - \ln(x^2+1)} = \frac{x^2}{x^2+1}$, so $w = C\frac{x^2+1}{x^2} = C\left(1 + \frac{1}{x^2}\right)$; choose $C = 1$.
7. $z = \int \left(1 + \frac{1}{x^2}\right)dx = x - \frac{1}{x} + d$; choose $d = 0$.
8. $y_2 = x \cdot \left(x - \frac{1}{x}\right) = x^2 - 1$. Check: $(x^2+1)\cdot 2 - 2x \cdot 2x + 2(x^2-1) = 0$ ✓; $W[x, x^2-1] = x^2 + 1 \neq 0$ ✓.

**ANSWER**
F.S.S. $= \{x,\ x^2 - 1\}$; general solution $y = c_1 x + c_2(x^2 - 1)$

**EXAMPLE 39 · WEEK+6.PDF P.32 (HOMEWORK)** Given $\varphi_p(x) = x$ solves $y'' - 2xy' + 2y = 0$, find the second linearly independent solution.

1. $y = xz$: $y' = z + xz',\ y'' = 2z' + xz''$.
2. $(2z' + xz'') - 2x(z + xz') + 2xz = xz'' + (2 - 2x^2)z' = 0$.
3. $w = z'$: $\frac{w'}{w} = \frac{2x^2-2}{x} = 2x - \frac{2}{x} \Rightarrow \ln|w| = x^2 - 2\ln|x| \Rightarrow w = \frac{e^{x^2}}{x^2}$.
4. $z = \int \frac{e^{x^2}}{x^2}\,dx$. This integral has no elementary antiderivative, so we leave it as an integral.

**ANSWER**
$y_2 = x\displaystyle\int \frac{e^{x^2}}{x^2}\,dx$; general solution $y = c_1 x + c_2\, x\displaystyle\int \frac{e^{x^2}}{x^2}\,dx$

**Note**

Leaving an integral is acceptable **only** when it cannot be done with elementary functions (like $e^{x^2}$). If it can be done — e.g. $\int \frac{(x+1)^2}{x^2}dx$ — do it ("Solve it!" as the lecturer wrote on the midterm).

**EXERCISES 12 #1 · WEEK+6.PDF P.37** Obtain the general solution of $y'' - \frac{4x}{2x-1}y' + \frac{4}{2x-1}y = 0$ for $x > 1$, after finding an exponential solution $\varphi_p = e^{\lambda x}$.

1. Put $y = e^{\lambda x}$ and multiply by $(2x-1)e^{-\lambda x}$: $\lambda^2(2x-1) - 4x\lambda + 4 = 0$, i.e. $x(2\lambda^2 - 4\lambda) + (4 - \lambda^2) = 0$ for all $x$.
2. Both brackets must vanish: $2\lambda(\lambda - 2) = 0$ and $\lambda^2 = 4 \Rightarrow \lambda = 2$. So $y_1 = e^{2x}$.
3. $y = e^{2x}z$: $y' = e^{2x}(2z + z')$, $y'' = e^{2x}(4z + 4z' + z'')$. Substituting and dividing by $e^{2x}$: $z'' + \left(4 - \frac{4x}{2x-1}\right)z' = 0$, i.e. $(2x-1)w' + (4x-4)w = 0$.
4. $\frac{w'}{w} = -\frac{4x-4}{2x-1} = -2 + \frac{2}{2x-1} \Rightarrow w = (2x-1)e^{-2x}$.
5. $z = \int (2x-1)e^{-2x}dx = -xe^{-2x}$ (integration by parts; check by differentiating), so $y_2 = e^{2x} \cdot (-xe^{-2x}) = -x$; drop the sign.

**ANSWER**

$y = c_1e^{2x} + c_2x$

**Where students lose points in reduction of order**

- Wrong $y''$: it is $\varphi_p''z + 2\varphi_p'z' + \varphi_pz''$ — the middle term has a **2**.
- Algebra slip in the $z'$ coefficient (expand everything, then collect).
- Forgetting $y_2 = y_1z$ — giving $z$ or $w$ as the answer.
- Stopping at $y_2$: the question usually also asks for the **general solution** $c_1y_1 + c_2y_2$.
- Leaving an integral that can be computed.

## In one minute

- Linear n-th order: $a_0y^{(n)} + \cdots + a_ny = f$, written $L(D)y = f$; homogeneous if $f \equiv 0$.
- IVP needs n conditions at one point; continuous coefficients and $a_0 \neq 0$ on $I \Rightarrow$ unique solution on all of $I$.
- Superposition: combinations of solutions of a **linear homogeneous** equation are solutions.
- Dependent = some non-trivial combination is identically 0. Wronskian = determinant of functions and derivatives.
- For solutions: independent $\Leftrightarrow W \neq 0$ (check one point). $x|x|, x^2$: $W \equiv 0$ but independent — they are not solutions.
- General solution = $c_1\varphi_1 + \cdots + c_n\varphi_n$ from a fundamental set; non-homogeneous: $\varphi_c + \varphi_p$.

- Reduction of order: $y = y_1 z$, $w = z'$, reduced equation $a_0 y_1 w' + (2a_0 y_1' + a_1 y_1) w = 0$; then $y_2 = y_1 \int w$.

CHAPTER 15

# Which method? Recognising the equation

The exam never tells you which method to use. This chapter is a checklist you can run in your head in under a minute for any first-order equation.

Summary of Chapters 4–14

→ **The one habit that wins points**

Before you integrate anything, **write down the type** and the reason: "Not exact, because $M_y \neq N_x$", "Bernoulli with $n = 3$", "homogeneous: $f(tx, ty) = f(x, y)$". Graders give method points even when the algebra slips, and naming the type makes you check it.

## Step 0: get the equation into a standard shape

Write it in both forms, because different tests need different forms:

$$\frac{dy}{dx} = f(x, y) \qquad \text{and} \qquad M(x, y)\,dx + N(x, y)\,dy = 0$$

DERIVATIVE FORM AND DIFFERENTIAL FORM

To switch: $\dfrac{dy}{dx} = -\dfrac{M}{N}$. Move everything to one side and simplify first. Many equations become obviously separable or linear after one line of algebra.

## The checklist, in order

Ask the questions in this order. Stop at the first **yes**.

| # | Ask yourself | If yes | Ch. |
|---|---|---|---|
| 1 | Can I write $y' = g(x)\,h(y)$? | **Separable.** Divide by $h(y)$, integrate both sides. Then check $h(y) = 0$. | 4 |
| 2 | Is $y$ (and $y'$) only to the first power, never multiplied together? $y' + p(x)y = q(x)$ | **Linear.** $\mu = e^{\int p\,dx}$, then $(\mu y)' = \mu q$. | 7 |
| 3 | In $M\,dx + N\,dy = 0$: is $M_y = N_x$? | **Exact.** Build $F$ with $F_x = M$, $F_y = N$. Answer $F = C$. | 5 |
| 4 | Not exact, but $\frac{1}{N}(M_y - N_x)$ has only $x$? Or $\frac{1}{M}(M_y - N_x)$ has only $y$? | **Integrating factor** $\mu(x) = e^{\int f}$ or $\mu(y) = e^{-\int g}$. Then solve as exact. | 6 |
| 5 | $y' + p(x)y = q(x)\,y^n$ with $n \neq 0, 1$? | **Bernoulli.** $u = y^{1-n}$ gives a linear equation in $u$. | 8 |

| # | Ask yourself | If yes | Ch. |
|---|---|---|---|
| 6 | Does $y'$ depend only on $\frac{y}{x}$? Is $f(tx, ty) = f(x, y)$? Do all terms of $M$ and $N$ have the same total degree? | **Homogeneous.** $y = ux$, $y' = u + xu'$ gives a separable equation. | 9 |
| 7 | $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$? | Compute $\Delta = a_1b_2 - a_2b_1$. $\Delta \neq 0$: shift $x = X + h$, $y = Y + k$ to make it homogeneous. $\Delta = 0$: use $z = a_1x + b_1y$. | 10 |
| 8 | $y' = p(x)y^2 + q(x)y + r(x)$ and a solution $y_1$ is given? | **Riccati.** $y = y_1 + \frac{1}{u}$ gives a linear equation in $u$. | 11 |
| 9 | $y = xy' + f(y')$? | **Clairaut.** General $y = cx + f(c)$. Singular: $x = -f'(p)$, $y = xp + f(p)$, then eliminate $p$. | 11 |
| 10 | None of these, but a hint or a repeated piece appears (like $e^y$, $\sin y$, $\ln y$)? | **Substitute** that piece as $u$. Its derivative is usually already in the equation. You often land on Bernoulli or linear. | 10 |

**Two quick tricks before you give up**

- **Flip the roles.** If the equation is not linear in $y$ but $x$ appears only to the first power, look at $\frac{dx}{dy}$. It may be linear (or Bernoulli) in $x$.
- **More than one door can be open.** A separable equation can also be exact, and a homogeneous one can also be Bernoulli. Any correct method gets full marks, so pick the one with the easiest integrals.

## Signatures: what each type looks like

| You see... | Think... |
|---|---|
| $y' = \frac{\text{only } x}{\text{only } y}$ | Separable |
| $y' + (\dots)\,y = (\dots)$, $y$ to the power 1 | Linear |
| $y' + (\dots)\,y = (\dots)\,y^n$ | Bernoulli |
| $y'$ built from $\frac{y}{x}$, or $x^2 + y^2$, $xy$, $y^2$ all degree 2 | Homogeneous |
| $y^2$ on the right and a known solution | Riccati |

| You see... | Think... |
|---|---|
| $M\,dx + N\,dy = 0$ with matching $M_y, N_x$ | Exact |
| Same, but $M_y \neq N_x$ | Integrating factor |
| $y = xp + f(p)$ | Clairaut |
| Straight-line coefficients $ax + by + c$ | Shift (Ch. 10) |
| $y''$ and "one solution is $y_1$" | Reduction of order |

## Recognition drill

Cover the right column and name the method. Each one is a real example from the slides or the midterm.

| Equation | Method and the reason |
|---|---|
| $(y^2+1)\,dx+(x^2+1)\,dy=0$ | **Separable**: $\frac{dy}{y^2+1}=-\frac{dx}{x^2+1}$. |
| $(\tan y+2x)\,dx+(x-x^2\tan y)\,dy=0$ | **Integrating factor**: $M_y=\sec^2 y\neq N_x=1-2x\tan y$. $\frac{1}{M}(M_y-N_x)=\tan y$ only has $y$, so $\mu(y)=e^{-\int\tan y\,dy}=\cos y$. (Midterm Q2.) |
| $e^y y'+e^y=e^{3y+2x}$ | **Substitution, then Bernoulli**: put $u=e^y$, so $u'=e^y y'$ and $u'+u=u^3e^{2x}$, Bernoulli with $n=3$. (Midterm Q3.) |
| $y'=\dfrac{y(y+2x)}{3x^2}$ | **Homogeneous** (or Bernoulli with $n=2$): the right side is $\frac{1}{3}\left(\frac{y}{x}\right)^2+\frac{2}{3}\frac{y}{x}$. Don't forget $y=0$. (Midterm Q4.) |
| $(x^2+x)y''-2xy'+2y=0$, $y_1=x$ | **Reduction of order**: $y=xz$, then $w=z'$ gives a first-order separable equation. (Midterm Q5.) |
| $\dfrac{dy}{dx}=\dfrac{y^2}{1-3xy}$ | **Linear in $x$**: flip to $\frac{dx}{dy}+\frac{3}{y}x=\frac{1}{y^2}$. |
| $(1+3x\sin y)\,dx-x^2\cos y\,dy=0$ | **Substitution** $u=\sin y$: $du=\cos y\,dy$ is already there, and it becomes linear in $u$. |
| "Rate of conversion is proportional to the square of the amount of A" | **Model first**: $\frac{dA}{dt}=-kA^2$, which is separable. Use the data to find $C$ and then $k$. (Midterm Q1.) |

**"Find all solutions" means: don't lose any**

Whenever you divide by something containing $y$ ($h(y)$ when separating, $y^n$ in Bernoulli, $u$ or $x$ in homogeneous), that value is thrown away. Check each one in the original equation. If it works and isn't covered by any value of $C$, write it next to the general solution as a **singular solution**. Midterm Q4 lost $y=0$ for many students.

## Second order: only one method in this course

If you see $y''$ and the question gives you one solution $y_1$ (often "it is known that $f(x)=x$ is a solution"), it is **reduction of order** (Chapter 14): put $y=y_1z$, and the $z$ terms cancel. With $w=z'$ you get a first-order separable equation. Then the general solution is $y=c_1y_1+c_2y_2$.

**Write this at the top of every answer**

1. The type, and the test that proves it.
2. The substitution or integrating factor, with its formula.
3. The new equation (linear / separable / exact) and its solution.
4. Back-substitute, and write the answer clearly with the arbitrary constant.
5. Any solutions lost by dividing, if the question says "all solutions".

## In one minute

- Put the equation in both forms: $y'=f(x,y)$ and $M\,dx+N\,dy=0$.

- Check in order: separable → linear → exact → integrating factor → Bernoulli → homogeneous → shift → Riccati / Clairaut → substitution.
- Not linear in $y$? Try linear in $x$.
- A strange repeated piece ($e^y$, $\sin y$) is a hint: substitute it.
- Second order with one known solution means reduction of order.
- Always name the type and the test, and look for lost solutions.

CHAPTER 16

# The 2023 midterm, fully solved

The real MAT 2039 midterm of 17.11.2023: five questions, 20 points each. For each one you get how to recognise it, a clean solution that follows the lecturer's handwritten steps (corrected where the handwriting has a mistake), and the traps that cost points.

From 2023-2024 midterm.pdf, pages 1–5

## The exam at a glance

| Q | Points | Type | Method |
|---|---|---|---|
| 1 | 20 | Word problem: rate $\propto$ (amount)$^2$ | Model $\frac{dA}{dt} = -kA^2$, separable, two data points |
| 2 | 20 | $M\,dx + N\,dy = 0$, not exact | Integrating factor $\mu(y)$, then exact |
| 3 | 20 | Equation with $e^y$ | Substitution $z = e^y$ ⇒ Bernoulli ⇒ linear |
| 4 | 20 | "Find **all** solutions" | Homogeneous $y = vx$ (or Bernoulli) + singular solution |
| 5 | 20 | One solution given | Reduction of order $y = xz$, then general solution |

**About the source**

The PDF contains the printed exam and the lecturer's handwritten solutions to Questions 1, 2, 3 and 5. **There is no handwritten solution for Question 4**; the solution below is ours (checked with sympy). The handwritten solution of Question 5 contains an algebra error — the correct version is given here and the difference is explained.

## Question 1 (20 points) — rate proportional to the square

*During a chemical reaction, substance A is converted into substance B at a rate that is proportional to the square of the amount of A. Initially, 50 grams of A is present, and after 2 hours only 10 grams of A remain unconverted. How much of A is present after 4 hours?*

**How to recognise it**

A story with "rate ... proportional to ..." and two measurements = Week 5 rate problem. The key words are "**square** of the amount" ⇒ $A^2$, and "**converted**" ⇒ A decreases ⇒ minus sign. (This is word for word Exercises 10.3 #1 from Week+5.pdf p.22.)

**MIDTERM 2023 Q1 · 2023-2024 MIDTERM.PDF P.1–2** Find $A(4)$.

1. Model: $\frac{dA}{dt} = -kA^2$, $A(0) = 50$, $A(2) = 10$, find $A(4)$.
2. Separable; write it in differential form. For $A \neq 0$: $\int \frac{dA}{A^2} = -\int k\,dt$.
3. $-\frac{1}{A} = -kt - C \Rightarrow A = \dfrac{1}{kt + C}$.
4. $A(0) = \frac{1}{k\cdot 0 + C} = 50 \Rightarrow C = \frac{1}{50}$, so $A(t) = \dfrac{50}{50kt + 1}$.
5. $A(2) = 10 \Rightarrow \dfrac{50}{100k + 1} = 10 \Rightarrow 5 = 100k + 1 \Rightarrow 100k = 4 \Rightarrow k = \frac{1}{25}$.
6. $A(t) = \dfrac{50}{2t + 1} \Rightarrow A(4) = \dfrac{50}{9}$.

**ANSWER**
$A(4) = \frac{50}{9} \approx 5.56$ grams

**Where students lose points**

- Writing $\frac{dA}{dt} = -kA$ (ignoring "square") and getting an exponential: $A(4) = 50\left(\frac{1}{5}\right)^2 = 2$ g — wrong.
- Making the integration constant inconsistent: from $-\frac{1}{A} = -kt - C$ you must get $\frac{1}{A} = kt + C$, not $kt - C$ (or keep the sign and let the data fix it).
- Answering with $A(t)$ or $k$ only; the question asks for a number (with units).

## Question 2 (20 points) — integrating factor

*Find the general solution of* $(\tan y + 2x)\,dx + (x - x^2 \tan y)\,dy = 0$.

**How to recognise it**

Form $M\,dx + N\,dy = 0 \Rightarrow$ test exactness first. If $M_y \neq N_x$, check whether $\frac{M_y - N_x}{N}$ depends only on $x$ or $\frac{M_y - N_x}{M}$ only on $y$. The trig identity $\sec^2 y = \tan^2 y + 1$ (printed as a hint on the exam sheet) is exactly what makes the quotient simplify.

**MIDTERM 2023 Q2 · 2023-2024 MIDTERM.PDF P.2–3** Solve $(\tan y + 2x)\,dx + (x - x^2 \tan y)\,dy = 0$.

1. $M = \tan y + 2x$, $N = x - x^2 \tan y$. $M_y = \sec^2 y$, $N_x = 1 - 2x \tan y$. Since $M_y \neq N_x$, it is **not exact**.
2. $\dfrac{M_y - N_x}{M} = \dfrac{\sec^2 y - 1 + 2x \tan y}{\tan y + 2x} = \dfrac{\tan^2 y + 2x \tan y}{\tan y + 2x} = \dfrac{\tan y(\tan y + 2x)}{\tan y + 2x} = \tan y$. (depends only on $y$; uses $\sec^2 y - 1 = \tan^2 y$)
3. Integrating factor: $\mu(y) = e^{-\int \tan y\,dy} = e^{-\int \frac{\sin y}{\cos y} dy} = e^{\ln|\cos y|} = |\cos y|$. (On an interval where $\cos y \neq 0$, any constant multiple of an integrating factor also works, so take $\mu(y) = \cos y$.)

4. Multiply both sides by $\cos y$: $(\sin y + 2x\cos y)\,dx + (x\cos y - x^2\sin y)\,dy = 0$. Call these $\tilde{M}, \tilde{N}$.
5. Check: $\tilde{M}_y = \cos y - 2x\sin y = \tilde{N}_x \Rightarrow$ the new equation is **exact**.
6. $F_x = \tilde{M} \Rightarrow F(x,y) = \int (\sin y + 2x\cos y)\,dx + g(y) = x\sin y + x^2\cos y + g(y)$.
7. $F_y = \tilde{N} \Rightarrow x\cos y - x^2\sin y + g'(y) = x\cos y - x^2\sin y \Rightarrow g'(y) = 0 \Rightarrow g(y) = 0$.
8. General solution: $F(x,y) = C$.

**ANSWER**
$x\sin y + x^2\cos y = C$, $C$ arbitrary constant

**Where students lose points**

- Sign of the integrating factor exponent: with $\frac{M_y - N_x}{M} = g(y)$ the factor is $e^{-\int g\,dy}$. Using $e^{+\int \tan y\,dy} = \sec y$ does **not** make the equation exact — always re-check $\tilde{M}_y = \tilde{N}_x$.
- Dividing by $N$ instead of $M$: $\frac{M_y - N_x}{N}$ is not a function of $x$ alone here, so that route is a dead end.
- Forgetting "$= C$": $F(x,y)$ alone is not a solution.

## Question 3 (20 points) — substitution, then Bernoulli

*Find the general solution of* $e^y\,y' + e^y = e^{3y+2x}$.

**How to recognise it**

You see $e^y\,y'$, which is exactly $\frac{d}{dx}(e^y)$. That shouts "put $z = e^y$". The right side $e^{3y}e^{2x} = z^3e^{2x}$ then gives a Bernoulli equation with $n = 3$. (The hint printed under this question, $\sec^2 y = \tan^2 y + 1$, is not used here — it belongs to Question 2.)

**MIDTERM 2023 Q3 · 2023-2024 MIDTERM.PDF P.3–4** Solve $e^y\,y' + e^y = e^{3y+2x}$.

1. Rewrite: $e^y y' + e^y = e^{3y}e^{2x} = (e^y)^3 e^{2x}$.
2. Let $z = e^y$, so $z' = e^y y'$. Then $z' + z = z^3e^{2x}$: **Bernoulli** with $n = 3$.
3. $u = z^{1-n} = z^{-2}$, $\frac{du}{dx} = -2z^{-3}\frac{dz}{dx}$. Multiply both sides by $\frac{-2}{z^3}$: $-\frac{2}{z^3}\frac{dz}{dx} - \frac{2}{z^2} = -2e^{2x}$.
4. That is $\frac{du}{dx} - 2u = -2e^{2x}$: **linear**.
5. $\mu(x) = e^{-\int 2\,dx} = e^{-2x}$: $e^{-2x}\frac{du}{dx} - 2ue^{-2x} = -2 \Rightarrow \frac{d}{dx}\left(e^{-2x}u\right) = -2$.
6. Integrate: $e^{-2x}u = -2x + C \Rightarrow u = -2xe^{2x} + Ce^{2x}$.
7. Back: $u = \frac{1}{z^2} = e^{-2y}$.

**ANSWER**
$e^{-2y} = -2xe^{2x} + Ce^{2x}$ (equivalently $y = -x - \frac{1}{2}\ln(C - 2x)$), $C$ arbitrary

**Note**

Since $z = e^y > 0$, the Bernoulli "extra" solution $z = 0$ is impossible, so nothing is lost by dividing by $z^3$.

**Where students lose points**

- Choosing $u = z^{-3}$ or $u = z^2$: for Bernoulli the new variable is $u = z^{1-n} = z^{-2}$.
- Forgetting to multiply the **right side** by $-2/z^3$ too (it becomes $-2e^{2x}$).
- Stopping at $u$ or $z$: go back to $y$.

## Question 4 (20 points) — find all solutions

*Find all solutions of* $y' = \dfrac{y(y+2x)}{3x^2}$.

**How to recognise it**

The right side is $\frac{y^2+2xy}{3x^2}$: every term has total degree 2 on top and bottom ⇒ a function of $\frac{y}{x}$ only ⇒ **homogeneous**, use $y = vx$. "Find **all** solutions" is the signal that some solution gets lost when you divide — you must find it and list it.

**MIDTERM 2023 Q4 · 2023-2024 MIDTERM.PDF P.1 (NO HANDWRITTEN SOLUTION IN THE PDF)**

Find all solutions of $y' = \frac{y(y+2x)}{3x^2}$, $x \neq 0$.

1. $y' = \frac{1}{3}\left(\frac{y}{x}\right)^2 + \frac{2}{3} \cdot \frac{y}{x}$ depends only on $\frac{y}{x}$: homogeneous. Put $y = vx$, $y' = v + xv'$.
2. $v + xv' = \frac{v^2+2v}{3} \Rightarrow xv' = \frac{v^2-v}{3} = \frac{v(v-1)}{3}$. (separable)
3. Before dividing, note the constant solutions: $v(v-1) = 0 \Rightarrow v = 0$ (i.e. $y = 0$) and $v = 1$ (i.e. $y = x$).
4. For $v \neq 0, 1$: $\displaystyle\int \frac{dv}{v(v-1)} = \int \frac{dx}{3x}$. Partial fractions: $\frac{1}{v(v-1)} = \frac{1}{v-1} - \frac{1}{v}$.
5. $\ln\left|\frac{v-1}{v}\right| = \frac{1}{3}\ln|x| + \ln C \Rightarrow \frac{v-1}{v} = Kx^{1/3}$, $K \neq 0$.
6. $1 - \frac{1}{v} = Kx^{1/3} \Rightarrow v = \dfrac{1}{1-Kx^{1/3}} \Rightarrow y = \dfrac{x}{1-Kx^{1/3}}$.
7. Lost solutions: $K = 0$ gives $y = x$, so $y = x$ is already in the family (allow $K = 0$). But $y = 0$ is **not** obtained for any $K$ ⇒ it is a **singular solution** and must be listed separately.

**ANSWER**

$y = \dfrac{x}{1 - Cx^{1/3}}$ ($C$ arbitrary) **and** the singular solution $y \equiv 0$

**→ Alternative: Bernoulli**

The same equation is Bernoulli: $y' - \frac{2}{3x}y = \frac{1}{3x^2}y^2$ ($n = 2$). With $u = y^{-1}$: $u' + \frac{2}{3x}u = -\frac{1}{3x^2}$, $\mu = x^{2/3}$, $\left(x^{2/3}u\right)' = -\frac{1}{3}x^{-4/3} \Rightarrow x^{2/3}u = x^{-1/3} + C \Rightarrow y = \frac{x}{1+Cx^{1/3}}$ — the same family (rename $C \to -C$). Dividing by $y^2$ loses $y = 0$, which again must be added.

**! Where students lose points**

- Not mentioning $y = 0$ — in a "find all solutions" question this is exactly where points are lost.
- Listing $y = x$ as "singular": it is in the family ($C = 0$), so it is not singular. Say so.
- Algebra in $v + xv' = \ldots$: subtract $v$ correctly: $\frac{v^2+2v}{3} - v = \frac{v^2-v}{3}$.

## Question 5 (20 points) — reduction of order

*Find a second solution of $(x^2 + x)y'' - 2xy' + 2y = 0$ if it is known that $f(x) = x$ is a solution. Write the general solution.*

**★ How to recognise it**

Second-order linear homogeneous, one solution given ⇒ reduction of order (Theorem 17, Week+6.pdf p.31). Put $y = xz$. It is the same pattern as Example 38, $(x^2 + 1)y'' - 2xy' + 2y = 0$.

**MIDTERM 2023 Q5 · 2023-2024 MIDTERM.PDF P.4–5 (CORRECTED)** Find $y_2$ and the general solution of $(x^2 + x)y'' - 2xy' + 2y = 0$, given $y_1 = x$.

1. $y_2(x) = f(x)z(x) = xz$. Then $y_2' = z + xz'$, $y_2'' = z' + z' + xz'' = 2z' + xz''$.
2. Substitute: $(x^2 + x)(2z' + xz'') - 2x(z + xz') + 2xz = 0$.
3. Collect: $(x^3 + x^2)z'' + \left(2x^2 + 2x - 2x^2\right)z' + (-2x + 2x)z = 0 \Rightarrow (x^3 + x^2)z'' + 2x\,z' = 0$. ($(x^2 + x) \cdot 2z' = (2x^2 + 2x)z'$; the $2x^2$ cancels with $-2x \cdot xz'$)
4. $w = z'$: $x^2(x + 1)w' + 2xw = 0$, i.e. $w' + \frac{2}{x(x+1)}w = 0$ (linear and separable).
5. For $w \neq 0$: $\int \frac{dw}{w} = -\int \frac{2}{x(x+1)}dx$. Partial fractions: $\frac{2}{x(x+1)} = \frac{2}{x} - \frac{2}{x+1}$.
6. $\ln|w| = -2\ln|x| + 2\ln|x+1| + \ln C \Rightarrow w = K\frac{(x+1)^2}{x^2}$; $w = 0$ is also a solution, so $w = \alpha\frac{(x+1)^2}{x^2}$. Choose $\alpha = 1$.
7. $z = \int \frac{(x+1)^2}{x^2}dx = \int \left(1 + \frac{2}{x} + \frac{1}{x^2}\right)dx = x + 2\ln|x| - \frac{1}{x}$ (constant chosen 0).
8. $y_2 = xz = x^2 + 2x\ln|x| - 1$. Check: $W[x, y_2] = x\,y_2' - y_2 = (x+1)^2 \neq 0$ on any interval where $x^2 + x \neq 0$.

**ANSWER**

$y_2 = x^2 + 2x\ln|x| - 1$; general solution $y = c_1 x + c_2\left(x^2 + 2x\ln|x| - 1\right)$

**The handwritten solution has an error**

In the handwritten solution (p.4) the $z'$ coefficient is written as $(2x - 2x^2)$ instead of $2x$: the $2x^2$ from $(x^2 + x) \cdot 2z'$ was dropped. This leads to $x^2(x+1)w' - 2x(x-1)w = 0$, $w = \frac{(x+1)^4}{x^2}$ and $y_2 = x \int \frac{(x+1)^4}{x^2} dx$, which is **not** a solution of the equation (substituting it leaves $2(x+1)^4 \neq 0$). The method and all later steps are fine; only that coefficient is wrong. The correct $w$ is $\frac{(x+1)^2}{x^2}$. Also note the lecturer's "Solve it!": the integral must be evaluated.

**Where students lose points**

- The $z'$ coefficient (see the note above!). Quick check: it must equal $2a_0 y_1' + a_1 y_1 = 2(x^2 + x) - 2x \cdot x = 2x$.
- Answering only $y_2$ and forgetting "Write the general solution": $y = c_1 x + c_2 y_2$.
- Leaving $\int \frac{(x+1)^2}{x^2} dx$ unevaluated — it is elementary, so compute it.
- Forgetting $y_2 = xz$ (giving $z$ as the second solution).

## In one minute

- Q1: "$\propto$ square" $\Rightarrow \frac{dA}{dt} = -kA^2 \Rightarrow A = \frac{50}{2t+1}$, $A(4) = \frac{50}{9}$ g.
- Q2: $\frac{M_y - N_x}{M} = \tan y \Rightarrow \mu = \cos y \Rightarrow x \sin y + x^2 \cos y = C$.
- Q3: $z = e^y \Rightarrow$ Bernoulli $n = 3 \Rightarrow u = z^{-2} \Rightarrow e^{-2y} = (C - 2x)e^{2x}$.
- Q4: homogeneous $\Rightarrow y = \frac{x}{1 - Cx^{1/3}}$ plus singular $y = 0$.
- Q5: $y = xz \Rightarrow (x^3 + x^2)z'' + 2xz' = 0 \Rightarrow y_2 = x^2 + 2x \ln|x| - 1$, $y = c_1 x + c_2 y_2$.
- Always verify: the $z$-term must vanish in reduction of order; the new equation must be exact after $\mu$.

# Cheat sheet

Read this the night before the exam.

## Basics & IVPs

- **ODE**: one independent variable; **PDE**: partial derivatives. **Order** = highest derivative (not power).
- **Linear**: $a_0(x)y^{(n)} + \cdots + a_n(x)y = f(x)$, $a_0 \neq 0$. No $yy'$, $(y')^2$, $\sin y$, $\sqrt{y}$.
- **Solution** on $I$: $n$-times differentiable, satisfies the DE for *all* $x \in I$. Explicit $y = \varphi(x)$ / implicit $\Phi(x, y) = 0$.
- **General**: $n$ arbitrary constants. **Particular**: none. **Singular**: not obtainable from the general solution for any constants (e.g. $y \equiv 0$ for $(y')^2 = 4y$, general $(x + c)^2$).
- **Thm 1**: $f$ and $\frac{\partial f}{\partial y}$ continuous on $R : |x - x_0| \leq a, |y - y_0| \leq b \Rightarrow$ unique solution of $y' = f,\ y(x_0) = y_0$ on *some* $|x - x_0| < h$.
- Fails? Theorem is silent: $y' = x\sqrt{y},\ y(0) = 0$ has $y \equiv 0$ and $y = \frac{x^4}{16}$.
- Local only: $y' = 1 + y^2,\ y(0) = 0 \Rightarrow y = \tan x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

## Separable

- $f(y)y' = g(x)$ or $y' = g(x)h(y)$. For $h(y) \neq 0$: $\int \frac{dy}{h(y)} = \int g(x)dx + C$.
- **Then check** $h(y) = 0$: each root $y_1$ gives $y \equiv y_1$ (lost solution). Singular if no $C$ gives it.
- $y' = -2xy^2$: $y = \frac{1}{x^2 + C}$, $y = 0$ singular. $y' = 2xy$: $y = Ke^{x^2}$, $y = 0$ is $K = 0$ (not singular).
- Differential form: dividing by $x + 1$ can lose the line $x = -1$.
- $\int \frac{dy}{y^2 - 1} = \frac{1}{2} \ln \left|\frac{y-1}{y+1}\right|$; $\int \sec y\, dy = \ln|\sec y + \tan y|$; $\int \ln x = x \ln x - x$.

## Exact

- $M\,dx + N\,dy = 0$ exact $\iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ (Thm 2). Then $F_x = M,\ F_y = N$, answer $F(x, y) = C$.
- 1st way: $F = \int M\,dx + g(y)$, $F_y = N \Rightarrow g'(y)$ (only $y$!).
- 2nd way: $F = \int N\,dy + h(x)$, $F_x = M \Rightarrow h'(x)$.
- Ex. 12: $2xe^y dx + (x^2e^y + \cos y)dy = 0 \Rightarrow x^2e^y + \sin y = C$.

## Integrating factors

- $\mu \neq 0$ with $\mu M\,dx + \mu N\,dy = 0$ exact. PDE: $N\mu_x - M\mu_y = \mu(M_y - N_x)$.
- **Case 1**: $\frac{1}{N}(M_y - N_x) = f(x) \Rightarrow \mu(x) = e^{\int f(x)dx}$.
- **Case 2**: $\frac{1}{M}(M_y - N_x) = g(y) \Rightarrow \mu(y) = e^{-\int g(y)dy}$.
- Steps: not exact $\to$ factor $M_y - N_x \to$ divide by $N$, then $M \to \mu \to$ re-check $\tilde{M}_y = \tilde{N}_x \to$ build $F$.
- $x^m y^n$ / $(xy)^n$ given: multiply, match coefficients (Ex. 14: $\mu = xy^2$).
- $e^{-2\ln|x|} = x^{-2}$, $e^{5\ln|y|} = y^5$, $e^{\ln|\sin y|} = \sin y$.

## Linear (1st order)

- Standard form $y' + p(x)y = q(x)$ (divide by the coefficient of $y'$ first!).
- $\mu = e^{\int p(x)dx}$, $(\mu y)' = \mu q$, $y = \frac{1}{\mu}\left(\int \mu q\,dx + C\right)$. Contains **all** solutions.
- Domain: $p, q$ continuous on $I \ni x_0$; avoid zeros of the leading coefficient, $\tan$ at $\frac{\pi}{2} + k\pi$, $\cot$ at $k\pi$.
- Ex. 15: $y' + xy = xe^{x^2/2},\ y(0) = 1 \Rightarrow y = \frac{1}{2}e^{x^2/2} + \frac{1}{2}e^{-x^2/2}$.
- Not linear in $y$? Try $\frac{dx}{dy} + p(y)x = q(y)$: $y^2dx + (3xy - 1)dy = 0 \Rightarrow xy^3 = \frac{y^2}{2} + C$.

## Bernoulli

- Form: $y' + p(x)y = q(x)y^n$ ($n = 0, 1$: already linear).
- $u = y^{1-n}$, $u' = (1 - n)y^{-n}y'$. Multiply the equation by $(1 - n)y^{-n}$:
- $\boxed{u' + (1 - n)p\,u = (1 - n)q}$ — linear, $\mu = e^{\int (1-n)p\,dx}$, $(\mu u)' = \mu(1 - n)q$.
- $y = 0$ **is a solution when** $n > 0$ — list it.
- Hidden Bernoulli: $z = e^y$ ($e^y y'$ present), $z = \sin y$ ($\cos y\, y'$), $u = y^n$ ($ny^{n-1}y'$). Midterm Q3: $e^y y' + e^y = e^{3y+2x} \to z' + z = e^{2x}z^3$.
- Power of $x$ instead of $y$? Flip to $\frac{dx}{dy}$, unknown $x(y)$.

## Homogeneous & shifts

- Homogeneous: $y' = g(y/x) \iff f(\lambda x, \lambda y) = f(x, y) \iff M, N$ homogeneous of the **same** degree ($F(\lambda x, \lambda y) = \lambda^n F$).
- $y = ux$, $\boxed{y' = u + xu'} \Rightarrow xu' = g(u) - u \Rightarrow \int \frac{du}{g(u) - u} = \ln|x| + C$.
- Roots $u_0$ of $g(u) = u \Rightarrow$ lines $y = u_0 x$: in the family (some constant) or extra? In $M\,dx + N\,dy = 0$ form also test $x = 0$.
- Midterm Q4: $y' = \frac{y(y+2x)}{3x^2} \Rightarrow y = \frac{x}{1 + Cx^{1/3}}$ and $y = 0$.
- $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$: $\Delta = a_1b_2 - a_2b_1$. $\Delta \neq 0$: $x = u + h$, $y = v + k$ ($(h, k)$ = intersection) $\Rightarrow$ homogeneous. $\Delta = 0$: $z = a_1x + b_1y \Rightarrow$ separable.

## Riccati & Clairaut

- Riccati: $y' = py^2 + qy + r$. Needs a particular solution $y_1$ (try const, $ax + b$, $a/x$, $x^\beta$).
- $y = y_1 + \frac{1}{u} \Rightarrow \boxed{u' + (2py_1 + q)u = -p}$ (linear). $y = y_1$ is also a solution.
- $r = 0$: Bernoulli $n = 2$. ($y = y_1 + w$: Bernoulli in $w$.)
- Clairaut: $y = xp + f(p)$, $p = y'$. Differentiate: $\frac{dp}{dx}(x + f'(p)) = 0$.
- General: $\boxed{y = cx + f(c)}$. Singular: $x = -f'(p)$, $y = xp + f(p)$, eliminate $p$.
- $y = xp + p^2$: $y = cx + c^2$; singular $y = -\frac{x^2}{4}$.

## Orthogonal trajectories & toolkit

- 1) Differentiate $\Phi(x,y,c)=0$; 2) **eliminate** $c \Rightarrow y'=f(x,y)$; 3) solve $y'=-\frac{1}{f(x,y)}$.
- $x^2+y^2=c^2 \to y=kx$; $y=kx^2 \to y^2+\frac{x^2}{2}=C$; $y=ce^x \to y^2=-2x+C$; $y^2=2cx+c^2$ self-orthogonal.
- $\int \tan = -\ln|\cos|$, $\int \cot = \ln|\sin|$, $\int \frac{f'}{f}=\ln|f|$, $\int xe^{-x}=-(x+1)e^{-x}$, $\int \frac{\ln x}{x^2}=-\frac{\ln x+1}{x}$.
- $e^{-\ln f}=\frac{1}{f}$, $e^{a\ln x}=x^a$. $\ln|z|=\ln|x|+\ln|C| \Rightarrow z=Kx \Rightarrow$ add lost $z=0 \Rightarrow D\in\mathbb{R}$.

## Word problems → DE

- "rate of change of A" = $\frac{dA}{dt}$; "$\propto$" = $k\cdot$; decreasing ⇒ $-k$, $k>0$
- $\propto$ amount: $A'=-kA \Rightarrow A=A_0e^{-kt}$; $\propto$ square: $A'=-kA^2 \Rightarrow A=\frac{1}{kt+C}$; $\propto \sqrt{A}$: $2\sqrt{A}=-kt+C$
- Half-life $h$: $N=N_0(\frac{1}{2})^{t/h}$; doubles every $d$: $N=N_0 2^{t/d}$
- Data: "initially" ⇒ $t=0$ gives $C$; 2nd reading gives $k$; "cooling at 5°/min" ⇒ $T'(0)=-5$
- Midterm '23: $A(0)=50, A(2)=10 \Rightarrow A=\frac{50}{2t+1}$, $A(4)=\frac{50}{9}$

## Cooling · falling · mixing

- Cooling: $T'=-k(T-T_m) \Rightarrow T=T_m+(T_0-T_m)e^{-kt}$; $e^{-kt_1}=\frac{T(t_1)-T_m}{T_0-T_m}$
- Falling: $mv'=mg-kv \Rightarrow v=(v_0-\frac{mg}{k})e^{-kt/m}+\frac{mg}{k}$; terminal $\frac{mg}{k}$; $x=\int v\,dt$
- Mixture: $x'=r_{\text{in}}-r_{\text{out}}$, $r_{\text{in}}=\text{flow}\cdot c_{\text{in}}$, $r_{\text{out}}=\text{flow}\cdot\frac{x}{V(t)}$, $V=V_0+(\text{in}-\text{out})t$
- Ex. 30: $x'=6-\frac{3}{500}x$, $x=1000(1-e^{-3t/500})$

## Linear n-th order theory

- $a_0y^{(n)}+\cdots+a_ny=f$; IVP: $y,y',\ldots,y^{(n-1)}$ at one $x_0$
- $a_i, f$ continuous, $a_0\neq 0$ on $I$ ⇒ unique solution on all of $I$
- Superposition: needs **linear + homogeneous**
- $W=\begin{vmatrix} f_1 & f_2 \\ f_1' & f_2' \end{vmatrix}$. For **solutions**: indep ⇔ $W\neq 0$; $W\equiv 0$ or never 0
- $x|x|, x^2$: $W\equiv 0$ but independent (not solutions)
- General: $c_1\varphi_1+\cdots+c_n\varphi_n$ (F.S.S.); non-hom.: $\varphi_c+\varphi_p$

## Reduction of order

- Know $y_1$: $y=y_1z$, $y''=y_1''z+2y_1'z'+y_1z''$; $z$-term must cancel
- $w=z'$: $a_0y_1w'+(2a_0y_1'+a_1y_1)w=0$ (separable) → $w$, $C=1$
- $y_2=y_1\int w\,dx$; answer $y=c_1y_1+c_2y_2$
- Ex. 38: $(x^2+1)y''-2xy'+2y=0$, $y_1=x \Rightarrow y_2=x^2-1$
- Mid '23: $(x^2+x)y''-2xy'+2y=0$, $y_1=x \Rightarrow w=\frac{(x+1)^2}{x^2}$, $y_2=x^2+2x\ln|x|-1$

CHAPTER 18

# Self-test

Circle your answers, then check the answers chapter. No peeking.

**1** **The integrating factor $\mu = e^{-\int \frac{3}{x}dx}$ simplifies to**

**A** $-3\ln x$ **B** $x^3$

**C** $\dfrac{1}{x^3}$ **D** $e^{-3x}$

---

**2** $\displaystyle\int x\,e^{-x}\,dx = ?$

**A** $(x-1)e^{-x}+C$ **B** $\frac{x^2}{2}e^{-x}+C$

**C** $-xe^{-x}+C$ **D** $-(x+1)e^{-x}+C$

---

**3** **You reach $\ln|z| = \ln|x| + \ln|C|$ ($C \neq 0$) after dividing by $z$, and $z = 0$ also solves the equation. How does the lecturer write the final answer?**

**A** $z = Dx$, $D \in \mathbb{R}$ **B** $z = Cx$, $C > 0$ only

**C** $z = e^x + C$ **D** $z = x + C$

---

**4** **What is the order of $\left(\frac{dy}{dx}\right)^3 = y^2$?**

**A** 6 **B** 1

**C** 2 **D** 3

---

**5** **$y = (x+c)^2$ is the general solution of $(y')^2 - 4y = 0$. The solution $y \equiv 0$ is …**

**A** not a solution **B** a singular solution

**C** part of the general solution **D** the particular solution with $c = 0$

---

**6** **$x^2 + y^2 = 4$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$. On which interval are its explicit solutions valid?**

**A** $x \in \mathbb{R}$ **B** $-2 \le x \le 2$

**C** $-2 < x < 2$ **D** $x > 0$

---

**7** **For $y' = x\sqrt{y},\ y(0) = 0$, which statement is true?**

**A** The only solution is $y = \frac{x^4}{16}$

**B** Theorem 1 guarantees a unique solution

**C** Theorem 1 does not apply, and $y \equiv 0$ and $y = \frac{x^4}{16}$ are both solutions

**D** No solution exists

---

**8** **Why does $y' + p(x)y = q(x),\ y(x_0) = y_0$ (with $p, q$ continuous) have a unique solution near $x_0$ ?**

**A** Because $q(x)$ is never zero

**B** Because $f = q(x) - p(x)y$ and $\frac{\partial f}{\partial y} = -p(x)$ are continuous

**C** Because the equation is separable

**D** Because $\frac{\partial f}{\partial x} = q'(x)$ exists

---

**9** **Which equation is separable?**

**A** $\frac{dy}{dx} = x^2 + y$

**B** $\frac{dy}{dx} - x^2y^2 = x^2$

**C** $\frac{dy}{dx} = x + y$

**D** $\frac{dy}{dx} = \sin(x + y)$

---

**10** **Which constant functions solve $\frac{dy}{dx} = (y^2 - 1)\ln x$?**

**A** $y = 1$ and $y = -1$

**B** none

**C** only $y = 1$

**D** $y = 0$

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**11** **$M(x, y)\,dx + N(x, y)\,dy = 0$ is exact (Theorem 2) if and only if ...**

**A** $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$

**B** $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

**C** $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

**D** $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$

---

**12** **Building $F$ the 1st way: after $F = \int M\,dx + g(y)$, what is the next step?**

**A** Set $g(y) = 0$

**B** Differentiate $F$ with respect to $y$ and set it equal to $N$

**C** Differentiate $F$ with respect to $x$ and set it equal to $N$

**D** Integrate $N$ with respect to $x$

---

**13** **An integrating factor for $(x^2 + y^2 + x)\,dx + xy\,dy = 0$ is ...**

**A** $\frac{1}{x}$

**B** $x$

**C** $y$

**D** $e^x$

---

**14** **Example 14: $\mu = x^m y^n$ is an integrating factor for $y(3x - 2y^2)\,dx + x(3x - 5y^2)\,dy = 0$ when ...**

**A** $m = -2,\ n = -3$

**B** $m = 1,\ n = -2$

**C** $m = 1,\ n = 2$

**D** $m = 2,\ n = 1$

---

**15** **The integrating factor of $y' + \frac{2}{x}y = x^2$ ($x > 0$) is ...**

**A** $2\ln x$

**B** $x^2$

**C** $\frac{1}{x^2}$

**D** $e^{2x}$

---

**16** General solution of $y' + \frac{1}{x}y = 3x$ $(x > 0)$?

**A** $y = x^2 + C$
**B** $y = x^2 + \frac{C}{x}$
**C** $y = x^3 + \frac{C}{x}$
**D** $y = \frac{x^2 + C}{x^2}$

---

**17** For $y' + \dfrac{2}{x}y = x^2y^4$, which substitution makes the equation linear?

**A** $u = y^{-3}$
**B** $u = y^3$
**C** $u = y^{-4}$
**D** $u = y^4$

---

**18** A student solves $y' + p(x)y = q(x)y^2$ with $u = 1/y$ and gets a one-parameter family. In a find all solutions question, what must still be added?

**A** $x = 0$
**B** $y = 1$
**C** Nothing — the family is complete
**D** $y = 0$

---

**19** With $y = ux$, $\dfrac{dy}{dx} = ?$

**A** $u + x\dfrac{du}{dx}$
**B** $u\dfrac{du}{dx} + x$
**C** $x\dfrac{du}{dx}$
**D** $\dfrac{du}{dx}$

---

**20** Solving $y' = \dfrac{y(y + 2x)}{3x^2}$ with $y = ux$ gives the family $y = \dfrac{x}{1 + Cx^{1/3}}$. Which solution must be listed separately?

**A** $x = 0$
**B** $y = -2x$
**C** $y = x$
**D** $y = 0$

---

**21** For the Riccati equation $y' = py^2 + qy + r$ with known solution $y_1$, the substitution $y = y_1 + \frac{1}{u}$ gives

**A** $u' + (py_1 + q)u = -p$
**B** $u' - (2py_1 + q)u = p$
**C** $u' + qu = r$
**D** $u' + (2py_1 + q)u = -p$

---

**22** $y' = y^2 - \dfrac{y}{x} + \dfrac{\alpha}{x^2}$ has a particular solution $y_1 = x^\beta$. Then $\beta = ?$

**A** $-2$
**B** $-1$
**C** $1$
**D** $2$

---

**23** The general solution of the Clairaut equation $y = xp + f(p)$, $p = y'$, is

**A** $y = xf'(c)$
**B** $y = cx + f(c)$
**C** $y = cx$
**D** $y = f(x)$

---

**24** Differentiating $y = xp + f(p)$ with respect to $x$ leads to

**A** $p = x + f'(p)$

**B** $\dfrac{dp}{dx} = f(p)$

**C** $x\,f'(p) = p$

**D** $\big(x + f'(p)\big)\dfrac{dp}{dx} = 0$

---

**25** For the family $y = ce^{x}$, the slope with $c$ eliminated is

**A** $y' = e^{x}$

**B** $y' = y$

**C** $y' = -\dfrac{1}{y}$

**D** $y' = ce^{x}$

---

**26** Example 26: the orthogonal trajectories of the parabolas $y = kx^2$ are

**A** $y^2 - \dfrac{x^2}{2} = C$

**B** $y = Cx$

**C** $y^2 + \dfrac{x^2}{2} = C$

**D** $x^2 + y^2 = C$

---

**27** A tank initially contains 100 lt. Liquid enters at 3 lt/min and leaves at 5 lt/min. The volume at time $t$ is

**A** $100 - 2t$

**B** $100$

**C** $100 + 2t$

**D** $100 - 5t$

---

**28** Using $\frac{dN}{dt} = kN$ (no minus sign) for a decay with half-life 138 years, the constant is

**A** $k = \frac{138}{\ln 2}$

**B** $k = -138 \ln 2$

**C** $k = -\frac{\ln 2}{138}$

**D** $k = \frac{\ln 2}{138}$

---

**29** The general solution of $y'' + y = x$ is

**A** $c_1 \cos x + c_2 \sin x + x$

**B** $x + c_1$

**C** $c_1 \cos x + c_2 \sin x$

**D** $c_1 \cos x + c_2 \sin x - x$

---

**30** In reduction of order with $y = xz$, the second derivative is

**A** $y'' = z' + xz''$

**B** $y'' = 2z' + xz''$

**C** $y'' = xz''$

**D** $y'' = 2z + xz''$

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## Mock midterm (90 minutes)

Same shape as the real exam: 5 questions × 20 points. Solve on paper.

**1** A cup of coffee at 90 °C is placed in a room whose temperature is kept at 20 °C. After 10 minutes the coffee is at 60 °C. Assuming Newton's law of cooling, when will the coffee reach 30 °C? **(20 pts)**

**2** Find the general solution of $(x^2 + y^2 + x)\,dx + xy\,dy = 0$. **(20 pts)**

**3** Find the general solution of $\cos(y)\,y' + \sin(y) = x\sin^2(y)$. (Hint: let $z = \sin y$.) **(20 pts)**

**4** Find all solutions of $\frac{dy}{dx} = 2x\left(y^2 - 1\right)$. **(20 pts)**

**5** Given that $y_1 = x^2$ is a solution of $x^2y'' - 3xy' + 4y = 0$ for $x > 0$, find a second linearly independent solution and write the general solution. **(20 pts)**

CHAPTER 19

# Answers

Got one wrong? Read why, then go back to that chapter.

**1** **C** $e^{-3\ln x} = e^{\ln x^{-3}} = x^{-3}$. Forgetting the minus sign gives $x^3$. (Week+3.pdf p.40)

**2** **D** By parts with $f = x$, $g' = e^{-x}$: $-xe^{-x} + \int e^{-x}dx = -xe^{-x} - e^{-x}$. Differentiate to check. (Integration toolkit (by parts; needed in problem m-p-b-01))

**3** **A** $|z| = |Cx| \Rightarrow z = \pm Cx = Kx$ ($K \neq 0$); adding the lost solution $z = 0$ allows $K = 0$, so any real $D$. (Week+3.pdf p.24)

**4** **B** Order = highest derivative that occurs. Only $y'$ appears; the cube is a power, not a third derivative. (Week+1-3.pdf p.8, 11)

**5** **B** $y \equiv 0$ satisfies the equation, but $(x + c)^2$ equals 0 for all $x$ for no $c$ ($c = 0$ gives $x^2$, not 0). So it is singular. (Week+1-3.pdf p.20)

**6** **C** At $x = \pm 2$, $y = 0$ and the equation (division by $y$) is undefined, so the endpoints are excluded. (Week+1-3.pdf p.15)

**7** **C** $\partial f/\partial y = \frac{x}{2\sqrt{y}}$ is not continuous at $y = 0$, so the theorem is silent; indeed two solutions pass through $(0, 0)$. (Week+1-3.pdf p.25)

**8** **B** Exactly the hypotheses of Theorem 1: $f$ and $f_y = -p(x)$ are continuous on any rectangle around $(x_0, y_0)$. (Week+1-3.pdf p.26)

**9** **B** $\frac{dy}{dx} = x^2 + x^2y^2 = x^2(1 + y^2)$ factors as $g(x)h(y)$. The other right-hand sides do not factor. (Week+1-3.pdf p.33)

**10** **A** $h(y) = y^2 - 1 = 0$ at $y = \pm 1$; then $y' = 0 = 0 \cdot \ln x$. The slide warns: be careful for $y = \mp 1$. (Week+1-3.pdf p.33, 37)

**11** **B** $M = F_x$, $N = F_y$, so $M_y = F_{xy} = F_{yx} = N_x$. Cross the letters: $M$ by $y$, $N$ by $x$. (Week 2-2.pdf p.4)

**12** **B** $F_y = N$ determines $g'(y)$. ($F_x = M$ is already true by construction.) (Week 2-2.pdf p.6)

**13** **B** $M_y - N_x = 2y - y = y$; $\frac{y}{N} = \frac{y}{xy} = \frac{1}{x} = f(x)$. So $\mu = e^{\ln x} = x$. (modeled on Week 2-2.pdf p.13)

**14** **C** Matching coefficients gives $3(n + 1) = 3(m + 2)$ and $-2(n + 3) = -5(m + 1)$, so $n - m = 1$, $5m - 2n = 1$: $m = 1,\ n = 2$. (Week 2-2.pdf p.18–19)

**15** **B** $\mu = e^{\int p} = e^{\int \frac{2}{x}dx} = e^{2\ln x} = x^2$. Then $y = \frac{x^3}{5} + \frac{C}{x^2}$. (Week 2-2.pdf p.23)

**16** **B** $\mu = x$: $(xy)' = 3x^2 \Rightarrow xy = x^3 + C \Rightarrow y = x^2 + \frac{C}{x}$. The constant is divided by $\mu$ too. (Week 2-2.pdf p.24–25)

**17** **A** $n = 4$, so $u = y^{1-n} = y^{-3}$. Using $y^{-n}$ instead of $y^{1-n}$ is the classic slip. (Week+3.pdf p.3)

**18** **D** Dividing by $y^2$ needs $y \neq 0$. Since $n = 2 > 0$, $y \equiv 0$ satisfies $0 + p \cdot 0 = q \cdot 0$ but no constant in $1/y = \ldots$ produces it. (Week+3.pdf p.4)

19 **A** Product rule: $\frac{d}{dx}(ux) = u'x + u \cdot 1$. Dropping the $+u$ is the most common error. (Week+3.pdf p.23)

20 **D** The $u$-equation $xu' = \frac{1}{3}u(u-1)$ has constant solutions $u = 0$ and $u = 1$. $u = 1$ ($y = x$) is $C = 0$; $u = 0$ ($y = 0$) is not in the family. $x = 0$ is outside the domain of this derivative form. (Week+4.pdf p.18–19)

21 **D** After cancelling $y_1' = py_1^2 + qy_1 + r$: $-\frac{u'}{u^2} = (2py_1 + q)\frac{1}{u} + \frac{p}{u^2}$; multiply by $-u^2$. (Week+4.pdf p.3–4)

22 **B** Plugging in: $\beta x^{\beta-1} = x^{2\beta} - x^{\beta-1} + \alpha x^{-2}$. The $x^{2\beta}$ term must match $x^{\beta-1}$: $2\beta = \beta - 1$, $\beta = -1$ (and then $\alpha = -1$). (Week+4.pdf p.7)

23 **B** Differentiating gives $\frac{dp}{dx}(x + f'(p)) = 0$; $\frac{dp}{dx} = 0$ means $p = c$, and you just replace $p$ by $c$. (Week+4.pdf p.10–11)

24 **D** $y' = p + x\frac{dp}{dx} + f'(p)\frac{dp}{dx}$ and $y' = p$, so $p$ cancels. (Week+4.pdf p.9–10)

25 **B** $y' = ce^x$ and $ce^x = y$. $y' = -\frac{1}{y}$ is the trajectories' equation, the next step. (Week+4.pdf p.27)

26 **C** $m_1 = \frac{2y}{x}$, so $y' = -\frac{x}{2y} \Rightarrow 2y\,dy = -x\,dx \Rightarrow y^2 = -\frac{x^2}{2} + C$: ellipses. (Week+4.pdf p.26)

27 **A** $V(t) = V_0 + (\text{in} - \text{out})t = 100 + (3-5)t = 100 - 2t$; so $r_{\text{out}} = \frac{5x}{100-2t}$. (Week+5.pdf p.29)

28 **C** $N_0 e^{138k} = \frac{N_0}{2} \Rightarrow 138k = -\ln 2$. Without a minus in the model, the data make $k$ negative. (Week+5.pdf p.19)

29 **A** $\varphi_c = c_1 \cos x + c_2 \sin x$ and $\varphi_p = x$ ($0 + x = x$); general solution $= \varphi_c + \varphi_p$. (Week+6.pdf p.24–25)

30 **B** $y' = z + xz'$, so $y'' = z' + (z' + xz'') = 2z' + xz''$. (Week+6.pdf p.33)

## Mock midterm: full solutions

**QUESTION 1** A cup of coffee at 90 °C is placed in a room whose temperature is kept at 20 °C. After 10 minutes the coffee is at 60 °C. Assuming Newton's law of cooling, when will the coffee reach 30 °C?

Newton's law: $\frac{dT}{dt} = -k(T-20)$, solution $T = 20 + 70e^{-kt}$; the 10-minute reading gives $e^{-10k}$.

1. $\frac{dT}{dt} = -k(T-20)$, $T(0) = 90$, $T(10) = 60$.
2. Separable: $T(t) = 20 + Ce^{-kt}$; $T(0) = 90 \Rightarrow C = 70$.
3. $T(10) = 60 \Rightarrow 70e^{-10k} = 40 \Rightarrow e^{-10k} = \frac{4}{7} \Rightarrow k = \frac{1}{10}\ln\frac{7}{4}$.
4. $T(\bar{t}) = 30 \Rightarrow 70e^{-k\bar{t}} = 10 \Rightarrow e^{-k\bar{t}} = \frac{1}{7} \Rightarrow \bar{t} = \frac{\ln 7}{k} = \frac{10\ln 7}{\ln(7/4)}$.

**ANSWER**
$\bar{t} = \frac{10\ln 7}{\ln(7/4)} \approx 34.8$ minutes

**QUESTION 2** Find the general solution of $(x^2 + y^2 + x)\,dx + xy\,dy = 0$.

Not exact. Compute $\frac{1}{N}(M_y - N_x)$: it depends on $x$ only, so use Case 1 of Theorem 4.

1. $M = x^2 + y^2 + x$, $N = xy$. $\frac{\partial M}{\partial y} = 2y \neq \frac{\partial N}{\partial x} = y$. Not exact.

2. $M_y - N_x = y$. $\frac{1}{M}(M_y - N_x) = \frac{y}{x^2+y^2+x}$ is not a function of $y$ only; $\frac{1}{N}(M_y - N_x) = \frac{y}{xy} = \frac{1}{x} = f(x)$.
3. $\mu(x) = e^{\int f(x)dx} = e^{\ln|x|} = x$ (take $x > 0$).
4. Multiply: $(x^3 + xy^2 + x^2)\,dx + x^2 y\,dy = 0$. $\tilde{M}_y = 2xy = \tilde{N}_x$: exact.
5. $F = \int (x^3 + xy^2 + x^2)\,dx + g(y) = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} + g(y)$.
6. $F_y = x^2 y + g'(y) \overset{!}{=} x^2 y \Rightarrow g'(y) = 0 \Rightarrow g(y) = 0$.

**ANSWER**
$\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2y^2}{2} = C$, $C$ arbitrary constant (integrating factor $\mu = x$).

**QUESTION 3** Find the general solution of $\cos(y)\,y' + \sin(y) = x\sin^2(y)$. (Hint: let $z = \sin y$.)

$z = \sin y$ gives $z' = \cos y\, y'$, and the equation becomes Bernoulli in $z$ with $n = 2$.

1. Let $z = \sin y$. Then $\dfrac{dz}{dx} = \cos y \dfrac{dy}{dx}$, so the equation is $z' + z = x\,z^2$: a **Bernoulli equation** with $p = 1$, $q = x$, $n = 2$.
2. $z = 0$ (i.e. $\sin y = 0$, $y = k\pi$) solves it: $0 + 0 = 0$. For $z \neq 0$: $u = z^{1-2} = z^{-1}$, $u' = -z^{-2}z'$.
3. Multiply $z' + z = xz^2$ by $-z^{-2}$: $-z^{-2}z' - z^{-1} = -x$, i.e. $u' - u = -x$ (linear).
4. Integrating factor $\mu = e^{-\int dx} = e^{-x}$: $\dfrac{d}{dx}\left(e^{-x}u\right) = -xe^{-x}$.
5. By parts: $\int -xe^{-x}dx = xe^{-x} + e^{-x} = (x+1)e^{-x}$. So $e^{-x}u = (x+1)e^{-x} + C$, $u = x + 1 + Ce^x$.
6. Back-substitute $u = \dfrac{1}{z} = \dfrac{1}{\sin y}$.

**ANSWER**
$\sin y = \dfrac{1}{x + 1 + Ce^x}$, $C$ arbitrary; and the constant solutions $y = k\pi$ ($\sin y = 0$).

**QUESTION 4** Find all solutions of $\frac{dy}{dx} = 2x\left(y^2 - 1\right)$.

Separable with $h(y) = y^2 - 1$. Solve for $y \neq \mp 1$, then check $y = 1$ and $y = -1$ separately.

1. $\frac{dy}{dx} = \underbrace{2x}_{g(x)}\underbrace{(y^2-1)}_{h(y)}$: separable.
2. For $y \neq \mp 1$: $\int \frac{dy}{y^2-1} = \int 2x\,dx$.
3. Partial fractions $\frac{1}{y^2-1} = \frac{1/2}{y-1} - \frac{1/2}{y+1}$: $\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = x^2 + C$.
4. $\left|\frac{y-1}{y+1}\right| = e^{2C}e^{2x^2} \Rightarrow \frac{y-1}{y+1} = Ke^{2x^2}$, $K = \pm e^{2C} \neq 0$.
5. Solve for $y$: $y - 1 = Ke^{2x^2}(y+1) \Rightarrow y = \frac{1+Ke^{2x^2}}{1-Ke^{2x^2}}$.
6. If $y^2 - 1 = 0$: $y = 1$ and $y = -1$ give $y' = 0 = 2x \cdot 0$. Both are solutions (lost solutions).
7. $K = 0$ gives $y = 1$, so $y = 1$ is contained in the family. $\frac{1+A}{1-A} = -1$ would need $1 = -1$: no $K$ gives $y = -1$.

**ANSWER**

$y = \frac{1+Ke^{2x^2}}{1-Ke^{2x^2}}$, $K$ arbitrary (contains $y = 1$ for $K = 0$), and the singular solution $y = -1$.

**QUESTION 5** Given that $y_1 = x^2$ is a solution of $x^2y'' - 3xy' + 4y = 0$ for $x > 0$, find a second linearly independent solution and write the general solution.

Reduction of order: $y = x^2 z$; the $z$-term cancels and you get a first-order equation for $w = z'$.

1. $y = x^2 z$: $y' = 2xz + x^2 z'$, $y'' = 2z + 4xz' + x^2 z''$.
2. $x^2(2z + 4xz' + x^2 z'') - 3x(2xz + x^2 z') + 4x^2 z = x^4 z'' + x^3 z' = 0$.
3. $w = z'$: $xw' + w = 0 \Rightarrow \frac{w'}{w} = -\frac{1}{x} \Rightarrow w = \frac{1}{x}$.
4. $z = \int \frac{dx}{x} = \ln x \Rightarrow y_2 = x^2 \ln x$; $W[x^2, x^2 \ln x] = x^3 \neq 0$.

**ANSWER**

$y_2 = x^2 \ln x$; $y = c_1 x^2 + c_2 x^2 \ln x$